Chapter 2

Groups

2.1 The Category Grp

2.1.1 Groups

Definition 2.1.1. A *group* is a set G together with a binary operation $\cdot : G \times G \to G$ such that

- $\cdot$ is associative
- there exists an identity element $1 \in G$ such that $1 \cdot g = g \cdot 1 = g$ for all $g \in G$
- for every $g \in G$, there exists an inverse element $h \in G$ such that $g \cdot h = h \cdot g = 1$

Example. The sets $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ are groups under addition. The sets $\mathbb{Q} \setminus \{0\}$, $\mathbb{R} \setminus \{0\}$, and $\mathbb{C} \setminus \{0\}$ are groups under multiplication. The set $\mathbb{Z}/n\mathbb{Z}$ of integers modulo n forms a group under addition modulo n.

Example. The set of $n \times n$ real matrices forms a group under multiplication.

Example. The *trivial group*, denoted $*$ (also commonly denoted 1), has one element $* = \{1\}$ with multiplication defined as $1 \cdot 1 = 1$.

Example. The quaternion group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ is defined according to the rules $i^2 = j^2 = k^2 = ijk = -1$. This gives the following multiplication table

$\cdot$	1	-1	i	$-i$	j	$-j$	k	$-k$
1	1	-1	i	$-i$	j	$-j$	k	$-k$
-1	-1	1	$-i$	i	$-j$	j	$-k$	k
i	i	$-i$	-1	1	k	$-k$	$-j$	j
$-i$	$-i$	i	$-$	-1	$-k$	k	j	$-j$
j	j	$-j$	$-k$	k	-1	1	i	$-i$
$-j$	$-j$	j	k	$-k$	1	-1	$-i$	i
k	k	$-k$	j	$-j$	$-i$	i	-1	1
$-k$	$-k$	k	$-j$	j	i	$-i$	1	-1

Example. If $\mathscr{C}$ is a category and $A \in \operatorname{ob} \mathscr{C}$, then $\operatorname{Aut}_{\mathscr{C}}(A)$ forms a group under composition (it is associative because morphism composition is associative, the identity is id_A, and inverses exist because automorphisms by definition have inverses). In particular, we write $S_n = \operatorname{Aut}_{\mathsf{Set}}([n])$ for the group of permutations of $[n]$, the symmetric group on n letters.

Remark 2.1.2. Every group G can be viewed as $\operatorname{Aut}_{\mathscr{C}}(A)$ for some category $\mathscr{C}$ and $A \in \operatorname{ob} \mathscr{C}$. We let $\mathscr{C}G$ be the category with one object $*$ and morphisms $f_g : * \to *$ for each $g \in G$, with composition defined as $f_h \circ f_g = f_{h \cdot g}$. The identity is then f_1, and each f_g is an automorphism with inverse $f_{g^{-1}}$. Then G is essentially "the same" (in a sense we will make precise) as the group $\operatorname{Aut}_{\mathscr{C}G}(*)$.

The identity of a group is unique: if 1 and $1'$ are both identities, then $1 = 1 \cdot 1' = 1'$ (as $1 = 1 \cdot 1'$ viewing $1'$ as the identity, and $1 \cdot 1' = 1'$ viewing 1 as the identity). Inverses are also unique: if h_1 and h_2 are both inverses to g, then

$$h_1 = h_1 \cdot 1 = h_1 \cdot (g \cdot h_2) = (h_1 \cdot g) \cdot h_2 = 1 \cdot h_2 = h_2.$$

Thus, we can write g^{-1} for *the* inverse of g. More generally, for $n \in \mathbb{Z}$, we write

$$g^n = \begin{cases} \underbrace{g \cdot g \cdots g}_{n \text{ times}} & n > 0 \\ 1 & n = 0 \\ \underbrace{g^{-1} \cdot g^{-1} \cdots g^{-1}}_{|n| \text{ times}} & n < 0 \end{cases}.$$

Groups also have a "cancellation law": if $g \cdot h_1 = g \cdot h_2$, then

$$h_1 = 1 \cdot h_1 = (g^{-1} \cdot g) \cdot h_1 = g^{-1} \cdot (g \cdot h_1) = g^{-1} \cdot (g \cdot h_2) = (g^{-1} \cdot g) \cdot h_2 = 1 \cdot h_2 = h_2.$$

Similarly, if $h_1 \cdot g = h_2 \cdot g$,then $h_1 = h_2$. In particular, if $g \cdot h = h$ for some $h \in G$, then $g \cdot h = 1 \cdot h$, so $g = 1$. Additionally, if $hg = 1$, then $hg = g^{-1}g$, so $h = g^{-1}$, meaning it suffices to check cancellation on one side.

2.1.2 Homomorphisms

The morphisms in the category of groups are the "structure-preserving" maps:

Definition 2.1.3. Let G be a group under $\cdot_G$ and H be a group under $\cdot_H$. A *(group) homomorphism* $\phi : G \to H$ is a set map from G to H that satisfies

$$\phi(g \cdot_G h) = \phi(g) \cdot_H \phi(h).$$

Example. For any groups G and H, there is a homomorphism $\phi : G \to H$ defined by $\phi(g) = 1_H$. This is a homomorphism, as

$$\phi(g_1 g_2) = 1_H = 1_H \cdot 1_H = \phi(g_1) \cdot \phi(g_2).$$

Example. The map $\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ given by reducing modulo n is a homomorphism.

While these only seem to preserve the multiplication of groups, they indeed preserve all the structure of a group:

Proposition 2.1.4. *Let G and H be groups, and $\phi : G \to H$ a homomorphism. Then*

(1) $\phi(1_G) = 1_H$

(2) $\phi(g^{-1}) = \phi(g)^{-1}$ *for all* $g \in G$

(3) $\phi(g^n) = \phi(g)^n$ *for all* $g \in G$ *and* $n \in \mathbb{Z}$

Proof.

(1) We have that

$$\phi(1_G) = \phi(1_G \cdot_G 1_G) = \phi(1_G) \cdot_H \phi(1_G),$$

so cancellation gives $1_H = \phi(1_G)$.

(2) We have that

$$1_H = \phi(1_G) = \phi(g^{-1} \cdot_G g) = \phi(g^{-1}) \cdot_H \phi(g),$$

so that $\phi(g^{-1}) = \phi(g)^{-1}$.

(3) If $n > 0$,

$$\phi(g^n) = \phi(\underbrace{g \cdots g}_{n}) = \underbrace{\phi(g) \cdots \phi(g)}_{n} = \phi(g)^n.$$

If $n = 0$, this is true by part (a). If $n < 0$,

$$\phi(g^n) = \phi(\underbrace{g^{-1} \cdots g^{-1}}_{|n|}) = \underbrace{\phi(g)^{-1} \cdots \phi(g)^{-1}}_{|n|} = \phi(g)^n.$$

■

Definition 2.1.5. The objects in the category Grp are groups, and the morphisms are homomorphisms.

Notation. If G is a group, we write gh for $g \cdot h$.

2.1.3 Morphisms in Grp

Proposition 2.1.6. *Let G and H be groups and let $\phi : G \to H$ be a homomorphism.*

(1) ϕ is an epimorphism in Grp *if and only if ϕ is surjective as a set function*

(2) ϕ is a monomorphism in Grp *if and only if ϕ is injective as a set function*

(3) ϕ is an isomorphism in Grp *if and only if ϕ is bijective as a set function*

Proof.

(1) If ϕ is surjective as a set function and $\alpha \circ \phi = \beta \circ \phi$, then the fact that ϕ is a Set-epimorphism means $\alpha = \beta$, meaning ϕ is a Grp-epimorphism. We save the proof of the reverse direction for when we have developed more machinery (§2.4.3).

(2) If ϕ is injective as a set function and $\phi \circ \alpha = \phi \circ \beta$, then the fact that ϕ is a Set-monomorphism means $\alpha = \beta$, meaning ϕ is a Grp-monomorphism. Conversely, if ϕ is not injective, say $\phi(g) = \phi(h)$, we can define $\alpha : \mathbb{Z} \to G$ by $n \mapsto g^n$ and $\beta : \mathbb{Z} \to G$ by $n \mapsto h^n$. Then,

$$(\phi \circ \alpha)(n) = \phi(g^n) = \phi(g)^n = \phi(h)^n = \phi(h^n) = (\phi \circ \beta)(n),$$

so ϕ is not a monomorphism.

(3) If ϕ is bijective as a set function, it has an inverse set-function ψ. We'll show that ψ is a homomorphism. If $h_1, h_2 \in H$, we can write $h_1 = \phi(g_1)$ and $h_2 = \phi(g_2)$ by the surjectivity of ϕ. Now,

$$\psi(h_1h_2) = \psi(\phi(g_1)\phi(g_2)) = \psi(\phi(g_1g_2)) = g_1g_2 = \psi(\phi(g_1))\psi(\phi(g_2)) = \psi(h_1)\psi(h_2).$$

Thus, ψ is a homomorphism, and so an inverse for ϕ in Grp, meaning ϕ is an isomorphism in Grp. Conversely, if ϕ is an isomorphism in Grp, then it has an inverse homomorphism, which is in particular an inverse set function, meaning ϕ is bijective.

■

If G is a group, we'll write $\mathrm{End}(G)$ and $\mathrm{Aut}(G)$ to mean $\mathrm{End}_{\mathsf{Grp}}(G)$ and $\mathrm{Aut}_{\mathsf{Grp}}(G)$ respectively. The above result in particular implies that endomorphisms are automorphisms if and only if they are bijective.

Proposition 2.1.7. *Let G be a group and $g \in G$. The map $\varphi_g : h \mapsto ghg^{-1}$ is an automorphism of G.*

Proof. Note that φ_g is a homomorphism, as

$$\varphi_g(h_1h_2) = gh_1h_2g^{-1} = gh_1g^{-1}gh_2g^{-1} = \varphi_g(h_1)\varphi_g(h_2).$$

Now, $\varphi_g \circ \varphi_{g^{-1}} = \varphi_{g^{-1}} \circ \varphi_g = \mathrm{id}$, so φ_g is an isomorphism. ■

Automorphisms of this form are said to be *inner automorphisms* of G. Automorphisms not of this form are said to be *outer automorphisms*. Note that all automorphisms need not be inner: for example, the map $n \mapsto -n$ is an automorphism of $\mathbb{Z}$, but not an inner automorphism, as $\varphi_k(n) = k + n - k = n$ for any k.

Inner automorphisms of a group are, in a way, the "intrinsic" automorphisms, since they can be realized from "within" the group. We can see this in a category-theoretic way as well. Recall that every group G has an associated category $\mathscr{C}G$ with one object $*$ and morphisms f_g for each $g \in G$. A homomorphism $\phi : G \to H$ gives rise to a functor $\Phi : \mathscr{C} \to \mathscr{G}$ by $\Phi(*) = *$ and $\Phi(f_g) = f_{\phi(g)}$. This is a functor, as

$$\Phi(\mathrm{id}_{\mathscr{C}G}) = \Phi(f_{1_G}) = f_{\phi(1_G)} = f_{1_H} = \mathrm{id}_{\mathscr{C}H}$$

and

$$\Phi(f_g \circ f_h) = \Phi(f_{gh}) = f_{\phi(gh)} = f_{\phi(g)\phi(h)} = f_{\phi(g)} \circ f_{\phi(h)} = \Phi(f_g) \circ \Phi(f_h).$$

Conversely, every functor $\Phi : \mathscr{C}G \to \mathscr{C}H$ gives rise to a homomorphism $\phi : G \to H$ given by $f_{\phi(g)} = \Phi(f_g)$ (the same calculation shows that this is indeed a homomorphism). Thus, we can think of homomorphisms $G \to H$ as functors

$\mathscr{C}G \to \mathscr{C}H$. Now, for $\Phi, \Psi \in [\mathscr{C}G, \mathscr{C}H]$ (corresponding to homomorphisms ϕ and ψ), we can consider what it means to have a natural transformation α between them. The data of α is a map $\alpha_* : * \to *$ such that

$$\begin{tikzcd} * \arrow[r, "\Phi(f)"] \arrow[d, "\alpha_*"'] & * \arrow[d, "\alpha_*"] \\ * \arrow[r, "\Psi(f)"'] & * \end{tikzcd}$$

commutes for all $f \in \mathrm{Hom}_{\mathscr{C}G}(*, *)$. Writing $\alpha_* = f_a$ for some $a \in H$ and $f = f_g$ for some $g \in G$, we must have

$$\begin{tikzcd} * \arrow[r, "f_{\phi(g)}"] \arrow[d, "f_a"'] & * \arrow[d, "f_a"] \\ * \arrow[r, "f_{\psi(g)}"'] & * \end{tikzcd}$$

This means $a\phi(g) = \psi(g)a$ for all $g \in G$, so that $\psi = \varphi_a \circ \phi$. Thus, there is a natural transformation $\Phi \to \Psi$ if and only if there is an inner automorphism φ_a of G such that $\psi = \varphi_a \circ \phi$. In particular, all natural transformations between homomorphisms are natural isomorphisms, and the endomorphisms $G \to G$ that are naturally isomorphic to the identity map are precisely the inner automorphisms.

2.1.4 Universality in Grp

The trivial group $*$ is a zero object in Grp. For a group G, the map $* \to G$ that sends $1 \in *$ to $1 \in G$ is a homomorphism. Additionally, since homomorphisms must preserve the identity, it is the only possible homomorphism $* \to G$, so $*$ is initial. Additionally, the map $G \to *$ that sends everything to 1 is a homomorphism, and the only possible homomorphism, so $*$ is final.

If G and H are groups, we can put a group structure on their Cartesian product $G \times H$ by

$$(g_1, h_1)(g_2, h_2) = (g_1 g_2, h_1 h_2).$$

This construction is called the *direct product*. More generally, if $(G_i)_{i \in I}$ is a family of groups, we form their direct product by putting a group structure on $\prod_{i \in I} G_i$ by

$$(f_1 f_2)(i) = f_1(i) f_2(i) \quad \forall i \in I.$$

There are natural projection maps $\pi_j : \prod_{i \in I} G_i \to G_j$ given by $f \mapsto f(j)$. These are homomorphisms by the definition of the direct product.

We can also define the direct sum

$$\bigoplus_{i \in I} G_i = \left\{ f \in \prod_{i \in I} G_i : f(i) = 1_{G_i} \text{ for cofinitely many } i \in I \right\}$$

with the same group structure. We have inclusion maps $\iota_j : G_j \to \bigoplus_{i \in I} G_i$ that send $g \in G_j$ to

$$f(i) = \begin{cases} g & i = j \\ 1_{G_i} & i \neq j \end{cases}.$$

Proposition 2.1.8. *The direct product is the product in the category* Grp.

Proof. Let $(G_i)_{i \in I}$ be a family of groups, $G = \prod_{i \in I} G_i$, and suppose there are maps $\alpha_i : H \to G_i$ for some group H. If there is a map $\phi : H \to G$ that commutes with the maps to the G_i, then

$$\phi(h)(i) = \pi_i \circ \phi(h) = \alpha_i(h),$$

so ϕ is unique if it exists. Additionally, this map is a homomorphism:

$$\phi(gh)(i) = \alpha_i(gh) = \alpha_i(g)\alpha_i(h) = \phi(g)(i) \cdot \phi(h)(i) = \big(\phi(g)\phi(h)\big)(i).$$

$\blacksquare$

We would hope that $G = \bigoplus_{i\in I} G_i$ is the coproduct in Grp. If we had maps $\alpha_i : G_i \to H$, a unique map $\phi : G \to H$ that commutes with the maps from the G_i would satisfy

$$\phi(f) = \phi\left(\prod_{i\in I} \iota_i(f(i))\right) = \prod_{i\in I} \phi\circ\iota_i(f(i)) = \prod_{i\in I} \alpha_i(f(i)).$$

Note that all products are finite, as $f(i) = 1_{G_i}$ for all but finitely many i. For this to be a homomorphism, we would need the $\alpha_i(f(i))$ to commute with each other, which need not be true. For instance, we can define $\alpha_1 : \mathbb{Z}/4\mathbb{Z} \to Q_8$ by $n \mapsto i^n$, and $\alpha_2 : \mathbb{Z}/4\mathbb{Z} \to Q_8$ by $n \mapsto j^n$, and the map $\phi : \mathbb{Z}/4\mathbb{Z} \oplus \mathbb{Z}4\mathbb{Z} \to Q_8$ is not a homomorphism: for example,

$$\phi(1,0)\phi(0,1) = ij \neq ji = \phi(0,1)\phi(1,0),$$

but if ϕ were a homomorphism, we would have $\phi(1,0)\phi(0,1) = \phi(1,1) = \phi(0,1)\phi(1,0)$. However, we can salvage this.

Definition 2.1.9. A group G is said to be *abelian* if $gh = hg$ for all $g, h \in G$. The objects in the category AbGrp are abelian groups, and the morphisms are group homomorphisms.

Proposition 2.1.10. *The direct sum is the coproduct in the category* AbGrp.

Proof. Let $(G_i)_{i\in I}$ be a family of abelian groups, $G = \bigoplus_{i\in I} G_i$, and suppose there are maps $\alpha_i : G_i \to H$ for some group H. If there is a map $\phi : G \to H$ that commutes with the maps from the G_i, then

$$\phi(f) = \phi\left(\prod_{i\in I} \iota_i(f(i))\right) = \prod_{i\in I} \phi\circ\iota_i(f(i)) = \prod_{i\in I} \alpha_i(f(i)),$$

where the products are finite since cofinitely many of the $f(i)$ are 1_{G_1}. Thus, ϕ is unique if it exists. Additionally, this map is a homomorphism:

$$\phi(f_1f_2) = \prod_{i\in I} \alpha_i\big((f_1f_2)(i)\big) = \prod_{i\in I} \alpha_i(f_1(i))\alpha_i(f_2(i)) = \prod_{i\in I} \alpha_i(f_1(i)) \prod_{i\in I} \alpha_i(f_2(i)) = \phi(f_1)\phi(f_2).$$

■

Notation. We typically write abelian groups additively. That is, we denote the group operation by $+$ rather than by $\cdot$ (thus, we don't drop the operation in abelian groups), and the identity by 0 rather than 1.

We'll typically only consider direct sums in the context of abelian groups. Coproducts do indeed exist in Grp, but we will hold off on this discussion until §2.6.1.

2.2 Subgroups

2.2.1 Subgroups

Definition 2.2.1. Let G be a group. A nonempty subset $H \subseteq G$ is said to be a *subgroup* of G if $h_1h_2 \in H$ and $h_1^{-1} \in H$ for all $h_1, h_2 \in H$. We write $H \leq G$.

Example. Any group G contains the trivial subgroup $1 = \{1_G\}$. Additionally, G is always a subgroup of itself.

Example. For $m \in \mathbb{Z}$, the set $m\mathbb{Z} \subseteq \mathbb{Z}$ of multiples of m forms a subgroup of $\mathbb{Z}$.

Example. The set $\{\pm 1, \pm i\}$ is a subgroup of Q_8.

Subgroups are essentially groups in their own right that happen to be contained in other groups. Note that subgroups must contain the identity: they must contain some element g, and so also contain g^{-1}, and thus contain $gg^{-1} = 1$.

Proposition 2.2.2. *If G is a group, a subset $H \subseteq G$ is a subgroup of G if and only if H is nonempty and $gh^{-1} \in H$ for all $g, h \in H$.*

Proof. If $H \leq G$, then H is nonempty and $h^{-1} \in H$ for all $h \in H$, meaning $gh^{-1} \in H$ for all $g, h \in H$.

Conversely, if H is nonempty and $gh^{-1} \in H$ for $g, h \in H$, then there is some element $a \in H$. Then, $1 = aa^{-1} \in H$, and $g^{-1} = 1g^{-1} \in H$ for all $g \in H$. Additionally, for $g, h \in H$, $h^{-1} \in H$, so $gh = g(h^{-1})^{-1} \in H$. Thus, H is a subgroup. ■

Proposition 2.2.3. *Let G be a group and $H, H' \leq G$ be subgroups.*

(1) $H \cap H' \leq G$

(2) $H \cup H' \leq G$ *if and only if* $H \subseteq H'$ *or* $H' \subseteq H$

Proof.

(1) Since $1 \in H, H'$, $1 \in H \cap H'$, so $H \cap H'$ is nonempty. If $g, h \in H \cap H'$, then $g, h \in H$, so $gh^{-1} \in H$. Similarly, $gh^{-1} \in H'$, so $gh^{-1} \in H \cap H'$, so $H \cap H' \leq G$.

(2) If (WLOG) $H \subseteq H'$, $H \cup H' = H' \leq G$. Conversely, if $H \not\subseteq H'$ and $H' \not\subseteq H$, we can find $h \in H \setminus H'$ and $h' \in H' \setminus H$. Now, $hh' \notin H$ (if $hh' = g \in H$, then $h' = h^{-1}g \in H$) and similarly $hh' \notin H'$, so $hh' \notin H \cup H'$. Thus, $H \cup H' \not\leq G$.

■

2.2.2 Kernels and Images

Definition 2.2.4. Let G and H be groups, and let $\phi : G \to H$ be a homomorphism.

(1) the *kernel* of ϕ is $\ker \phi = \{g \in G : \phi(g) = 1_H\}$

(2) the *image* of ϕ is $\operatorname{im} \phi = \{\phi(g) : g \in G\}$

Proposition 2.2.5. *Let G and H be groups, and let $\phi : G \to H$ be a homomorphism.*

(1) $\ker \phi \leq G$

(2) $\operatorname{im} \phi \leq H$

Proof.

(1) As $\phi(1_G) = 1_H$, $1_G \in \ker \phi$, so $\ker \phi$ is nonempty. If $g, h \in \ker \phi$, then $\phi(gh^{-1}) = \phi(g)\phi(h)^{-1} = 1_H 1_H^{-1} = 1_H$, so $gh^{-1} \in \ker \phi$.

(2) As $\phi(1_G) = 1_H$, $1_H \in \operatorname{im} \phi$, so $\operatorname{im} \phi$ is nonempty. If $\phi(g), \phi(h) \in \operatorname{im} \phi$, then $\phi(g)\phi(h)^{-1} = \phi(gh^{-1}) \in \operatorname{im} \phi$.

■

Proposition 2.2.6. *Let G and H be groups, and let $\phi : G \to H$ be a homomorphism.*

(1) ϕ is injective if and only if $\ker \phi = 1$

(2) ϕ is surjective if and only if $\operatorname{im} \phi = H$

Proof.

(1) Suppose ϕ is injective. If $g \in \ker \phi$, then $\phi(g) = 1_H = \phi(1_G)$, so $g = 1_G$. Thus, $\ker \phi = 1$. Conversely, if $\ker \phi = 1$ and $\phi(g) = \phi(h)$, then $\phi(gh^{-1}) = 1_H$, so $gh^{-1} \in \ker \phi = 1$, meaning $g = h$. Thus, ϕ is injective.

(2) If ϕ is surjective, then any $h \in H$ can be written as $\phi(g)$ for $g \in G$, so $h \in \operatorname{im} \phi$, meaning $H = \operatorname{im} \phi$. Conversely, if $H = \operatorname{im} \phi$, then any $h \in H = \operatorname{im} \phi$ can be written as $\phi(g)$ for some $g \in G$, so ϕ is surjective.

■

If $\phi : G \to H$ and $K = \ker \phi$, we have a natural injection $\iota : K \to G$ given by including elements of K in G, and the commuting diagram

$$\begin{tikzcd}
 & G \arrow[dr, "\phi"] & \\
K \arrow[ur, hook, "\iota"] \arrow[dr] & & H \\
 & * \arrow[ur] &
\end{tikzcd}$$

Additionally, if $\psi : Z \to G$ satisfies

$$\begin{tikzcd}
& G \arrow[dr, "\phi"] & \\
Z \arrow[ur, "\psi"] \arrow[dr] & & H \\
& * \arrow[ur] &
\end{tikzcd}$$

then $\phi(\psi(Z)) = 1$, so that $\psi(Z) \subseteq \ker \phi$. We therefore get a map $h : Z \to K$ given by $h(z) = \psi(z)$ such that $\iota \circ h = \psi$. That is, we have the diagram

$$\begin{tikzcd}
& & G \arrow[dr, "\phi"] & \\
Z \arrow[r, "h"] \arrow[urr, bend left, "\psi"] \arrow[drr, bend right] & K \arrow[ur, hook, "\iota"] \arrow[dr] & & H \\
& & * \arrow[ur] &
\end{tikzcd}$$

Additionally, if $\iota \circ h = \psi$, then we must have $h(z) = \psi(z)$, so this map h is unique. Thus, K (together with the map $\iota : K \hookrightarrow G$) is the fibered product $G \times_H *$. This is the universal property of the kernel. We can also consider the *cokernel* of the map ϕ as the fibered coproduct $H \amalg_G *$. We will provide a concrete construction of the cokernel in §2.3.3.

Remark 2.2.7. The (co)kernel of $\phi : G \to H$ is the (co)equalizer of

$$\begin{tikzcd}
G \arrow[r, shift left, "\phi"] \arrow[r, shift right, "1"'] & H
\end{tikzcd}$$

2.2.3 Generating Subgroups

If G is a group and $X \subseteq G$, it is natural to ask about the smallest subgroup of G containing X. We can define two reasonable such subgroups:

$$H_1 = \bigcap_{X \subseteq H \leq G} H \qquad H_2 = \{x_1^{\pm 1} \cdots x_n^{\pm 1} : x_i \in X\}.$$

These are both subgroups of G, and they both have the property that they are contained in any subgroup containing X. This in particular means that they are equal. We denote this group by $\langle X \rangle$. In the case $X = \{g\}$, $\langle g \rangle = \langle X \rangle = \{g^n : n \in \mathbb{Z}\}$.

Definition 2.2.8. A group G is said to be *cyclic* if $G = \langle g \rangle$ for some $g \in G$. More generally, a subgroup $H \leq G$ is said to be cyclic if $h = \langle g \rangle$ for some $g \in G$.

Proposition 2.2.9. *Cyclic groups are abelian.*

Proof. Let $G = \langle g \rangle$ be cyclic, and let $g^n, g^m \in G$ be arbitrary elements. Then,

$$g^n g^m = g^{n+m} = g^{m+n} = g^m g^n,$$

so they commute. ■

Note that $\mathbb{Z}$ is cyclic, as $\mathbb{Z} = \langle 1 \rangle$. Similarly, $\mathbb{Z}/n\mathbb{Z}$ is cyclic. These are in fact the only possible cyclic groups, as we will soon see.

Definition 2.2.10. Let G be a group. The *order of* G, denoted $|G|$, is the cardinality of the underlying set.

Definition 2.2.11. Let G be a group and $g \in G$. The *order of* g, denoted $|g|$, is the least positive integer n such that $g^n = 1$, if such an n exists. Otherwise, $|g| = \infty$.

It is logical to use the same terminology and notation for these definitions for the following reason.

Proposition 2.2.12. *Let G be a group and $g \in G$. Then $|g| = |\langle g \rangle|$.*

Proof. If $|g| = \infty$, then the powers $1, g, g^2, \ldots$ are distinct: if $g^n = g^m$ for $n < m$, then $g^{m-n} = 1$, contradicting the fact that there are no positive integers k such that $g^k = 1$. Thus, $|g| = \infty = |\langle g \rangle|$. Now, suppose $|g| = k$ is finite. By reducing exponents modulo k, we have that $\langle g \rangle = \{1 = g^0, g, \ldots, g^{k-1}\}$. Additionally, if $g^n = g^m$ for $0 \leq n < m < k-1$, then $g^{m-n} = 1$ and $0 < m - n < k$, contradicting the minimality of k. Thus, $g^0, g, \ldots, g^{k-1}$ are distinct, so $|\langle g \rangle| = k$. ■

Proposition 2.2.13. *Let $g \in G$. Then $g^n = 1$ if and only if $|g|$ divides n.*

Proof. If $|g|$ divides n, say $n - |g| \cdot k$, then $g^n = g^{|g| \cdot k} = (g^{|g|})^k = 1^k = 1$. Conversely, if $g^n = 1$, write $n = q|g| + r$ for $0 \leq r < |g|$. Then
$$1 = g^n = g^{q|g|+r} = (g^{|g|})^q g^r = g^r,$$
so the minimality of $|g|$ means $r = 0$. ■

Proposition 2.2.14. *Let G be a cyclic group. If $|G| = \infty$, then $G \cong \mathbb{Z}$. Otherwise, if $|G| = n$ is finite, $G \cong \mathbb{Z}/n\mathbb{Z}$.*

Proof. Let $G = \langle g \rangle$. If $|G| = \infty$, then the map $\mathbb{Z} \to G$ given by $k \mapsto g^k$ is a surjective homomorphism. Additionally, it is injective because the exponents of g are distinct, as verified above. If $|G| = n$, then the map $\mathbb{Z}/n\mathbb{Z} \to G$ given by $k \mapsto g^k$ is a surjective group homomorphism because working modulo n is equivalent to the relation $g^n = 1$. Additionally, the powers $g^0, g^1, \ldots, g^{n-1}$ are distinct, so the map is injective. ■

We will be able to provide a better justification for why the map $k \mapsto g^k$ is well-defined once we have developed the theory of *quotient groups*.

2.3 Quotient Groups

2.3.1 Quotient Groups

Suppose we have an equivalence relation $\sim$ on a group G and we wish to define a group operation on $G/\sim$ by $[g_1][g_2] = [g_1 g_2]$. In order for this to be well-defined, we would need $g_1 h_1 \sim g_2 h_2$ whenever $g_1 \sim g_2$ and $h_1 \sim h_2$. We'll call equivalence relations with this property *multiplicative*. In particular, using $g \sim g$, we need $gh_1 \sim gh_2$ whenever $h_1 \sim h_2$ (we'll say an equivalence relation satisfying this is *left multiplicative*). Similarly, we need $g_1 h \sim g_2 h$ whenever $g_1 \sim g_2$ (we'll say such $\sim$ are *right multiplicative*)[1]. A multiplicative equivalence relation is thus left and right multiplicative. Conversely, an equivalence relation that is both left and right multiplicative is in fact multiplicative, as if $g_1 \sim g_2$ and $h_1 \sim h_2$, then $g_1 h_1 \sim g_1 h_2 \sim g_2 h_2$.

Proposition 2.3.1. *Let G be a group and $\sim$ an equivalence relation on G.*

(1) If $\sim$ is left multiplicative, then $H = \{h \in G : 1 \sim h\}$ is a subgroup of G, and $g_1 \sim g_2$ iff $g_2^{-1} g_1 \in H$.

(2) If $\sim$ is right multiplicative, then $H = \{h \in G : 1 \sim h\}$ is a subgroup of G, and $g_1 \sim g_2$ iff $g_2 g_1^{-1} \in H$.

Proof. We show the first statement, as the second is similar. Note $1 \sim 1$, so $1 \in H$, meaning H is nonempty. If $h_1 \in H$, then $1 \sim h_1$, so multiplying by h_1^{-1} gives $h_1^{-1} \sim 1$. Thus, $h_1^{-1} \in H$. Additionally, if $h_1, h_2 \in H$, then $1 \sim h_1$ and $1 \sim h_2$, so multiplying the second relation by h_1 gives $1 \sim h_1 \sim h_1 h_2$. Thus, $h_1 h_2 \in H$, so $H \leq G$.

If $g_1 \sim g_2$, then $g_2^{-1} g_1 \sim 1$, so that $g_2^{-1} g_1 \in H$. Conversely, if $g_2^{-1} g_1 \in H$, then $g_2^{-1} g_1 \sim 1$, so $g_1 \sim g_2$. ■

On the other hand, if $H \leq G$, the relation $\sim_H$ given by $g_1 \sim_H g_2$ iff $g_2^{-1} g_1 \in H$ is a left multiplicative equivalence relation[2]:

- Reflexive: $h^{-1} h = 1 \in H$ for all $h \in G$, so $h \sim_H h$.
- Symmetric: if $h_1 \sim_H h_2$, then $h_2^{-1} h_1 \in H$, so $h_1^{-1} h_2 = \left(h_2^{-1} h_1\right)^{-1} \in H$, so $h_2 \sim_H h_1$.
- Transitive: if $h_1 \sim_H h_2$ and $h_2 \sim_H h_3$, then $h_2^{-1} h_1, h_3^{-1} h_2 \in H$, so $h_3^{-1} h_1 = \left(h_3^{-1} h_2\right)\left(h_2^{-1} h_1\right) \in H$, meaning $h_1 \sim_H h_3$.
- Left Multiplicative: if $h_1 \sim_H h_2$, then $(gh_2)^{-1}(gh_1) = h_2^{-1} g^{-1} g h_1 = h_2^{-1} h_1 \in H$, so $gh_1 \sim gh_2$.

[1] None of this terminology is standard
[2] The notation $\sim_H$ is also not standard

We also have that $1 \sim h$ iff $h \sim 1$ iff $1^{-1}h \in H$ iff $h \in H$, so that $H = [1]_{\sim_H}$. In particular, if $H \neq K$, then the relations $\sim_H$ and $\sim_K$ are distinct, as $[1]_{\sim_H} = H \neq K = [1]_{\sim_K}$. Similarly, the relation $\sim^H$ defined by $g_1 \sim^H g_2$ iff $g_2 g_1^{-1} \in H$ is a right multiplicative equivalence relation, $H = [1]_{\sim^H}$, and the relations $\sim^H$ and $\sim^K$ are distinct for $H \neq K$.

Now, $[g]_{\sim_H}$ is the set of all g' such that $g^{-1}g' \in H$, which is precisely the set $\{gh : h \in H\}$. The set $[g]_{\sim^H}$ is the set $\{hg : h \in H\}$.

Definition 2.3.2. Let G be a group and $H \leq G$. A *left coset* of H is a set

$$gH = \{gh : h \in H\},$$

and a *right coset* of H is a set

$$Hg = \{hg : h \in H\}.$$

The set of all left cosets of H is denoted G/H, and the set of all right cosets of H is denoted $H \setminus G$.

Now, an equivalence relation $\sim$ is multiplicative if and only if it is both left and right multiplicative. If $H = [1]_\sim$, then $\sim$ is left multiplicative iff $\sim = \sim_H$ and right multiplicative iff $\sim = \sim^H$. Thus, $\sim$ is multiplicative iff $\sim = \sim_H = \sim^H$. For $\sim_H = \sim^H$, we need $gH = [g]_{\sim_H} = [g]_{\sim^H} = Hg$ for all $g \in G$.

Definition 2.3.3. Let G be a group. A subgroup $H \leq G$ is *normal* if $gH = Hg$ for all $g \in G$. We write $H \trianglelefteq G$.

Remark 2.3.4. If G is abelian, then all subgroups of G are normal. However, nonabelian groups may contain subgroups that are not normal. For example, let $G = \mathrm{GL}_2(\mathbb{R})$ be the group of invertible 2×2 real matrices under matrix multiplication, and let $H \leq G$ be the subgroup of upper triangular matrices. Then $H \ntrianglelefteq G$, as

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} H = \left\{ \begin{pmatrix} 0 & c \\ a & b \end{pmatrix} : a, b, c \in \mathbb{R} \right\}$$

$$H \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \left\{ \begin{pmatrix} b & a \\ c & 0 \end{pmatrix} : a, b, c \in \mathbb{R} \right\}.$$

Remark 2.3.5. If $K \leq H \leq G$, we write $K \trianglelefteq H$ to mean that K is normal in H. That is, that $gK = Kg$ for all $g \in H$. This need not imply that $K \trianglelefteq G$ (for instance, we have $K \trianglelefteq K$, even if $K \ntrianglelefteq G$).

Thus, if H is a normal subgroup, then $\sim_H = \sim^H$ is multiplicative, and so we can define a natural group structure on $G/\sim_H = G/H$ (and conversely, every multiplicative equivalence relation is obtained from a normal subgroup of G).

Definition 2.3.6. Let G be a group and $H \trianglelefteq G$. The *quotient group* G/H is the set $G/H = \{gH : g \in G\}$ with the group structure

$$(g_1 H)(g_2 H) = (g_1 g_2)H.$$

It is easy to see that the identity is $1H = H$, that the inverse of gH is $g^{-1}H$, and that associativity is inherited from G's associativity, so this indeed does form a group.

Remark 2.3.7. One may hope, by analogy of notation, that $G \cong G/H \times H$. Unfortunately, this is not true. For instance, if $G = \mathbb{Z}/4\mathbb{Z}$ and $H = \{0, 2\} \cong \mathbb{Z}/2\mathbb{Z}$, then $G/H \cong \mathbb{Z}/2\mathbb{Z}$, but it is not true that $\mathbb{Z}/4\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ (for instance, the former contains an element of order 4, while the latter doesn't). However, it is sometimes possible to piece together H and G/H to recover G, as we will see in §2.4.7.

Now, we have a natural group homomorphism $\pi : G \to G/H$ given by $g \mapsto gH$. The kernel of this map is $\{g : gH = H\} = H$, so every normal subgroup can be realized as the kernel of some homomorphism. The converse is also true.

Lemma 2.3.8. *Let G be a group and $H \leq G$. The following are equivalent:*

(1) $gH = Hg$ for all $g \in G$

(2) $gHg^{-1} \subseteq H$ for all $g \in G$

(3) $gHg^{-1} = H$ for all $g \in G$

Proof.

(1)⇒(2) Suppose $gH = Hg$ for all $g \in G$. Fix $h \in H$ and $g \in G$. Then, $gh =\in gH = Hg$, so we can write $gh = h'g$ for some $h' \in H$. Now, $ghg^{-1} = h'gg^{-1} = h' \in H$, so $gHg^{-1} \subseteq H$.

(2)⇒(3) Suppose that $gHg^{-1} \subseteq H$ for all $g \in G$. Fix $g \in G$, so that $gHg^{-1} \subseteq H$. Additionally, $g^{-1}Hg = g^{-1}H(g^{-1})^{-1} \subseteq H$, so $H \subseteq gHg^{-1}$ (if $h \in H$, then $g^{-1}hg \in H$, say $g^{-1}hg = h'$, so that $h = gh'g^{-1} \in gHg^{-1}$). Thus, we have $gHg^{-1} = H$.

(3)⇒(1) Suppose that $gHg^{-1} = H$ for all $g \in G$, and fix $g \in G$. Then $gH = Hg$ (if $gh \in gH$, then $ghg^{-1} \in H$, say $ghg^{-1} = h'$, so that $gh = h'g$, meaning $gH \subseteq Hg$. Similarly, $Hg \subseteq gH$).

■

Thus, if $\phi : G \to H$ is a group homomorphism and $K = \ker \phi$, then $K \trianglelefteq G$: if $g \in G$ and $k \in K$, then $\phi(gkg^{-1}) = \phi(g)\phi(k)\phi(g)^{-1} = \phi(g)\phi(g)^{-1} = 1$, so $gkg^{-1} \in K$, meaning $gKg^{-1} \subseteq K$.

In contrast to kernels, images need not be normal. For example, if $H \ntrianglelefteq G$ and $\phi : H \hookrightarrow G$ is the inclusion homomorphism, then $H = \operatorname{im} \phi$ is not normal in G.

2.3.2 Lagrange's Theorem

We now turn to determining properties about these objects.

Lemma 2.3.9. *Let G be a group and $H \leq G$. Then $g_1 \sim_H g_2$ if and only if $g_1^{-1} \sim^H g_2^{-1}$.*

Proof. If $g_1 \sim_H g_2$, then $g_2^{-1}g_1 \in H$, so that $(g_2^{-1})(g_1^{-1})^{-1} \in H$, meaning $g_1^{-1} \sim g_2^{-1}$. Conversely, if $g_1^{-1} \sim^H g_2^{-1}$, then $g_2^{-1}g_1 = g_2^{-1}(g_1^{-1})^{-1} \in H$, so $g_1 \sim_H g_2$. ■

Thus, the map $g \mapsto g^{-1}$ provides a bijection between equivalence classes of $\sim_H$ and $\sim^H$. In particular, they have the same number of equivalence classes. That is, the number of left cosets of H is equal to the number of right cosets of H.

Definition 2.3.10. Let G be a group and $H \leq G$. The *index of H in G*, denoted $[G : H]$ is the number of left (equivalently, right) cosets of H.

Now, since the (left) cosets of H are the equivalence classes of $\sim_H$, they partition G. Additionally, for each $g \in G$, $|gH| = |H|$ (the map $H \to gH$ given by $h \mapsto gh$ is a bijection with inverse $h \mapsto g^{-1}h$). Thus, cosets partition G into $[G : H]$ sets of size $|H|$, so we have deduced the following:

Theorem 2.3.11 (Lagrange)**.** *Let G be a finite group and $H \leq G$. Then*

$$|G| = [G : H] \cdot |H|.$$

In particular, $|H|$ divides $|G|$.

Corollary 2.3.12. *Let G be a finite group and $g \in G$. Then $|g|$ divides $|G|$.*

Proof. $|g| = |\langle g \rangle|$. ■

Example. The set $(\mathbb{Z}/p\mathbb{Z})^\times = \{1, \ldots, p-1\}$ forms a group under multiplication modulo p when p is prime. If $x \in (\mathbb{Z}/p\mathbb{Z})^\times$, then $|x|$ divides $|(\mathbb{Z}/p\mathbb{Z})^\times| = p - 1$, so $x^{p-1} = 1$. This is known as *Fermat's Little Theorem.*

Since the index of a subgroup is independent of whether we consider left or right cosets, we can deduce results like the following.

Proposition 2.3.13. *Let G be a group and $H \leq G$ with $[G : H] = 2$. Then $H \trianglelefteq G$.*

Proof. The left cosets of H partition G and there are exactly two left cosets of H, so $G/H = \{H, G \setminus H\}$. Similarly, $H \setminus G = \{H, G \setminus H\}$. Now, if $g \in H$, then $gH = H = Hg$. If $g \notin H$, then $gH \neq H$, so $gH = G \setminus H$, and similarly $Hg = G \setminus H$, meaning $gH = Hg$. Thus, H is normal in G. ■

2.3.3 The First Isomorphism Theorem

Perhaps the most important result in group theory is the following.

Theorem 2.3.14 (First Isomorphism Theorem)**.** *Let G and H be groups, and let $\phi : G \to H$ be a homomorphism. Then*

$$G/\ker\phi \cong \operatorname{im}\phi$$

via the map $\Phi(g\ker\phi) = \phi(g)$.

Proof. We first verify that Φ is well-defined. Suppose that $g\ker\phi = h\ker\phi$, so that $h^{-1}g \in \ker\phi$. Then $1 = \phi(h^{-1}g) = \phi(h)^{-1}\phi(g)$, so that $\phi(g) = \phi(h)$. Thus, Φ is indeed well-defined.

Now, Φ is a homomorphism, as

$$\Phi\big((g\ker\phi)(h\ker\phi)\big) = \Phi\big((gh)\ker\phi\big) = \phi(gh) = \phi(g)\phi(h) = \Phi(g\ker\phi)\Phi(h\ker\phi).$$

Additionally, if $\phi(g) \in \operatorname{im}\phi$, then $\phi(g) = \Phi(g\ker\phi)$, so Φ is surjective. Finally, if $\Phi(g\ker\phi) = 1$, then $\phi(g) = 1$, meaning $g \in \ker\phi$. Thus, $g\ker\phi = 1\ker\phi$, so Φ is injective, and thus an isomorphism. ■

Example. We can now provide a more concrete proof that cyclic groups of order $n < \infty$ are isomorphic to $\mathbb{Z}/n\mathbb{Z}$. If $|g| = n$ and $\phi : \mathbb{Z} \to \langle g\rangle$ is given by $k \mapsto g^k$, then $\ker\phi = n\mathbb{Z}$, as $g^k = 1$ iff $n \mid k$. Additionally, ϕ is surjective, so $\mathbb{Z}/n\mathbb{Z} \cong \langle g\rangle$.

Another result of a similar flavor is the following:

Theorem 2.3.15 (Universal Property of Quotients)**.** *If H is a group, $N \trianglelefteq H$, and $\phi : H \to Z$ is a homomorphism such that $\phi(N) = 1$ (that is, $N \leq \ker\phi$), then there exists a unique homomorphism $\Phi : H/N \to Z$ such that*

$$\begin{tikzcd} H \arrow[r, "\phi"] \arrow[dr, two heads, "\pi_N"'] & Z \\ & H/N \arrow[u, dashed, "\exists!\Phi"'] \end{tikzcd}$$

commutes.

Proof. If $\Phi \circ \pi_N = \phi$, then we must have $\Phi(gN) = \phi(g)$, so such a map is unique if it exists. This map Φ is a well-defined homomorphism by the same arguments as the first isomorphism theorem. ■

Put another way, we have the commuting diagram

$$\begin{tikzcd} & H \arrow[dr, two heads, "\pi_N"] & \\ N \arrow[ur, hook] \arrow[dr] & & H/N \\ & * \arrow[ur] & \end{tikzcd}$$

and whenever we have

$$\begin{tikzcd} & H \arrow[dr, "\phi"] & \\ N \arrow[ur, hook] \arrow[dr] & & Z \\ & * \arrow[ur] & \end{tikzcd}$$

we get a unique map $\Phi : H/N \to Z$ such that

$$\begin{tikzcd} & H \arrow[dr, two heads, "\pi_N"] \arrow[drr, bend left, "\phi"] & & \\ N \arrow[ur, hook] \arrow[dr] & & H/N \arrow[r, dashed, "\exists!\Phi"] & Z \\ & * \arrow[ur] \arrow[urr, bend right] & & \end{tikzcd}$$

Thus, H/N (together with the map $\pi_N : H \to H/N$) is the fibered coproduct $H \amalg_N *$, and so the cokernel of the inclusion map $N \hookrightarrow H$. This fact will be the basis of constructing the cokernel of a general map $G \to H$.

Given a group G and a subset $X \subseteq G$, we can construct $\langle X \rangle$, the smallest subgroup of G containing X by taking the intersection of all subgroups containing X. We can similarly construct the smallest *normal* subgroup containing X as the intersection

$$\bigcap_{X \subseteq N \trianglelefteq G} N.$$

This is indeed a normal subgroup, as

$$g\left(\bigcap_{X \subseteq N \trianglelefteq G} N\right)g^{-1} \subseteq \bigcap_{X \subseteq N \trianglelefteq G} gNg^{-1} \subseteq \bigcap_{X \subseteq N \trianglelefteq G} N$$

for all $g \in G$. Thus, it is the smallest normal subgroup containing X. In the case $X \leq G$ is a subgroup, we call this the *normal closure* of X.

Proposition 2.3.16 (Cokernels Exist in Grp)**.** *Let G and H be groups, $\phi : G \to H$ be a group homomorphism, and $N \trianglelefteq H$ be the normal closure of* $\operatorname{im}\phi$*. Then H/N (together with the projection map $\pi_N : H \to H/N$) is the cokernel of ϕ.*

Proof. Suppose we have a map $\psi : H \to Z$ with

$$\begin{tikzcd} & H \arrow[dr, "\psi"] & \\ G \arrow[ur, "\phi"] \arrow[dr] & & Z \\ & * \arrow[ur] & \end{tikzcd}$$

Then $\operatorname{im}\phi \subseteq \ker\psi$, so $N \leq \ker\psi$, since kernels are normal. By the universal property of quotients, there exists a unique $\Phi : H/N \to Z$ such that

$$\begin{tikzcd} H \arrow[r, "\psi"] \arrow[dr, two heads, "\pi_N"'] & Z \\ & H/N \arrow[u, dashed, "\exists!\Phi"'] \end{tikzcd}$$

Putting this in context,

$$\begin{tikzcd} & H \arrow[dr, "\pi_N"] \arrow[rr, bend left, "\psi"] & & \\ G \arrow[ur, "\phi"] \arrow[dr] & & H/N \arrow[r, dashed, "\exists!\Phi"] & Z \\ & * \arrow[ur] \arrow[rr, bend right] & & \end{tikzcd}$$

Thus, H/N with the map $\pi_N : H \to H/N$ is the fibered coproduct $H \amalg_G *$, and so is the cokernel of the map ϕ. ■

If we are working in AbGrp, all subgroups are automatically normal, so the cokernel takes the nicer form of $H/\operatorname{im}\phi$.

2.3.4 Constructing Subgroups

Let G be a group with $H, K \leq G$. We can try to construct the smallest subgroup containing H and K in a simple way. One way is to consider

$$HK = \{hk : h \in H, k \in K\},$$

but this is unfortunately not necessarily a subgroup of G. We can however learn more about this subset of G.

Proposition 2.3.17. *Let G be a group and $H, K \leq G$. Then*

$$|HK| = \frac{|H| \cdot |K|}{|H \cap K|}.$$

Proof. The map $H \times K \to HK$ given by $(h,k) \mapsto hk$ is surjective. We'll consider the number of pairs that map to the same fixed hk. If $hk = h_0k_0$, then $h^{-1}h_0 = kk_0^{-1}$ lies in $H \cap K$. Conversely, for each $a \in H \cap K$, the map $(ha, a^{-1}k)$ maps to hk. Thus, $|H \cap K|$ elements of $H \times K$ map to the same element of HK, so

$$|HK| = \frac{|H \times K|}{|H \cap K|} = \frac{|H| \cdot |K|}{|H \cap K|}.$$

■

More generally, if $A, B \subseteq G$, we can form the product

$$A \cdot B = \{ab : a \in A, b \in B\}.$$

This is commonly called the *minkowski product*. We can extend to products of multiple sets as

$$A_1 \cdots A_n = \{a_1 \cdots a_n : a_i \in A_i\}.$$

It's not hard to see that

$$(A_1 \cdots A_n)(B_1 \cdots B_m) = A_1 \cdots A_n B_1 \cdots B_m,$$

so that we in particular have an associative way of multiplying sets. Set multiplication also plays nice with unions and intersections:

$$A(B \cap C) \subseteq AB \cap AC \qquad A(B \cup C) = AB \cup AC$$
$$(A \cap B)C \subseteq AC \cap BC \qquad (A \cup B)C = AC \cup BC.$$

Note it is possible for equality to fail in the intersection case: for example, if $g \neq h$, then

$$G(\{g\} \cap \{h\}) = \varnothing \neq G = G\{g\} \cap G\{h\}.$$

However, we do have that $\{x\}(A \cap B) = \{x\}A \cap \{x\}B$.

We also have that

$$AB = \bigcup_{a \in A} aB = \bigcup_{b \in B} Ab.$$

In particular, if $K \trianglelefteq G$,

$$XK = \bigcup_{x \in X} xK = \bigcup_{x \in X} Kx = KX.$$

Conversely, if $XK = KX$ for all $X \subseteq G$, then in particular $gK = \{g\}K = K\{g\} = Kg$, so that $K \trianglelefteq G$.

If we define

$$H^{-1} = \{h^{-1} : h \in H\},$$

then the subgroup criterion requires that $H \neq \varnothing$ and $HH^{-1} \subseteq H$. All subgroups H also satisfy $HH = H$ and $H^{-1} = H$. We can see that $(HK)^{-1} = K^{-1}H^{-1}$. This makes it quite easy to verify when HK is a subgroup.

Proposition 2.3.18. *Let G be a group and $H, K \leq G$. If $H \trianglelefteq G$ or $K \trianglelefteq G$, then $HK \leq G$. If $H, K \trianglelefteq G$, $HK \trianglelefteq G$.*

Proof. Suppose that $H \trianglelefteq G$ or $K \trianglelefteq G$, so that $HK = KH$. Note that $1 = 1 \cdot 1 \in HK$, so HK is nonempty. Additionally,

$$(HK)(HK)^{-1} = HKK^{-1}H^{-1} = HKKH = HKH = HHK = HK.$$

Thus, $HK \leq G$. If both H and K are normal and $X \subseteq G$,

$$X(HK) = XHK = HXK = HKX = (HK)X,$$

so $HK \trianglelefteq G$. ■

More generally, $HK \leq G$ when $HK = KH$ (which is a weaker condition than being normal in G). In particular, if there is some $H, K \leq L \leq G$ with $K \trianglelefteq L$, then $HL \leq L \leq G$.

In the case one of H and K is normal in G, HK is the smallest subgroup containing both H and K (as it is a subgroup containing both H and K, and must be contained in every subgroup containing both H and K). We will investigate the structure of the subgroup HK further in §2.4.7.

2.3.5 Additional Isomorphism Theorems

Suppose that we have subgroups $H \leq G$ and $N \trianglelefteq G$. There are two ways we can consider what happens when we "quotient H by N". First, we can look at the smallest subgroup containing both H and N, and then quotient by N, resulting in the group HN/N. Alternatively, we can quotient H by the largest subgroup contained in both H and N, resulting in the group $H/H \cap N$. The content of the second isomorphism theorem is that these are in fact the same process.

Theorem 2.3.19 (Second Isomorphism Theorem)**.** *Let G be a group, $H \leq G$, and $N \trianglelefteq G$. Then $N \trianglelefteq HN$, $H \cap N \trianglelefteq H$, and*

$$\frac{HN}{N} \cong \frac{H}{H \cap N}.$$

Proof. Since $N \trianglelefteq G \supseteq HN$, $N \trianglelefteq HN$. Define a homomorphism $\phi : H \to HN/N$ by $h \mapsto (h1)N = hN$. If $h \in H$, $h \in \ker \phi$ if and only if $hN = N$ if and only if $h \in H \cap N$. Additionally, for $hn \in HN$, $(hn)N = hN = \phi(h)$, so ϕ is surjective. Thus, by the first isomorphism theorem,

$$\frac{H}{H \cap N} = \frac{H}{\ker \phi} \cong \operatorname{im} \phi = \frac{HN}{N}.$$

■

The second isomorphism theorem is sometimes also called the diamond isomorphism theorem, as we have the "diamond" of subgroups

$$\begin{tikzcd} & H \cap N \arrow[dl, dash] \arrow[dr, dash, dashed] & \\ H \arrow[dr, dash, dashed] & & N \arrow[dl, dash] \\ & HN & \end{tikzcd}$$

where downward lines represent inclusion, and the quotients along the solid lines are isomorphic.

Corollary 2.3.20 (Butterfly Lemma)**.** *Let G be a group, $H, K \leq G$, $h \trianglelefteq H$, and $k \trianglelefteq K$. Then,*

$$h(H \cap k) \trianglelefteq h(H \cap K) \qquad (h \cap K)k \trianglelefteq (H \cap K)k,$$

$$\frac{h(H \cap K)}{h(H \cap k)} \cong \frac{(H \cap K)k}{(h \cap K)k}.$$

Proof. We first verify that these are indeed reasonable quotients. Since $h \trianglelefteq H$, $h(H \cap K) \leq H$. Let $ab \in h(H \cap K)$. Since $ah = h = ha$ (as $a \in h$) and $h(H \cap k) = (H \cap k)h$ (as $h \trianglelefteq H \supseteq H \cap k$),

$$a\big(h(H \cap k)\big) = ah(H \cap k) = h(H \cap k) = (H \cap k)h = (H \cap k)ha = \big(h(H \cap k)\big)a.$$

Further, since $bh = hb$ (as $b \in H$ and $h \trianglelefteq H$), $bH = H = Hb$ (as $b \in H$), and $bk = kb$ (as $b \in K$ and $k \trianglelefteq k$),

$$b\big(h(H \cap k)\big) = bh(H \cap k) = hb(H \cap k) = h(bH \cap bk) = h(Hb \cap kb) = \big(h(H \cap k)\big)b.$$

Thus,

$$(ab)\big(h(H \cap k)\big) = a\big(h(H \cap k)\big)b = \big(h(H \cap k)\big)(ab),$$

so $h(H \cap K)$ is normal in $h(H \cap K)$. Similarly, $(h \cap K)k$ is normal in $(H \cap K)k$.

Since $H \cap k \subseteq H \cap K$, $(H \cap k)(H \cap K) = H \cap K$, so the second isomorphism theorem gives

$$\frac{h(H \cap K)}{h(H \cap k)} = \frac{\big(h(H \cap k)\big)(H \cap K)}{h(H \cap k)} \cong \frac{H \cap K}{(H \cap K) \cap h(H \cap k)} = \frac{H \cap K}{K \cap h(H \cap k)}.$$

Now, if an element of $h(H \cap k)$ lies in K, then the h term must lie in K (as the $H \cap k$ term automatically does), meaning that the element lies in $(h \cap K)(H \cap k)$. Conversely, $(h \cap K)(H \cap k) \subseteq K, h(H \cap k)$. Thus,

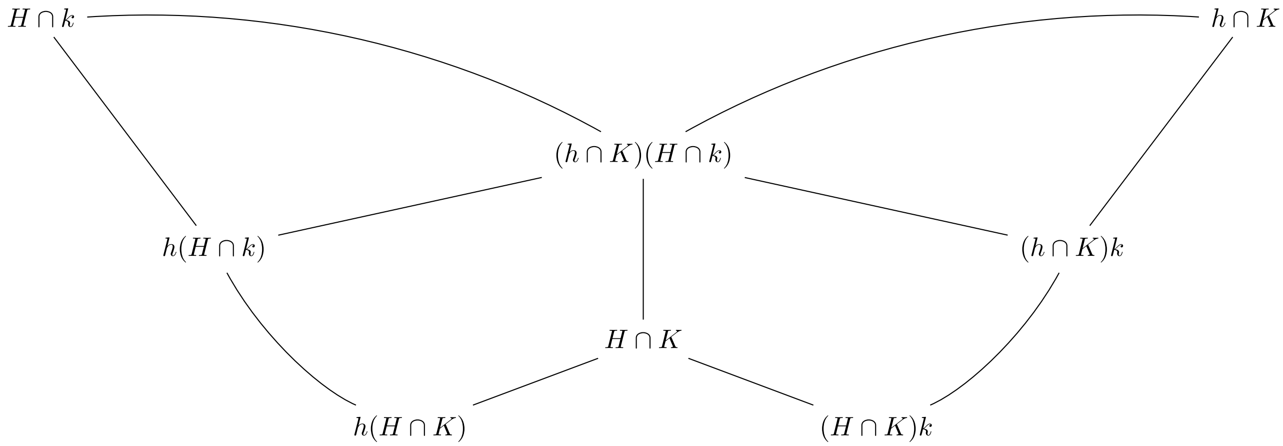
$$\frac{h(H \cap K)}{h(H \cap k)} \cong \frac{H \cap K}{(h \cap K)(H \cap k)}.$$

Symmetrically, we have

$$\frac{(H \cap K)k}{(h \cap K)k} \cong \frac{H \cap K}{(h \cap K)(H \cap k)}.$$

■

Remark 2.3.21. This is called the butterfly lemma because of the following diagram of the involved subgroups (where downward lines denote inclusion):



Theorem 2.3.22 (Third Isomorphism Theorem)**.** *Let G be a group, and $H, K \trianglelefteq G$ with $H \leq K$. Then $K/H \trianglelefteq G/H$, and*

$$\frac{G/H}{K/H} \cong \frac{G}{K}.$$

Proof. The surjective map $\pi : G \to G/K$ has kernel $K \supseteq H$, so the universal property of quotients gives a surjective map $\phi : G/H \to G/K$ with $gH \mapsto gK$. The kernel of this map is $\{kH : k \in K\} = K/H$. thus, by the first isomorphism theorem,

$$\frac{G/H}{K/H} = \frac{G/H}{\ker \phi} \cong \operatorname{im} \phi = \frac{G}{K}$$

■

The third isomorphism theorem therefore allows us to "cancel" quotients similar to the way one would cancel fractions.

Lemma 2.3.23. *Let G and H be groups, $\phi : G \to H$ be a homomorphism, $K \leq G$, and $L \leq H$.*

(1) $\phi(K) \leq H$

(2) If $K \trianglelefteq G$ and ϕ is surjective, then $\phi(K) \trianglelefteq H$

(3) $\phi^{-1}(L) \leq G$

(4) If $L \trianglelefteq H$, then $\phi^{-1}(L) \leq G$

Proof.

(1) Since $1 = \phi(1) \in \phi(K)$, $\phi(K)$ is nonempty. If $\phi(k_1), \phi(k_2) \in \phi(K)$, then $\phi(k_1)\phi(k_2)^{-1} = \phi(k_1 k_2^{-1}) \in \phi(K)$, so $\phi(K) \leq H$.

(2) Let $h = \phi(g) \in H$ and $\phi(k) \in \phi(K)$. Then $\phi(g)\phi(k)\phi(g)^{-1} = \phi(gkg^{-1})$. Since $K \trianglelefteq G$, $gkg^{-1} \in K$, so $\phi(gkg^{-1}) \in \phi(K)$. Thus, $h\phi(K)h^{-1} \subseteq \phi(K)$.

(3) Since $\phi(1) = 1 \in L$, $1 \in \phi^{-1}(L)$. Let $g_1, g_2 \in \phi^{-1}(L)$. Then $\phi(g_1 g_2^{-1}) = \phi(g_1)\phi(g_2)^{-1}$ lies in L, so $g_1 g_2^{-1} \in \phi^{-1}(L)$. Thus, $\phi^{-1}(L) \leq G$.

(4) Let $g \in G$ and $g_1 \in \phi^{-1}(L)$. Then $\phi(g g_1 g^{-1}) = \phi(g)\phi(g_1)\phi(g)^{-1}$ lies in L, since $L \trianglelefteq H$. Thus, $g\phi^{-1}(L)g^{-1} \subseteq \phi^{-1}(L)$, so $\phi^{-1}(L)$ is normal.

$\blacksquare$

Theorem 2.3.24 (Fourth Isomorphism Theorem)**.** *Let G be a group and $N \trianglelefteq G$. Then there is an order-preserving bijection between subgroups of G/N and subgroups of G containing N. Further, this bijection also provides a bijection between normal subgroups.*

Proof. Let $\pi : G \to G/N$ be the natural projection. For each $N \leq H \leq G$, $\pi(H) \leq G$. Conversely, for each $\bar{H} \leq G/N$, $\pi^{-1}(\bar{H}) \leq G$, and $N = \pi^{-1}(1) \subseteq \pi^{-1}(\bar{H})$. We'll show that these two maps are inverses.

Let $N \leq H \leq G$, and write $\bar{H} = \pi(H)$. We have that $\pi^{-1}(\bar{H}) \supseteq H$. Suppose that $g \in \pi^{-1}(\bar{H})$, so that $gN = \pi(g) \in \bar{H} = \pi(H)$. Thus, $gN = hN$ for some $h \in H$, so $h^{-1}g \in N$. Since $N \leq H$, this means that $g \in H$, so $\pi^{-1}(\bar{H}) = H$.

Now, let $\bar{H} \leq G/N$, and write $H = \pi^{-1}(\bar{H})$. We have that $\pi(H) \subseteq \bar{H}$. Let $hN \in \bar{H}$. Then $h \in H$, so $hN = \pi(h) \in \pi(H)$, so $\bar{H} \subseteq \pi(H)$.

Thus, $\pi^{-1}(\pi(H)) = H$ for all $N \leq H \leq G$, and $\pi(\pi^{-1}(\bar{H})) = \bar{H}$ for all $\bar{H} \leq G/N$. This is order preserving because π is in particular a set map (so $H \subseteq K$ means $\pi(H) \subseteq \pi(K)$, and $\bar{H} \subseteq \bar{K}$ means $\pi^{-1}(\bar{H}) \subseteq \pi^{-1}(\bar{K})$).

Finally, since the map π is surjective, if $N \leq H \trianglelefteq G$, then $\pi(H) \trianglelefteq G/N$, and if $\bar{H} \trianglelefteq G/N$, then $\pi^{-1}(\bar{H}) \trianglelefteq G$. Thus, the bijection preserves normality. $\blacksquare$

We'll typically use bars to denote the images under quotients. If $N \leq H, K \leq G$, then it is not hard to see $\overline{H \cap K} = \overline{H} \cap \overline{K}$ and $\overline{\langle H, K\rangle} = \langle \overline{H}, \overline{K}\rangle$, so the lattice of subgroups between N and G and the lattice of subgroups of G/N are the same. Thus, the fourth isomorphism theorem is sometimes called the *lattice isomorphism theorem.*

2.3.6 Commutator Subgroups

Definition 2.3.25. Let G be a group and $g, h \in G$. Their commutator is

$$[g, h] = ghg^{-1}h^{-1}.$$

The commutator subgroup $[G, G]$ of G is the subgroup

$$\big\langle [g, h] : g, h \in G \big\rangle$$

generated by all commutators in G.

The commutator subgroup measures the "failure to not be abelian". A group G is abelian if and only if every commutator is equal to 1, if and only if $[G, G] = 1$. More generally, if $\phi : G \to H$, then $\operatorname{im}\phi$ is abelian if and only if $\ker\phi \supseteq [G, G]$ (as $\operatorname{im}\phi$ is abelian if and only if ϕ sends every commutator to 1). As such, quotienting by the commutator subgroup provides the "most abelian" version of G.

Lemma 2.3.26. *If G is a group, then $[G, G] \trianglelefteq G$.*

Proof. Note that

$$[g, h]^{-1} = (ghg^{-1}h^{-1})^{-1} = hgh^{-1}g^{-1} = [h, g].$$

Thus, elements of $[G, G]$ are finite products of commutators. Now,

$$k[g, h]k^{-1} = kghg^{-1}h^{-1}k^{-1} = (kgk^{-1})(khk^{-1})(kgk^{-1})^{-1}(khk^{-1})^{-1} = [kgk^{-1}, khk^{-1}].$$

Thus, if $[g_1, h_1] \cdots [g_n, h_n]$ is an arbitrary element of $[G, G]$, and $k \in G$, then

$$k\big([g_1, h_1] \cdots [g_n, h_n]\big)k^{-1} = \big(k[g_1, h_1]k^{-1}\big) \cdots \big(k[g_n, h_n]k^{-1}\big) = [kg_1k^{-1}, kh_1k^{-1}] \cdots [kg_nk^{-1}, kh_nk^{-1}] \in [G, G],$$

so $[G, G] \trianglelefteq G$. $\blacksquare$

Definition 2.3.27. If G is a group, its *abelianization* G^{ab} is the group $G/[G, G]$.

Note that the abelianization of a group is necessarily itself abelian, as it sends commutators to 1.

Theorem 2.3.28 (Universal Property of Abelianization)**.** *Let G be a group, H an abelian group, and $\phi : G \to H$ a homomorphism. If $\pi : G \to G^{ab}$ is the projection map, then there exists a unique $\Phi : G^{ab} \to H$ such that*

$$\begin{tikzcd} G \arrow[r, "\phi"] \arrow[dr, "\pi"'] & H \\ & G^{ab} \arrow[u, dashed, "\exists!\Phi"'] \end{tikzcd}$$

commutes.

Proof. Since $\operatorname{im}\phi \le H$ is abelian, $[G,G] \le \ker\phi$. The universal property of quotients yields the desired Φ. ■

Now, if $\phi : G \to H$ is a group homomorphism, we can compose with the projection $H \to H^{ab}$ to get a homomorphism $G \to H^{ab}$. Since H^{ab} is abelian, this induces a map $\phi^{ab} : G^{ab} \to H^{ab}$. This map is determined by the commutative diagram

$$\begin{tikzcd} G \arrow[r, "\phi"] \arrow[d, two heads] & H \arrow[d, two heads] \\ G^{ab} \arrow[r, dashed, "\phi^{ab}"'] & H^{ab} \end{tikzcd}$$

In particular, we have the commutative diagram

$$\begin{tikzcd} G \arrow[r, "\phi"] \arrow[d, two heads] & H \arrow[r, "\psi"] \arrow[d, two heads] & K \arrow[d, two heads] \\ G^{ab} \arrow[r, dashed, "\phi^{ab}"] \arrow[rr, dashed, bend right, "(\psi\circ\phi)^{ab}"'] & H^{ab} \arrow[r, dashed, "\psi^{ab}"] & K^{ab} \end{tikzcd}$$

so that $\psi^{ab} \circ \phi^{ab} = (\psi \circ \phi)^{ab}$, so abelianization provides a functor $\mathsf{Grp} \to \mathsf{AbGrp}$.

There is a "forgetful functor" $U : \mathsf{AbGrp} \to \mathsf{Grp}$ that sends an abelian group $H \in \operatorname{ob}\mathsf{AbGrp}$ to the same group in $\operatorname{ob}\mathsf{Grp}$, and that sends homomorphisms between abelian groups to the corresponding homomorphisms between groups. Now, the universal property of abelianization tells us that the map $\tau_{G,H} : \phi \mapsto \phi\circ\pi$ between homomorphisms $G^{ab} \to H$ and homomorphisms $G \to H = U(H)$ is a bijection. It can be verified that the diagram

$$\begin{tikzcd} G_2^{ab} \arrow[dr, dashed, "f^{ab}"] \arrow[ddd, "\psi"'] & & & G_2 \arrow[lll, "\pi_{G_2}"'] \arrow[dl, "f"'] \arrow[ddd, "\tau_{G_2,H_2}(\psi)"] \\ & G_1^{ab} \arrow[d, "\phi"'] & G_1 \arrow[l, "\pi_{G_1}"'] \arrow[d, "\tau_{G_1,H_1}(\phi)"] & \\ & H_1 \arrow[r, leftrightarrow, "\mathrm{id}"'] \arrow[dl, dashed, "g"'] & U(H_1) \arrow[dr, dashed, "U(g)"] & \\ H_2 \arrow[rrr, leftrightarrow, "\mathrm{id}"'] & & & U(H_2) \end{tikzcd}$$

commutes when $\psi = g \circ \phi \circ f^{ab}$ (the inner square commutes by the definition of τ, the upper and lower trapezoids commute by the definition of abelianization, the left trapezoid commutes by the definition of ψ, and the right trapezoid commutes because the outer square and the rest of the diagram commute), which shows that *abelianization is the left adjoint of the forgetful functor* (recall that adjoints require the right trapezoid to commute).

We note that, in showing that $[G,G] \trianglelefteq G$, we showed that $\varphi_k([g,h]) = [kgk^{-1}, khk^{-1}]$, where φ_k denotes conjugation by k. We only used the fact that conjugation by k is a group homomorphism, and so we can in fact easily deduce a stronger result.

Proposition 2.3.29. *Let G and H be groups, and $\phi : G \to H$ be a homomorphism.*

(1) $\phi([G,G]) \subseteq [H,H]$

(2) If ϕ is surjective, then $\phi([G,G]) = [H,H]$

(3) $[G,G]$ is fixed by every automorphism of G

Proof.

(1) Since $\phi([g,h]) = [\phi(g), \phi(h)]$, the generators of $\phi([G,G])$ are contained in $[H,H]$, and so the whole image is contained in $[H,H]$.

(2) If $h_1, h_2 \in H$ with $h_1 = \phi(g_1)$ and $h_2 = \phi(g_2)$, then $\phi([g_1,g_2]) = [h_1,h_2]$, so that $[H,H]$ is contained in $\phi([G,G])$.

(3) Automorphisms are in particular surjective.

■

Thus, the commutator subgroup is highly unique, as no other subgroup can play the same role under an automorphism of G. Subgroups $H \leq G$ that are fixed under every automorphism of G are called *characteristic* subgroups, and we often write $H \operatorname{char} G$. Quite often, these subgroups are central to the structure of G, and are "canonically defined". We also note the following result.

Proposition 2.3.30. *Let $K \leq H \leq G$ be groups.*

(1) If $K \operatorname{char} H$ and $H \operatorname{char} G$ then $K \operatorname{char} G$.

(2) If $K \operatorname{char} H$ and $H \trianglelefteq G$, then $K \trianglelefteq G$.

Proof.

(1) If σ is an automorphism of G, then $\sigma|_H$ is an automorphism of H, and thus fixes K.

(2) For each $g \in G$, conjugation by g is an automorphism of H, and thus fixes K.

■

2.4 Group Actions

2.4.1 The Category $G - \mathsf{Set}$

Definition 2.4.1. Let G be a group and X be a set. A *(left) group action of G on X* is a binary operation $\cdot : G \times X \to X$ such that

- $g \cdot (h \cdot x) = (gh) \cdot x$ for all $g, h \in G$ and $x \in X$
- $1 \cdot x = x$ for all $x \in X$

We say G *acts on* X, or write $G \curvearrowright X$.

Remark 2.4.2. The requirement that $1 \cdot x = x$ is indeed necessary. Otherwise, for example, if we fix $y \in X$, the map $g \cdot x = y$ does satisfy $g \cdot (h \cdot x) = (gh) \cdot x$.

Example. The map $g \cdot x = x$ is a group action for all groups G and sets X.

Just as with group operations, we will often drop the "$\cdot$" in group actions. Now, for any $g \in G$, the map $g\_\_$ is a bijection $X \to X$ with inverse $g^{-1}\_\_$, as $g(g^{-1}\_\_) = (gg^{-1})\_\_ = 1\_\_ = \mathrm{id}_X$ and $g^{-1}(g\_\_) = (g^{-1}g)\_\_ = 1\_\_ = \mathrm{id}_X$. Additionally, if we write $\rho(g) = g\_\_$, then $\rho(gh) = (gh)\_\_ = g(h\_\_) = \rho(g) \circ \rho(h)$, so ρ is a group homomorphism $G \to \mathrm{Aut}_{\mathsf{Set}}(X)$. Conversely, given a homomorphism $\rho : G \to \mathrm{Aut}_{\mathsf{Set}}(X)$, defining $gx = \rho(g)(x)$ gives an an action of G on X. Thus, we have the following equivalent formulation of a group action.

Definition 2.4.3. Let X be a set and G be a group. A group action of G on X is a homomorphism $G \to \mathrm{Aut}_{\mathsf{Set}}(X)$. More generally, if $\mathscr{C}$ is a category and $X \in \mathrm{ob}\,\mathscr{C}$, an action of G on X is a homomorphism $G \to \mathrm{Aut}_{\mathscr{C}}(X)$. We'll write $G \curvearrowright_{\mathscr{C}} X$ if the category isn't clear.

We'll almost always consider actions in the category Set. Whenever we have two sets X and Y both acted upon by G, we can consider which maps $X \to Y$ preserve the additional structure added by an action of G.

Definition 2.4.4. Let G be a group and X and Y sets with $G \curvearrowright X, Y$. A map $f : X \to Y$ is said to be *G-equivariant* if

$$f(gx) = gf(x)$$

for all $g \in G$ and $x \in X$.

Definition 2.4.5. Let G be a group. The objects of the category $G - \mathsf{Set}$ are sets together with actions of G, and the morphisms are G-equivariant maps.

Remark 2.4.6. Homomorphisms $G \to \mathrm{Aut}_{\mathsf{Set}}(X)$ are precisely the same thing as functors $\mathscr{C}G \to \mathsf{Set}$ that send $*$ to X. Thus, we can think of the objects in the category $G - \mathsf{Set}$ as objects in the functor category $[\mathscr{C}G, \mathsf{Set}]$. If we have two functors $F, G : \mathscr{C}G \to \mathsf{Set}$ with $F(*) = X$ and $G(*) = Y$, then a natural transformation $F \to G$ is a map $\alpha_* : X \to Y$ such that

$$\begin{tikzcd} X \arrow[r, "F(f_g)"] \arrow[d, "\alpha_*"'] & X \arrow[d, "\alpha_*"] \\ Y \arrow[r, "G(f_g)"'] & Y \end{tikzcd}$$

commutes for all $g \in G$. Since $F(f_g)$ is the action of g on X, and $G(f_g)$ is the action of g on Y, this means that

$$\alpha_*(g\_\_) = g\alpha_*(\_\_).$$

Thus, the natural transformations in $[\mathscr{C}G, \mathsf{Set}]$ are precisely the G-equivariant maps in Set, so these maps are indeed the right morphisms to consider.

There is also a notion of a *right group action*, which is a map $X \times G \to X$ satisfying $x1 = x$ and $(xg)h = x(gh)$ for all $x \in X$ and $g, h \in G$. Given a left group action $\cdot_L$ of G on X, the map $\cdot R$ given by $x \cdot_R g = g^{-1} \cdot_L x$ is a right group action of G on X. Similarly, given a right group action $\cdot_R$ of G on X, the map $\cdot_L$ given by $g \cdot_L x = x \cdot_R g^{-1}$ is a left group action of G.

To translate the more general definition of a group homomorphism $G \to \mathrm{Aut}_{\mathscr{C}}(X)$ to work for right actions, we could require it to be a map ρ satisfying $\rho(gh) = \rho(h)\rho(g)$, which is precisely the same as $\rho(g) = \phi(g^{-1})$ for a homomorphism ϕ. Alternatively, we could take a homomorphism $G \to \mathrm{Aut}_{\mathscr{C}^{op}}(X)$, which is a little bit more satisfying. Similarly, a right action of G on $X \in \mathrm{ob}\,\mathsf{Set}$ corresponds to a contravariant functor $\mathscr{C}G \to \mathsf{Set}$, or an element of $[\mathscr{C}G^{op}, \mathsf{Set}]$.

Note that the convention that the composition $f \circ g$ means "apply g and then f" is fairly arbitrary: one could just as easily define $f \circ g$ to mean "apply f and then g". In this case, one typically writes functions on the right, as $x(f \circ g) = (xf)g$. If we view the elements of $\mathrm{Aut}_{\mathsf{Set}}(X)$ as right operators (i.e. composing in this way), then a right action of G is a homomorphism $G \to \mathrm{Aut}_{\mathsf{Set}}(X)$, and a left action of G is a homomorphism $G \to \mathrm{Aut}_{\mathsf{Set}^{\mathsf{op}}}(X)$.

We will stick with the typical convention of functions acting on the left, and only consider left group actions, since right group actions add no new information. We will however have cause to think about functions acting on the right when we consider modules over rings in .

The empty set together with the empty action of G is the initial object in $G - \mathsf{Set}$, and the singleton $\{*\}$ with the action $g* = *$ is the final object. If $G \curvearrowright (X_i)_{i \in I}$, we can act G on the disjoint union $\bigsqcup_{i \in I} X_i$ by acting it on each of the disjoint components. This forms the coproduct in $G - \mathsf{Set}$. We can also act G on the product $\prod_{i \in I} X_i$ by acting it pointwise, forming the product in $G - \mathsf{Set}$.

2.4.2 The Structures of Group Actions

Let G be a group. We can act G on its underlying set by left multiplication:

$$g \cdot h = gh.$$

That is, we define the homomorphism $\rho : G \to \mathrm{Aut}_{\mathsf{Set}}(G)$ by $\rho(g) : h \mapsto gh$. More generally, if $H \leq G$, we can act G on the set G/H of cosets of H by

$$g \cdot hH = (gh)H.$$

Recall that, for $g \in G$, the map $\varphi_g : h \mapsto ghg^{-1}$ is a group automorphism of G. Additionally,

$$\varphi_{gh}(k) = (gh)k(gh)^{-1} = ghkh^{-1}g^{-1} = \varphi_g \circ \varphi_h(k),$$

so the map $g \mapsto \varphi_g$ is a group homomorphism $G \to \mathrm{Aut}_{\mathsf{Grp}}(G)$. Thus, G can act on itself by conjugation. This is both an action of $G \in \mathrm{ob}\,\mathsf{Grp}$ and $G \in \mathrm{ob}\,\mathsf{Set}$.

If $H \leq G$, then $gHg^{-1} \leq G$, and so we can also act G on the set $\{H : H \leq G\}$ via conjugation. More generally, if $\alpha \in \mathrm{Aut}(G)$, then $\alpha(H) \leq G$, so we can act $\mathrm{Aut}(G)$ on $\{H : H \leq G\}$. In fact, if $G \curvearrowright X$, then we can act G on $\mathscr{P}(X)$ by $gA = \{ga : a \in A\}$.

Definition 2.4.7. Let G be a group, X a set, and $G \curvearrowright X$. Let $x \in X$.

- The *orbit* of x is the set
$$\mathcal{O}_x = \{gx : g \in G\}.$$

- The *stabilizer* of x is the set
$$G_x = \{g \in G : gx = x\}.$$

Proposition 2.4.8. *Let G be a group, X a set, $G \curvearrowright X$, and $x \in X$. Then $G_x \leq G$.*

Proof. Since $x = 1x$, $1 \in G_x$, so G_x is nonempty. If $g, h \in G_x$, then
$$(gh^{-1})x = (gh^{-1})(hx) = (gh^{-1}h)x = gx = x,$$
so $gh^{-1} \in G_x$. Thus, $G_x \leq G$. ∎

Write $x \sim y$ iff $x \in \mathcal{O}_y$ (that is, $x \sim y$ iff $x = gy$ for some $g \in G$). Since $x = 1x$, $\sim$ is reflexive. If $x = gy$, then $g^{-1}x = g^{-1}(gy) = (g^{-1}g)y = y$, so $\sim$ is symmetric. If $x = gy$ and $y = hz$, then $x = gy = g(hz) = (gh)z$, so $\sim$ is transitive. Thus, orbits are the equivalence classes of an equivalence relation, and so partition X. Unlike with cosets, orbits need not have the same size. However, we can determine the size of an orbit.

Theorem 2.4.9 (Orbit-Stabilizer)**.** *Let G be a group, X a set, $G \curvearrowright X$, and $x \in X$. Then there is a G-equivariant bijection between the set G/G_x of left cosets of G_x and the orbit $\mathcal{O}_x$ of x. In particular, $|\mathcal{O}_x| = [G : G_x]$ if the latter is finite, and $|G| = |\mathcal{O}_x| \cdot |G_x|$ if G is finite.*

Proof. Note that $gx = hx$ if and only if $h^{-1}gx = x$ if and only if $h^{-1}g \in G_x$ if and only if $gG_x = hG_x$. Thus, the map $f : G/G_x \to \mathcal{O}_x$ given by $gG_x \mapsto gx$ is a well-defined bijection. It is G-equivariant as
$$f\big(g(hG_x)\big) = f\big((gh)G_x\big) = (gh)x = g(hx) = gf(hG_x).$$
∎

Just as in Grp, a morphism in $G - \mathsf{Set}$ is an isomorphism if and only if it's bijective as a set map. Since the orbits partition X, every action of G on a set is the disjoint union of the actions of G on the orbits. Orbit-stabilizer tells us that the actions of G on orbits are isomorphic (in $G - \mathsf{Set}$) to the actions of G on its quotient groups. Thus, every action of G on a set is (isomorphic to) the coproduct (or disjoint union) of the actions of G on certain quotient groups (note that the empty set is the disjoint union of an empty collection, so this also holds for the empty action). These actions of G on quotient groups are what are known as *transitive* actions.

Definition 2.4.10. Let G be a group, X a set, and $G \curvearrowright X$. The action of G on X is said to be

- *transitive* if, for all $x, y \in X$, there is $g \in G$ with $y = gx$ (equivalently, there is only one orbit)
- *faithful* if, for any $g, h \in G$, $gx = hx$ for all $x \in X$ if and only if $x = y$ (equivalently, the homomorphism $G \to \mathrm{Aut}_{\mathsf{Set}}(X)$ is injective)
- *free* if, for any $g \neq h \in G$ and $x \in X$, $gx \neq hx$ (equivalently, $gx \neq x$ for all $g \neq 1 \in G$ and $x \in X$)
- *torsor* if it is both free and transitive

It is not hard to check that these properties are preserved under $G - \mathsf{Set}$ isomorphism, as one would expect. Note that the action of G on G/H is transitive, as $(hg^{-1})(gH) = hH$. Conversely, orbit-stabilizer tells us that every transitive action is $G-\mathsf{Set}$-isomorphic to the action of G on G/H for some $H \leq G$. Thus, since the orbits of an action partition the set, this means that every G-action is $G - \mathsf{Set}$-isomorphic to the coproduct of transitive G-actions.

Note that free actions are faithful (with the exception of the action of a nontrivial group on the empty set, which is free but not faithful), but faithful actions need not be free: for example, the action of S_n on $[n]$ is faithful but not free for $n > 2$. The action of G on itself by left multiplication is free (and thus faithful) in addition to being transitive. If $H \leq G$ is a nontrivial subgroup, the action of G on G/H is not free (take $1 \neq h \in H$, so that $hH = 1H$), but can be faithful (S_n acts faithfully and transitively on $[n]$, and so faithfully and transitively on S_n/G_1).

The action of G on itself by left multiplication is torsor, as it is free and transitive. The action of G on the empty set is also torsor. These are essentially the only two ways that an action can be torsor.

Proposition 2.4.11. *Let G be a group, X a nonempty set, and let the action $G \curvearrowright X$ be torsor. Then there is a G-equivariant bijection between G and X.*

Proof. Let $x \in X$. Since the action $G \curvearrowright X$ is transitive, $\mathcal{O}_x = X$, and we have a $G - \mathsf{Set}$ isomorphism between the action of G on X and the action of G on G/G_x. Since the action of G on X is free, $G_x = 1$, so $G/G_x = G$, meaning we have a $G - \mathsf{Set}$ isomorphism between the actions of G on itself (via left multiplication) and on X. We note that this isomorphism is given by $g \leftrightarrow gx$. ∎

More generally, if we have a (nonempty) free action, we can split it up into its orbits to see that it is the coproduct (i.e. disjoint union) of torsor actions, and so the coproduct (i.e. disjoint union) of some number of copies of the action of G on itself by left multiplication. Note that the empty action of G is the coproduct of an empty collection of $G \curvearrowright G$s, so all free actions come about in this way.

We now turn to examining the action of G by conjugation.

Definition 2.4.12. Let G be a group acting on itself by conjugation, and let $g \in G$.

- The *conjugacy class* of g is its orbit under the action of conjugation
$$\{hgh^{-1} : h \in G\}.$$
- The *centralizer* of g is its stabilizer under the action of conjugation
$$Z_G(g) = \{h \in G : hgh^{-1} = g\} = \{h \in G : gh = hg\}.$$
- If $H \leq G$, the *centralizer* of H is
$$Z_G(H) = \bigcap_{g \in H} Z_G(g) = \{h \in G : gh = hg \ \forall g \in H\}.$$
- The *center* of G is $Z(G) = Z_G(G) = \{h \in G : gh = hg \ \forall g \in G\}$.

Since $Z_G(h)$ is the stabilizer of h, $Z(G)$ consists of all $g \in G$ such that φ_g is the identity map. In particular, $Z(G)$ is the kernel of the map $g \mapsto \varphi_g$, and $G/Z(G) \cong \mathrm{Inn}(G)$, where $\mathrm{Inn}(G) \leq \mathrm{Aut}(G)$ is the group of inner automorphisms of G. Note also that $Z(G)$ equivalently consists of all $g \in G$ that commute with every element of G. By this definition, it is not hard to check that $Z(G) \operatorname{char} G$.

Note that $\varphi_h(g) = g$ if and only if $gh = hg$ if and only if $\varphi_g(h) = h$, so $Z_G(g)$ is equivalently the set of fixed points of φ_g, and $Z(G)$ is the set of points that are fixed by the action of G. Thus, $Z_G(g) = G$ if and only if φ_g fixes G if and only if $\varphi_g = \mathrm{id}$ if and only if $g \in Z(G)$. More generally, $Z_G(H) = G$ if and only if $H \leq Z(G)$.

Definition 2.4.13. Let G act on its subgroups via conjugation. The *normalizer* of $H \leq G$ is its stabilizer under the action
$$N_G(H) = \{g \in G : gHg^{-1} = H\}.$$

For any $h \in H$, $hHh^{-1} = H$, so $H \leq N_G(H)$. Additionally, by definition, $H \trianglelefteq N_G(H)$. Further, $H \trianglelefteq K$ if and only if $K \leq N_G(H)$, so in particular $N_G(H) = G$ if and only if $H \trianglelefteq G$. Now, if $K \leq N_G(H)$, then $KH = HK$, so we can extend the second isomorphism theorem: if $K \leq N_G(H)$, then $HK \leq G$, $H \trianglelefteq HK$, $H \cap K \trianglelefteq K$, and $HK/H \cong K/H \cap K$.

Now, the number of conjugates of H is $[G : N_G(H)]$ by the Orbit-Stabilizer theorem. This allows us to derive the following result.

Proposition 2.4.14. *Let G be a finite group, and $H < G$. Then there exists an element of G that is not conjugate to any element of H.*

Proof. Suppose that $\bigcup_{g \in G} gHg^{-1} = G$. Then, we have that
$$|G| - 1 = |G \setminus \{1\}| = \left| \bigcup_{g \in G} gHg^{-1} \setminus \{1\} \right| \leq [G : N_G(H)] \cdot (|H| - 1) \leq [G : H](|H| - 1) = |G| - [G : H],$$
so that $[G : H] \leq 1$ and $G = H$. ■

As we will see in , in the "worst case" that $H \cap gHg^{-1} = 1$ for all $g \in G \setminus H$, the set of all elements not conjugate to H in fact forms a subgroup of G.

2.4.3 Applications of Group Actions

As a first application of group actions, we verify the claim made in §2.1.3 that epimorphisms in Grp are surjective.

Proposition 2.4.15. *Let G and H be groups, and $\phi : G \to H$ be a* Grp *epimorphism. Then ϕ is surjective.*

Proof. Let H act on the set $H/\phi(G)$ of left cosets of $\phi(G)$ via left multiplication, and let $X = H/\phi(G) \cup \{*\}$, where $*$ is some arbitrary element. We can extend to an action on X by $h* = *$, so this gives us a homomorphism $\Phi : H \to S = \mathrm{Aut}_{\mathsf{Set}}(X)$.

Let $\sigma \in S$ be the map that swaps $\phi(G)$ and $*$, and fixes all other cosets of $\phi(G)$, and let $\Psi(h) = \sigma\Phi(h)\sigma^{-1}$. Now, if $h \in \phi(G)$, then $\Phi(h)$ fixes both $\phi(G)$ and $*$, meaning that $\Phi(h) = \sigma\Phi(h)\sigma^{-1} = \Psi(h)$. Thus, $\Phi \circ \phi = \Psi \circ \phi$, so $\Phi = \Psi$, since ϕ is an epimorphism. Now, for $h \in H$,

$$h\phi(G) = \Phi(h)(\phi(G)) = \Psi(h)(\phi(G)) = \sigma\Phi(h)\sigma^{-1}(\phi(G)) = \sigma\Phi(h)(*) = \sigma(*) = \phi(G),$$

so $h \in \phi(G)$. Thus, ϕ is surjective. ∎

We can also generalize the result of Proposition 2.3.13

Proposition 2.4.16. *Let G be a finite group, and let p be the smallest prime dividing $|G|$. If $H \leq G$ with $[G : H] = p$, then $H \trianglelefteq G$.*

Proof. Let $H \curvearrowright G/H$ by left multiplication, and note $|G/H| = [G : H] = p$. Since $H \in G/H$ is a fixed point, if $gH \neq H$, then $H \notin \mathcal{O}_{gH}$, so that $|\mathcal{O}_{gH}| \leq p - 1$. Now, by orbit-stabilizer, $|H| = |\mathcal{O}_{gH}| \cdot |G_{gH}|$, so $|\mathcal{O}_{gH}|$ divides $|G|$. Since p is the smallest prime dividing $|G|$ and $|\mathcal{O}_{gH}| < p$, we must have $|\mathcal{O}_{gH}| = 1$, so that $hgH = gH$ for all $h \in H$ and $g \in G \setminus H$. Thus, $g^{-1}hg \in H$ for all $g \in G \setminus H$ and $h \in H$, and so for all $g \in G$ (since it is always true when $g \in H$), so that $H \trianglelefteq G$. ∎

Suppose that a group G is acting on a finite set X. If the orbits of the action are $\mathcal{O}_1, \ldots, \mathcal{O}_n$, then $|X| = \sum_{i=1}^{n} |\mathcal{O}_i|$. Rearranging, suppose that $\mathcal{O}_1, \ldots, \mathcal{O}_k$ are the orbits of size > 1, and let $Z = \mathcal{O}_{k+1} \cup \cdots \cup \mathcal{O}_n$ consist of the fixed points of the action (i.e. the points $z \in X$ with $gz = z$ for all $g \in G$). Then, $|X| = |Z| + \sum_{i=1}^{k} |\mathcal{O}_i|$. If we fix some $x_i \in \mathcal{O}_i$, then

$$|X| = |Z| + \sum_{i=1}^{k} [G : G_{x_i}].$$

We have thus deduced the following

Proposition 2.4.17 (Class Equation)**.** *Let G be a group, X a finite set, and $G \curvearrowright X$. If $Z \subseteq X$ is the set of fixed points, $\mathcal{O}_1, \ldots, \mathcal{O}_k$ are the orbits of size > 1, and $x_i \in \mathcal{O}_i$, then*

$$|X| = |Z| + \sum_{i=1}^{k} [G : G_{x_i}].$$

In particular, if G is finite and $g_1, \ldots, g_k$ are representatives of the nontrivial conjugacy classes of G, then

$$|G| = |Z(G)| + \sum_{i=1}^{k} [G : Z_G(g_i)].$$

The class equation allows us to perform analysis on the number of fixed points.

Definition 2.4.18. Let p be a prime. A group G is said to be a *p-group* if $|G| = p^r$ for some integer $r \geq 0$.

Lemma 2.4.19. *Let G be a p-group acting on a finite set X, and let Z be the fixed points of the action. Then $|X| \equiv |Z| \pmod{p}$.*

Proof. By the class equation, write

$$|X| = |Z| + \sum_{i=1}^{k} [G : G_{x_i}].$$

Since each x_i lies in an orbit of size > 1, each of the G_{x_i} is a proper subgroup of G. Thus, $[G : G_{x_i}] > 1$ for each i. Since $[G : G_{x_i}]$ must divide $|G| = p^r$, this means that $[G : G_{x_i}]$ is a positive power of p, and so in particular divisible by p. Thus, reducing the class equation modulo p gives $|X| \equiv |Z| \pmod{p}$. ∎

This lemma allows us to prove some powerful results by acting p-groups on appropriate sets.

Proposition 2.4.20. *Let G be a p-group and $N \trianglelefteq G$. Then $Z(G) \cap N \neq 1$. In particular, $Z(G) \neq 1$.*

Proof. Let G act on N by conjugation. Since $Z(G)$ consists of the fixed points of G acting on itself by conjugation, $Z(G) \cap N$ consists of the fixed points of this action. Thus, $|Z(G) \cap N| \equiv |G| \equiv 0 \pmod{p}$. Now, since $1 \in Z(G) \cap N$, this means that $|Z(G \cap N)| > 0$, so $|Z(G) \cap N| \geq p > 1$. ■

Theorem 2.4.21 (Cauchy)**.** *Let G be a group and p a prime dividing $|G|$. Then G contains an element of order p.*

Proof. Let X be the set

$$X = \big\{(g_1, g_2, \ldots, g_p) : g_i \in G,\ g_1 g_2 \cdots g_p = 1\big\}.$$

Then $|X| = |G|^{p-1}$, since the value g_p is uniquely determined by the choices of $g_1, \ldots, g_{p-1}$. Let $\mathbb{Z}/p\mathbb{Z}$ act on X by shifting:

$$n \cdot (g_1, g_2, \ldots, g_p) = (g_{1+n}, g_{2+n}, \ldots, g_n).$$

The fixed points of this action are of the form $(g, \ldots, g)$ with $g^p = 1$. If Z is the set of fixed points, then $|Z| \equiv |X| = |G|^{p-1} \equiv 0 \pmod{p}$. Now, $(1, \ldots, 1)$ is a fixed point, so there must be at least one other fixed point, $(g, \ldots, g)$ with $g \neq 1$ and $g^p = 1$. Since p is a prime, this means g has order p. ■

Group actions are also quite useful in the realm of combinatorics via the following result.

Proposition 2.4.22 (Burnside's Lemma)**.** *Let G be a group acting on a finite set X, and let n be the number of orbits of the action. For each $g \in G$, let $\operatorname{Fix}(g) = \{x \in X : gx = x\}$. Then*

$$n = \frac{1}{|G|} \sum_{g \in G} |\operatorname{Fix}(g)|.$$

Proof. Consider the set $S = \{(g, x) \in G \times X : gx = x\}$. For each $g \in G$, there are $|\operatorname{Fix}(g)|$ choices of x such that $(g, x) \in S$, so that $|S| = \sum_{g \in G} |\operatorname{Fix}(g)|$. Alternatively, for each $x \in X$, there are $|G_x|$ choices of g such that $(g, x) \in S$, so that $|S| = \sum_{x \in X} |G_x|$.

Now, if the orbits of the action $G \curvearrowright X$ are $\mathcal{O}_1, \ldots, \mathcal{O}_n$, then

$$|S| = \sum_{x \in X} |G_x| = \sum_{i=1}^{n} \sum_{x \in \mathcal{O}_i} |G_x| = \sum_{i=1}^{n} \sum_{x \in \mathcal{O}_i} \frac{|G|}{|\mathcal{O}_i|} = \sum_{i=1}^{n} |G| = n \cdot |G|.$$

Thus,

$$n = \frac{1}{|G|} \sum_{g \in G} |\operatorname{Fix}(g)|.$$

■

Burnside's Lemma is useful in counting "up to symmetry". For example, suppose we want to create a bracelet with k beads, each of which can take one of n colors. We want to count the number of such bracelets, where two bracelets are considered the same if one can be rotated to get the other. This is the same as counting the number of orbits of $[n]^k$ under the shifting action of $\mathbb{Z}/k\mathbb{Z}$. Now, the number of fixed points under the action of $i \in \mathbb{Z}/k\mathbb{Z}$ is $n^{\gcd(i,k)}$ (if $i(x_0, \ldots, x_{k-1}) = (x_0, \ldots, x_{k-1})$, then $c_j = c_{j+mi}$ (where indices are taken modulo k) for any m, which is precisely the same as saying $c_j = c_\ell$ if $j \equiv \ell \pmod{\gcd(i,k)}$). Thus, the number of such bracelets is

$$\frac{1}{k} \sum_{i=0}^{k-1} n^{\gcd(i,k)}.$$

2.4.4 The Symmetric Group

Recall that the symmetric group S_n is the group of permutations $\sigma : [n] \to [n]$. We'll denote elements $\sigma \in S_n$ by writing the string $[\sigma(1) \cdots \sigma(n)]$. For example, $[32154]$ is the map

$$1 \mapsto 3 \qquad 2 \mapsto 2 \qquad 3 \mapsto 1 \qquad 4 \mapsto 5 \qquad 5 \mapsto 4.$$

We could alternatively consider the "cycles" formed by repeated applications of σ. For example, [32154] has the cycles

$$1 \mapsto 3 \mapsto 1 \qquad 2 \mapsto 2 \qquad 4 \mapsto 5 \mapsto 4.$$

Every permutation can be decomposed into these cycles[3]. We denote the cycle decomposition using parentheses:

$$[32154] = (13)(2)(45).$$

We'll often omit cycles with a single element, so the above permutation becomes (13)(45). We define the *cycle type* of a permutation to be the multiset of the lengths of the nontrivial cycles in the permutation's cycle notation. For instance, the cycle type of the above permutation is $\{2, 2\}$.

Definition 2.4.23. A *k-cycle* is a permutation whose cycle decomposition is of the form $(i_1 i_2 \cdots i_k)$. A *transposition* is another name for a 2-cycle.

If we notate permutations by cycle notation, then the cycle decomposition of a permutation is the same as writing it as a product of disjoint cycles. If two cycles permute a disjoint set of elements, then they commute. Thus, since a k-cycle has order k, the order of a permutation with cycle type $\{k_1, \dots, k_\ell\}$ is $\operatorname{lcm}(k_1, \dots, k_\ell)$, as raising a permutation to the mth power is the same as raising each of the cycles to the mth power.

Now,

$$\sigma(i_1 \cdots i_k)\sigma^{-1} = (\sigma(i_1) \cdots \sigma(i_k)),$$

so that

$$\sigma(i_1^1 \cdots i_{k_1}^1) \cdots (i_1^\ell \cdots i_{k_\ell}^\ell)\sigma^{-1} = \big(\sigma(i_1^1) \cdots \sigma(i_{k_1}^1)\big) \cdots \big(\sigma(i_1^\ell) \cdots \sigma(i_{k_\ell}^\ell)\big).$$

Since σ is a bijection, this means that the cycles on the right hand side are disjoint. Thus, permutations in the same conjugacy class must have the same cycle type. Conversely, if we have permutations

$$(i_1^1 \cdots i_{k_1}^1) \cdots (i_1^\ell \cdots i_{k_\ell}^\ell)(i_1^{\ell+1}) \cdots (i_1^m)$$

and

$$(j_1^1 \cdots j_{k_1}^1) \cdots (j_1^\ell \cdots j_{k_\ell}^\ell)(j_1^{\ell+1}) \cdots (j_1^m)$$

of the same cycle type (where we write out the cycles of length 1), then we can take σ to be the permutation $i_p^q \mapsto j_p^q$ to see that they are conjugate.

Since any permutation can be broken into its cycle decomposition, the k-cycles (for $k = 2, \dots, n$) generate S_n. There are, however, many smaller generating sets for S_n.

Proposition 2.4.24. *The following sets generate S_n:*

(1) The set of all transpositions

(2) The transpositions $(12), (13), \dots, (1n)$

(3) The elements (12) *and* $(23 \cdots n)$

(4) The transpositions $(12), (23), \dots, (n-1, n)$

Proof.

(1) It suffices to show that any k-cycle can be written as the product of transpositions. We have that

$$(i_1 \cdots i_k) = (i_1 i_2)(i_2 i_3) \cdots (i_{k-1} i_k),$$

where evaluation is done right to left.

(2) It suffices to show an arbitrary transposition can be written in terms of these permutations. We have that

$$(ij) = (1i)(1j)(1i).$$

(3) It suffices to show a transposition of the form $(1i)$ can be written in terms of these permutations. For $0 \le i \le n-2$,

$$(1, 2+i) = (23 \cdots n)^i (12)(23 \cdots n)^{-i}.$$

[3]Formally, the cycles of σ are the orbits of the $\mathbb{Z}$-action on $[n]$ given by $m \cdot i = \sigma^m(i)$

(4) It suffices to show that $(23\cdots n)$ can be written as a product of these transpositions. We have that

$$(23\cdots n) = (23)(34)\cdots(n-1,n).$$

$\blacksquare$

We now turn to examining the structure of $\mathrm{Aut}(S_n)$ for $n \geq 3$ (when $n = 1$ or 2, $\mathrm{Aut}(S_n) = 1$). We first study $\mathrm{Inn}(S_n)$, the group of inner automorphisms.

Proposition 2.4.25. *For $n \geq 3$, $\mathrm{Inn}(S_n) \cong S_n$.*

Proof. Recall that $\mathrm{Inn}(S_n) \cong S_n/Z(S_n)$ (as $Z(S_n)$ is the kernel of the map $\sigma \mapsto \varphi_\sigma$), so it suffices to show that $Z(S_n) = 1$. Let $\sigma \in S_n$ not be the identity, and take i with $i \neq \sigma(i)$. Take $j \neq i, \sigma(i)$. Then,

$$\big(\sigma(ij)\big)(i) = \sigma(j) \qquad \big((ij)\sigma\big)(i) = \sigma(i),$$

so $\sigma \notin Z(S_n)$. $\blacksquare$

Now, if $\varphi : S_n \to S_n$ is an automorphism and $\tau = \rho\sigma\rho^{-1}$, then $\varphi(\tau) = \varphi(\rho)\varphi(\sigma)\varphi(\rho)^{-1}$, meaning that φ sends elements in the same conjugacy class to elements in the same conjugacy class. Conversely, since φ^{-1} does the same, σ and τ are conjugate if and only if $\varphi(\sigma)$ and $\varphi(\tau)$ are conjugate. Additionally, since conjugacy classes in S_n are the same as cycle types, elements of the same conjugacy class have the same order. This allows us to prove many facts about automorphisms of S_n.

Lemma 2.4.26. *If $\varphi : S_n \to S_n$ is an automorphism that sends the conjugacy class of transpositions to itself, then φ is inner.*

Proof. We first show that $\varphi(1i) = (ab_i)$ for some distinct $a, b_2, \ldots, b_n$. Notice that the product of two distinct transpositions is either a 3-cycle (if they are not disjoint), or a permutation of cycle type $\{2, 2\}$ (if they are disjoint). In particular, since the products $(1i)(1j)$ have order 3, the products $\varphi(1i)\varphi(1j)$ must also have order 3. Since $\varphi(1i)$ and $\varphi(1j)$ are by assumption transpositions, this means they are not disjoint.

Now, the only way our claim could fail is if we have $\varphi(1i) = (ab)$, $\varphi(1j) = (bc)$, and $\varphi(1k) = (ac)$ for some a, b, c. However,

$$(1i)(1j)(1k) = (kji1)$$

has order 4, while

$$(ab)(bc)(ac) = (cb)$$

has order 2, so this is impossible. Thus, we must be able to write $\varphi(1i) = (ab_i)$ for some shared a. Since φ is a bijection, the b_i must all be distinct.

Now, if $\sigma = [ab_2 \cdots b_n]$, then

$$(\sigma^{-1}\varphi\sigma)(1i) = (1i).$$

Since transpositions of the form $(1i)$ generate S_n, this means that $\sigma^{-1}\varphi\sigma = \mathrm{id}$, so that $\varphi = \varphi_\sigma$ is inner. $\blacksquare$

Theorem 2.4.27. *Let $n \geq 3$ and $n \neq 6$. Then every automorphism of S_n is inner, and $\mathrm{Aut}(S_n) = \mathrm{Inn}(S_n) \cong S_n$.*

Proof. It suffices to show that every automorphism sends transpositions to transpositions. Now, it must send the conjugacy class of transpositions to another conjugacy class. Additionally, the elements in this conjugacy class must have order 2, so this other conjugacy class must consist of elements of cycle type $\{2, \ldots, 2\}$.

Now, the number of products of $k \leq n/2$ disjoint transpositions (that is, the number of permutations of cycle type $\{\underbrace{2, \ldots, 2}_{k}\}$) is $\frac{n!}{2^k k!(n-2k)!}$, as can be verified by a simple counting argument. The number of transpositions is $\binom{n}{2} = \frac{n(n-1)}{2}$. We thus seek solutions with $k \neq 1$ to

$$\frac{n(n-1)}{2} = \frac{n!}{2^k k!(n-2k)!},$$

or equivalently

$$\frac{(n-2)!}{(n-2k)!} = 2^{k-1}k!.$$

Expending out the expression on the left,

$$(n-2)(n-3)\cdots(n-2k+1) = 2^{k-1}k!.$$

Both sides have exactly $2k-2$ terms (where we ignore the factor of 1 in the $k!$ on the right). Additionally, if $n-2k+1 \geq 2$, then the terms on the left are greater than or equal to their counterparts on the right, and there are some strict inequalities due to the factors of 2 on the right, meaning there are no solutions to the equation. If $n-2k+1 = 1$, so that $n = 2k$, then we want to solve

$$(2k-2)(2k-3)\cdots(k+1) = 2^{k-1}.$$

One can check that $k = 2$ is not a solution, while $k = 3$ is. If $k \geq 4$, then $k+1 > 4$, so that

$$(2k-2)(2k-3)\cdots(k+1) > 2 \cdot 2 \cdots 2 \cdot 4 = 2^{k-1}.$$

Thus, the only nontrivial solution is $n = 6$ and $k = 3$. This means that, as long as $n \neq 6$, any automorphism of S_n must send transpositions to transpositions, and thus be inner. ■

S_6 indeed does have an outer automorphism, as we will construct now. There are $\binom{6}{2} = 15$ two-element subsets of $[6]$. We'll consider groupings of these subsets into 5 groups of 3 pairs, such that each of $1, 2, \ldots, 6$ appears once in each group. One can check that there are exactly 6 such groupings:

$$\begin{aligned}
\mathbf{A} &= \{12/34/56, 13/25/46, 14/26/35, 15/24/36, 16/23/45\} \\
\mathbf{B} &= \{12/34/56, 13/26/45, 14/25/36, 15/23/46, 16/24/35\} \\
\mathbf{C} &= \{12/35/46, 13/24/56, 14/25/36, 15/26/34, 16/23/45\} \\
\mathbf{D} &= \{12/35/46, 13/26/45, 14/23/56, 15/24/36, 16/25/34\} \\
\mathbf{E} &= \{12/36/45, 13/24/56, 14/26/35, 15/23/46, 16/25/34\} \\
\mathbf{F} &= \{12/36/45, 13/25/46, 14/23/56, 15/26/34, 16/24/35\}
\end{aligned}$$

Now, we have an action of S_6 on $\{\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}, \mathbf{E}, \mathbf{F}\}$ by permuting the elements in the groupings. This gives a homomorphism $\varphi : S_6 \to S_6$.

We first verify that φ is an automorphism. Since it is a map between finite sets, it suffices to show φ is injective. It is a fact (that we will prove in §2.4.5) that the only normal subgroups of S_6 are $1, A_6, S_6$, where A_6 is a subgroup that in particular contains the 3-cycle (123). Now,

$$\varphi(123) = (\mathbf{ADE})(\mathbf{BFC}),$$

so that (123) doesn't lie in $\ker\varphi$. Thus, $\ker\varphi = 1$. Additionally,

$$\varphi(12) = (\mathbf{AB})(\mathbf{CD})(\mathbf{EF}),$$

so φ is not inner.

Note that an outer automorphism must swap transpositions and permutations of cycle type $\{2, 2, 2\}$. Thus, if $\sigma, \tau \in \mathrm{Aut}(S_6)$ are outer, then $\sigma^{-1}\tau$ is inner (as it sends transpositions to transpositions), meaning $\mathrm{Inn}(S_6)$ has exactly 2 cosets in $\mathrm{Aut}(S_6)$. Thus, we have the following.

Theorem 2.4.28. $\mathrm{Inn}(S_6) \cong S_6$, *and* $[\mathrm{Aut}(S_6) : \mathrm{Inn}(S_6)] = 2$.

The reader can check for themself that the outer automorphism φ we constructed satisfies $\varphi \circ \varphi = \mathrm{id}$. This in fact allows us to determine the structure of $\mathrm{Aut}(S_6)$ using the results of §2.4.7.

2.4.5 The Alternating Group

We now turn to describing the subgroup $A_n \leq S_n$, known as the alternating group. We write

$$\Delta = \prod_{i<j}(x_i - x_j)$$

and

$$\sigma(\Delta) = \prod_{i<j}(x_{\sigma(i)} - x_{\sigma(j)}).$$

Since σ is a permutation, each $(x_i - x_j)$ factor appears exactly once in $\sigma(\Delta)$ as either $(x_i - x_j)$ or $(x_j - x_i)$. Thus, $\sigma(\Delta) = \pm\Delta$.

Definition 2.4.29. The *sign* of $\sigma \in S_n$ is $(-1)^\sigma = \sigma(\Delta)/\Delta$. σ is said to be *even* if $(-1)^\sigma = 1$, and *odd* if $(-1)^\sigma = -1$.

If
$$I(\sigma) = \{i < j : \sigma(i) > \sigma(j)\}$$
and $n(\sigma) = |I(\sigma)|$, then $(-1)^\sigma = (-1)^{n(\sigma)}$. Now, if $(i,j) \in I(\sigma\tau)$, then either τ swaps i and j and σ keeps them in order, or τ keeps i and j in order and σ swaps them. Thus,
$$I(\sigma\tau) = \{i < j : (i,j) \in I(\tau), (\tau(j), \tau(i)) \notin I(\sigma)\} \sqcup \{i < j : (i,j) \notin I(\tau), (\tau(i), \tau(j)) \in I(\sigma)\}.$$
Taking sizes,
$$\begin{aligned} n(\sigma\tau) &= \Big(n(\tau) - \#\{i < j : (i,j) \in I(\tau), (\tau(j), \tau(i)) \in I(\sigma)\}\Big) + \Big(n(\sigma) - \#\{i < j : (i,j) \in I(\tau), (\tau(j), \tau(i)) \in I(\sigma)\}\Big) \\ &= n(\sigma) + n(\tau) - 2\#\{i < j : (i,j) \in I(\tau), (\tau(j), \tau(i)) \in I(\sigma)\}. \end{aligned}$$
Thus,
$$n(\sigma\tau) \equiv n(\sigma) + n(\tau) \pmod 2,$$
so that
$$(-1)^{\sigma\tau} = (-1)^\sigma(-1)^\tau.$$
Thus, we have the following.

Proposition 2.4.30. *The sign is a group homomorphism $S_n \to \{\pm 1\}$.*

It is very possible that the sign is the trivial homomorphism $\sigma \mapsto 1$. Luckily, this is not the case.

Proposition 2.4.31. *If $\tau \in S_n$ is a transposition, then $(-1)^\tau = -1$.*

Proof. Pick a suitable σ so that $\tau = \sigma(12)\sigma^{-1}$. Then,
$$(-1)^\tau = (-1)^\sigma(-1)^{(12)}(-1)^{\sigma^{-1}} = (-1)^{(12)},$$
since the sign is a homomorphism to the abelian group $\{\pm 1\}$. Since (12) swaps the two minimal elements, the only way to get a pair $(i,j) \in I(12)$ is if $i = 1$ and $j = 2$. ■

Definition 2.4.32. The *alternating group* A_n is kernel of the sign homomorphism.

The fact that the sign is a homomorphism tells us that if we can write $\sigma = \tau_1 \cdots \tau_k = \tau_1' \cdots \tau_\ell'$, where the τ_i and τ_j' are transpositions, then $k \equiv \ell \pmod 2$. This tells us that a permutation lies in A_n if and only if it can be expressed as the product of an even number of transpositions.

Proposition 2.4.33. *The 3-cycles generate A_n.*

Proof. Since every element of A_n can be written as a product of an even number of transpositions, it suffices to show that a product of two transpositions can be written as a product of 3-cycles. Let τ and τ' be two distinct transpositions. If τ and τ' are not disjoint, say $\tau = (ij)$ and $\tau' = (jk)$, then
$$\tau\tau' = (ij)(jk) = (ijk).$$
If τ and τ' are disjoint, say $\tau = (ij)$ and $\tau' = (k\ell)$, then
$$\tau = (ij)(k\ell) = (ij)(jk)(jk)(k\ell) = (ijk)(jk\ell).$$
■

Since we can write k-cycles as
$$(i_1 \cdots i_k) = (i_1 i_2) \cdots (i_{k-1} i_k),$$
the sign of a k-cycle is $(-1)^{k-1}$. Thus, the sign of a permutation of cycle type $\{k_1, \dots, k_\ell\}$ is
$$(-1)^{k_1 + \cdots + k_\ell - \ell}.$$
Now, a permutation σ of cycle type $\{k_1, \dots, k_\ell\}$ lies in A_n if and only if $\sum_{i=1}^\ell k_i - 1$ is even. Thus, an even number of the $k_i - 1$ terms are odd, which is to say an even number of the k_i are even.

The first isomorphism theorem tells us that $S_n/A_n \cong \{\pm 1\}$, so that $[S_n : A_n] = 2$. Now, we know that the conjugacy classes in S_n correspond to cycle types. However, just because two elements are conjugate in S_n doesn't mean they are conjugate in A_n, so a natural question to ask is how conjugacy classes break up in A_n.

Proposition 2.4.34. *Let G be a finite group and $H \trianglelefteq G$ with $[G:H]=2$. If $[h]_G$ and $[h]_H$ denote the conjugacy classes of h in G and H respectively, then $[h]_G \neq [h]_H$ if and only if $Z_G(h) \leq H$. In this case, $[h]_H$ is half of $[h]_G$.*

Proof. Suppose that $[h]_H = [h]_G$. Let $g \in G \setminus H$ and take $k \in H$ such that $ghg^{-1} = khk^{-1}$. Then $k^{-1}g \in Z_G(h)$, so that $Z_G(h) \not\leq H$ in particular.

Now suppose that $[h]_H \neq [h]_G$, and let $g \in G$ with $ghg^{-1} \notin [h]_H$ (in particular, $g \notin H$). Then, for every $k \in Z_G(h)$, $k^{-1}g \notin H$. Since H only has two cosets, this means $k \in H$, so $Z_G(h) \leq H$.

In the action of G or H on H by conjugation, we have

$$\#[h]_G = [G : Z_G(h)] \qquad \#[h]_H = [H : Z_H(g)].$$

If $[h]_G \neq [h]_H$, then $Z_G(h) \leq H$, so that $Z_G(h) = Z_H(h)$. Thus, in this case,

$$\#[h]_G = [G : Z_G(h)] = [G : Z_H(h)] = [G : H][H : Z_H(h)] = 2 \cdot \#[h]_H.$$

■

Corollary 2.4.35. *Let $\sigma \in S_n$. Then $[\sigma]_{A_n} \neq [\sigma]_{S_n}$ if and only if the cycle decomposition of σ consists of odd cycles of distinct lengths.*

Proof. If σ contains an even cycle, then that even cycle (which doesn't lie in A_n) centralizes σ, so $Z_{A_n}(\sigma) \not\leq A_n$. If σ contains two odd cycles of the same length, say $(a_1 \cdots a_k)$ and $(b_1 \cdots b_k)$, then the permutation $(a_1 b_1) \cdots (a_k b_k)$ (which doesn't lie in A_n as k is odd) centralizes σ, so $Z_{A_n}(\sigma) \not\leq A_n$. Thus, if $Z_{A_n}(\sigma) \leq A_n$, then the cycle decomposition of σ consists of odd cycles of distinct lengths.

Now, if the cycle decomposition of σ consists of odd cycles of distinct length, then a permutation that centralizes σ must cyclically permute each of the cycles of σ (since the cycles of σ have distinct lengths), and is thus a product of the cycles of σ. Since the cycles of σ all have odd lengths, and thus lie in A_n, this means that any permutation centralizing σ lies in A_n. ■

Note that the "distinctness" condition on cycle lengths includes cycles of length 1. This in particular means that the conjugacy class of 3-cycles splits in A_4, but not in A_n for $n \geq 5$. Since 3-cycles generate A_n, this is a quite powerful fact.

Definition 2.4.36. A group G is said to be *simple* if the only normal subgroups of G are 1 and G.

Theorem 2.4.37. *A_n is simple for $n \geq 5$.*

Proof. Let $1 \neq N \trianglelefteq A_n$. Since the 3-cycles generate A_n and are all conjugate to each other in A_n (as $n \geq 5$), it suffices to show that N contains a 3-cycle. Let $\sigma \in N$ be a nonidentity permutation.

Suppose first that $\sigma^2 \neq \mathrm{id}$, and take i with $i, \sigma(i), \sigma^2(i)$ all distinct. Let τ be the 3-cycle $(i, \sigma^2(i), \sigma(i))$. The commutator

$$[\sigma, \tau] = \sigma(\tau\sigma^{-1}\tau^{-1})$$

lies in N because N is normal. However, we also have

$$[\sigma, \tau] = (\sigma\tau\sigma^{-1})\tau^{-1} = (\sigma(i), \sigma^3(i), \sigma^2(i))(i, \sigma(i), \sigma^2(i)) = (i, \sigma^3(i), \sigma^2(i)).$$

Thus, N contains a 3-cycle.

Now suppose that $\sigma^2 = \mathrm{id}$, so that σ is the product of an even number of transpositions. In particular, the conjugacy class of σ doesn't split in A_n. Now, take i with $\sigma(i) \neq i$ and $j \neq i, \sigma(i)$ with $\sigma(j) \neq j$, and write $\sigma = (i, \sigma(i))(j, \sigma(j))\tau$, where τ is some permutation fixing $i, \sigma(i), j\sigma(j)$ with $\tau^2 = \mathrm{id}$. Since

$$\sigma' = (ij)\sigma(ij)^{-1} = (j, \sigma(i))(i, \sigma(j))\tau$$

is conjugate to σ in S_n, it is conjugate to σ in A_n, and thus lies in N. This means N also contains

$$\sigma\sigma' = (i, \sigma(i))(j, \sigma(j))(j, \sigma(i))(i, \sigma(j)),$$

which permutes at most 4 elements. Since $n \geq 5$, $\sigma\sigma'$ has a fixed point, so we may assume without loss of generality that our original permutation σ had a fixed point k.

Now, since k is a fixed point of σ, it is in particular a fixed point of τ. Thus,

$$\rho = (ik)\sigma(ik)^{-1} = (k, \sigma(i))(j, \sigma(j))\tau.$$

Since ρ is conjugate to σ in S_n, it is conjugate to σ in A_n, and thus lies in N. Now,

$$\sigma\rho = (i, \sigma(i))(j, \sigma(j))\tau(k, \sigma(i))(j, \sigma(j))\tau = (i, \sigma(i), k)$$

is a 3-cycle, as desired. ■

We can now verify the claim made that the only normal subgroups of S_n are 1, A_n, and S_n. If $N \trianglelefteq S_n$, then $N \cap A_n \trianglelefteq A_n$. If $N \cap A_n = A_n$, then $N \cap A_n = A_n$ or S_n (since A_n has index 2 in S_n). Otherwise, $N \cap A_n = 1$, which means $|NA_n| = |N| \cdot |A_n|/|N \cap A_n| = |N| \cdot |A_n|$. Either $|N| = 1$, in which case $N = 1$, or $|N| = 2$, in which case N is not normal (as normal subgroups of order 2 must be fixed under conjugation, and thus contained in $Z(S_n) = 1$). A more elegant proof of this fact comes from the Jordan-Hölder theorem, which we will show in §2.5.1.

2.4.6 Sylow's Theorem

Sylow's Theorem is a series of statements regarding the structure of a finite group. It particularly concerns p-subgroups (that is, subgroups whose orders are powers of a prime p). We start with a fairly simple, yet important, result.

Proposition 2.4.38. *Let G be a group of order p a prime. Then $G \cong \mathbb{Z}/p\mathbb{Z}$.*

Proof. Let $1 \neq g \in G$. Since $|g| \neq 1$ divides $|G| = p$, we must have $|g| = p$, so that $G = \langle g \rangle$. Thus, $G \cong \mathbb{Z}/p\mathbb{Z}$ is cyclic. ■

The structure of larger p-groups is not as simple, but we can still gain insight.

Proposition 2.4.39. *Let G be a p-group with $|G| = p^r$. Then there is a subnormal series*

$$1 = G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_{r-1} \trianglelefteq G_r = G$$

with $|G_i| = p^i$ and $G_i \trianglelefteq G$ for all i.

Proof. We proceed by induction on r, with the base cases of 0 being trivial. Let $|G| = p^r$ for $r \geq 1$. Since G is a p-group, $Z(G)$ is nontrivial, and thus contains an element of order p by Cauchy's theorem. If $H \leq Z(G)$ has order p, then $H \trianglelefteq G$ and G/H has order p^{r-1}. By induction, we have a series

$$1 = H/H = G_1/H \trianglelefteq G_2/H \trianglelefteq \cdots \trianglelefteq G_{r-1}/H \trianglelefteq G_r/H,$$

where $|G_i/H| = p^{i-1}$ and $G_i/H \trianglelefteq G/H$. Thus, we have a series

$$1 = G_0 \trianglelefteq H = G_1 \trianglelefteq G_2 \trianglelefteq \cdots \trianglelefteq G_{r-1} \trianglelefteq G_r$$

with $|G_i| = p^i$ and $G_i \trianglelefteq G$. ■

We now turn to studying p-*sub*groups of finite groups. We will make heavy use of Lemma 2.4.19.

Lemma 2.4.40. *Let G be a finite group and $H \leq G$ a p-subgroup. Then $[G : H] \equiv [N_G(H) : H] \pmod p$.*

Proof. Act H on the set G/H of left cosets of H. The coset gH is a fixed point if and only if $hgH = gH$ for all $h \in H$, if and only if $g^{-1}hg \in H$ for all $h \in G$. Since H is finite, this happens if and only if $g^{-1}Hg = H$, if and only if $g \in N_G(H)$. Thus, the fixed points of the action are the cosets $N_G(H)/H$, and so there are $[N_G(H) : H]$ fixed points. Since H is a p-group, this means

$$[G : H] \equiv [N_G(H) : H] \pmod p.$$

■

Lemma 2.4.41. *Let G be a finite group and $H \leq G$ a p-subgroup with $p \mid [G : H]$. Then there is a $H \trianglelefteq K \leq G$ with $[K : H] = p$.*

Proof. We have that $0 \equiv [G : H] \equiv [N_G(H) : H] \pmod p$, so that p divides $|N_G(H)/H|$. By Cauchy's theorem, $N_G(H)/H$ has an element of order p, and thus a subgroup $1 \trianglelefteq \bar{K}$ of order p. This corresponds to a subgroup $H \trianglelefteq K \leq N_G(H) \leq G$ with $[K : H] = p$. ■

We are able to use our lemma to build up p-subgroups of any given group. This is the key point in the proof of Sylow's Theorem.

Definition 2.4.42. Let p be a prime and let G be a group of order $p^r m$, where $p \nmid m$. A *Sylow p-subgroup* of G is a subgroup of order p^r. The set of all Sylow p-subgroups of G is denoted $\mathrm{Syl}_p(G)$.

Theorem 2.4.43 (Sylow)**.** *Let p be a prime and G be a finite group of order $p^r m$ with $p \nmid m$.*

(1) G has a Sylow p-subgroup.

(2) If $H \leq G$ is a p-subgroup and $P \leq G$ is a Sylow p-subgroup, then there is $g \in G$ such that $H \leq gHg^{-1}$. In particular, any two Sylow p-subgroups are conjugates.

(3) $\#\,\mathrm{Syl}_p(G) \mid m$ and $\#\,\mathrm{Syl}_p(G) \equiv 1 \pmod p$.

Proof.

(1) Let H be a p-subgroup of G of maximal order. If $|H| \neq p^r$, then $p \mid [G : H]$, so we can find $H \leq K \leq G$ with $[K : H] = p$. Then, $|K| = p|H|$ is a p-subgroup of G of strictly larger order, contradicting the maximality of H. Thus, $|H| = p^r$.

(2) Let H act on the set G/P of left cosets of P. Since $|G/P| = [G : P] = m$ is not divisible by p, there must be at least one fixed point gP under the action of H. Now, $hgP = gP$ for all $h \in H$, so that $h \in gPg^{-1}$. Thus, $H \leq gPg^{-1}$.

(3) Let G act on $\mathrm{Syl}_p(G)$ by conjugation, and fix $P \in \mathrm{Syl}_p(G)$. By (2) above, the orbit of P is $\mathrm{Syl}_p(G)$. The stabilizer of P is $N_G(P)$, so $\#\,\mathrm{Syl}_p(G) = [G : N_G(P)]$. Now,

$$m = [G : P] = [G : N_G(P)] \cdot [N_G(P) : P],$$

so $\#\,\mathrm{Syl}_p(G) = [G : N_G(P)]$ divides m. Additionally, $m = [G : P] \equiv [N_G(P) : P] \pmod p$, so reducing the above equation modulo p and dividing by m gives $\#\,\mathrm{Syl}_p(G) = [G : N_G(P)] \equiv 1 \pmod p$.

■

Let $P \in \mathrm{Syl}_p(G)$. If $P \trianglelefteq G$, then P is the only Sylow p-subgroup of G, as all Sylow p-subgroups are conjugate. On the other hand, if P is the only Sylow p-subgroup of G, we must have $P \trianglelefteq G$ for the same reason. In fact, P must be characteristic in G, as its image under any automorphism is a Sylow p-subgroup.

Example. We can use Sylow's Theorem to show that no group of order 40 is simple. We know it must have a Sylow 5-subgroup, and that the number of subgroups must divide 8 and be 1 (mod 5). This is only possible if there is exactly one Sylow 5-subgroup, so it must be normal.

We conclude this section with the so-called Frattini Argument[4], which is a quite useful lemma.

Proposition 2.4.44 (Frattini Argument)**.** *Let $N \trianglelefteq G$ be finite, and let P be a Sylow p-subgroup of N. Then $G = N_G(P)N$.*

Proof. Let $g \in G$. Then $gPg^{-1} \leq gNg^{-1} = N$, so gPg^{-1} is a Sylow p-subgroup of N. Thus, there is $n \in N$ with $gPg^{-1} = nPn^{-1}$ (since all Sylow p-subgroups of N are conjugate in N). This means that $n^{-1}g \in N_G(P)$, so that $g \in NN_G(P)$. Order considerations show $G = N_G(P)N$ as well. ■

2.4.7 Semidirect Products

We know the direct product $N \times H$ of two groups N and H is still a group. We describe a slightly more general way to construct a group out of two given groups.

Definition 2.4.45. Let N and H be groups, and $\alpha : H \to \mathrm{Aut}(N)$ be a homomorphism. The *semidirect product* $N \rtimes_\alpha H$ is the set $N \times H$ equipped with the operation

$$(n, h) \cdot (n', h') = (n \cdot \alpha(h)(n'), hh').$$

[4] It is likely called an argument because the ideas in the proof are very useful

The reader can verify for themselves that the semidirect product does indeed form a group with identity $(1,1)$ and $(n,h)^{-1} = (\alpha(h^{-1})(n^{-1}), h^{-1})$. Now,

$$(n,1)(n',1) = (n \cdot \alpha(1)(n'), 1) = (nn', 1)$$

and

$$(1,h)(1,h') = (1 \cdot \alpha(h)(1), hh') = (1, hh'),$$

so $N \rtimes_\alpha H$ naturally contains subgroups isomorphic to N and H, which we'll denote by $N_\rtimes$ and $H_\rtimes$ respectively. Clearly $N_\rtimes \cap H_\rtimes = 1$. Additionally,

$$(n,1)(1,h) = (n \cdot \alpha(1)(1), h) = (n,h),$$

so $N \rtimes_\alpha H = N_\rtimes H_\rtimes$.

We can in fact realize the action of α from within $N \rtimes_\alpha H$:

$$(1,h)(n,1)(1,h)^{-1} = (1,h)(n,1)(1,h^{-1}) = (\alpha(h)(n), h)(1,h^{-1}) = (\alpha(h)(n), 1).$$

More generally,

$$\begin{aligned}(n',h)(n,1)(n',h)^{-1} &= \big((n',1)(1,h)\big)(n,1)\big((n',1)(1,h)\big)^{-1} = (n',1)(1,h)(n,1)(1,h^{-1})(n'^{-1},1)\\ &= (n',1)(\alpha(h)(n),1)(n'^{-1},1) = (n'\alpha(h)(n)n'^{-1},1) \in N_\rtimes,\end{aligned}$$

so $N_\rtimes \trianglelefteq N \rtimes_\alpha H$. Thus, a natural question to ask is if we can go the other way, and this is indeed true.

Proposition 2.4.46. *Let G be a group with $G = NH$, $N \cap H = 1$ and $N \trianglelefteq G$. Then $G \cong N \rtimes_\alpha H$, where $\alpha(h)(n) = hnh^{-1}$.*

Proof. Any element of $G = NH$ can be written in the form nh for $n \in N$ and $h \in H$. This is unique: if $nh = n'h'$, then $n'^{-1}n = h'h^{-1} \in N \cap H = 1$, so $n = n'$ and $h = h'$. Thus, we have a bijection $\phi : G \to N \rtimes_\alpha H$ given by $nh \mapsto (n,h)$. We show ϕ is a homomorphism (and thus an isomorphism):

$$\phi\big((nh)(n'h')\big) = \phi\big((nhn'h^{-1})(hh')\big) = (n(hn'h^{-1}), hh') = (n \cdot \alpha(h)(n'), hh') = (n,h)(n',h') = \phi(nh)\phi(n'h').$$

■

Corollary 2.4.47. *Let G be a finite group, $N \trianglelefteq G$, and $H \leq G$. If $|G| = |N| \cdot |H|$ and $|N|$ and $|H|$ are relatively prime, then $G \cong N \rtimes H$.*

Proof. It suffices to show $N \cap H = 1$, as we then have $G = NH$. But $|N \cap H|$ divides $|N|$ and $|H|$, and thus must be 1. ■

Many times, we can in fact write this as a direct product, provided the relevant subgroups are normal. This is due to the following simple fact.

Proposition 2.4.48. *Let G be a group and $N, M \trianglelefteq G$ with $N \cap M = 1$. Then elements of N and elements of M commute with each other.*

Proof. Let $n \in N$ and $m \in M$, and consider $[n,m] = nmn^{-1}m^{-1}$. Since $M \trianglelefteq G$, we have that $nmn^{-1} \in M$, so that $[n,m] \in M$. Similarly, since $N \trianglelefteq G$, $mnm^{-1} \in N$, so that $[n,m] \in N$. Thus, $[n,m] \in N \cap M = 1$, so $nm = mn$. ■

This means that, in this case, the action of conjugation is trivial, and so if $G = NM$, we have that $G \cong N \times M$. This generalizes to more than two subgroups.

Proposition 2.4.49. *Let $M_1, \ldots, M_n \trianglelefteq G$ with $G = M_1 \cdots M_n$. The following are equivalent.*

(1) The map $(m_1, \ldots, m_n) \mapsto m_1 \cdots m_n$ is an isomorphism $M_1 \times \cdots \times M_n \cong G$.

(2) For each $g \in G$, the representation $g = m_1 \cdots m_n$ with $m_i \in M_i$ is unique.

(3) For $1 \leq i \leq n$, $M_i \cap \prod_{j \neq i} M_j = 1$.

If any of these happen, we say that G is the internal direct product of the M_i.

Proof.

(1)⇒(2) This follows from the injectivity of the map in question.

(2)⇒(3) Without loss of generality, take $i = 1$. Elements of M_1 are (uniquely) of the form $m_1 \cdot 1 \cdots 1$, while elements of $M_2 \cdots M_n$ are (uniquely) of the form $1 \cdot m_2 \cdots m_n$. Thus, uniqueness means that the only possible element of $M_1 \cap \prod_{j=2}^{n} M_j$ is 1.

(3)⇒(1) We in particular have that $M_i \cap M_j = 1$ for $i \neq j$, meaning elements of M_i and M_j commute. Thus, the multiplication map is a group homomorphism. It is surjective by the assumption that $G = M_1 \cdots M_n$, so we only need to show injectivity. Suppose that $(m_1, \ldots, m_n) \mapsto 1$. Then, for all i, we have that $m_i = \prod_{j \neq i} m_j^{-1}$ lies in $M_i \cap \prod_{j \neq i} M_j = 1$. Thus, multiplication has trivial kernel, and so is injective.

■

For instance, we can always decompose finite minimal normal subgroups as direct products:

Proposition 2.4.50. *Let G be a group and $N \trianglelefteq G$ be a finite minimal normal subgroup. Then there exist simple $S_1 \cong \cdots \cong S_n \leq G$ such that N is the internal direct product of the S_i.*

Proof. Let $S \trianglelefteq N$ be minimal (which is possible since N is finite). For each $g \in G$, we have that $gSg^{-1} \subseteq gNg^{-1} = N$. Thus, $\langle gSg^{-1} : g \in G \rangle \trianglelefteq G$ is contained in N, and so we have $N = \langle gSg^{-1} : g \in G \rangle$. Pick a minimal set of conjugates $S_1, \ldots, S_n$ that generate N (and note that these are all isomorphic). Since each S_i is normal in N, this means that $N = S_1 \cdots S_n$, so it remains to show that this product is direct.

Since $S_i \cap \prod_{j \neq i} S_j$ is normal in N (as the intersection of two normal subgroups) and contained in S_i, it is either trivial or equal to S_i. But if $S_i \cap \prod_{j \neq i} S_j = S_i$, then in particular $S_i \subseteq \prod_{j \neq i} S_i$, and so $S_1, \ldots, S_n$ is not minimal. Thus, $S_i \cap \prod_{j \neq i} S_j = 1$, and so our product is direct.

Finally, we show that S_i is simple. Let $1 < M \trianglelefteq S_i$. Since $S_j \cap S_i = 1$ for $j \neq i$, we have that

$$S_j \subseteq Z_G(S_i) \subseteq Z_G(M) \subseteq N_G(M).$$

Since we by assumption have that $S_i \subseteq N_G(M)$, we therefore have that $N = \prod S_i \subseteq N_G(M)$. Thus, $M \trianglelefteq N$, and so $M = S_i$, as desired. ■

2.4.8 Group Extensions

Semidirect products arise as they are also a natural construction in *group extensions*.

Definition 2.4.51. A sequence of homomorphisms

$$\cdots \xrightarrow{d_{i-1}} G_i \xrightarrow{d_i} G_{i+1} \xrightarrow{d_{i+1}} G_{i+2} \xrightarrow{d_{i+2}} \cdots$$

is said to be *exact* if $\operatorname{im} d_i = \ker d_{i+1}$ for all i. A *short exact sequence* is an exact sequence of the form

$$1 \longrightarrow N \longrightarrow G \longrightarrow H \longrightarrow 1$$

Definition 2.4.52. Exact sequences

$$\cdots \xrightarrow{d_{i-1}} G_i \xrightarrow{d_i} G_{i+1} \xrightarrow{d_{i+1}} G_{i+2} \xrightarrow{d_{i+2}} \cdots$$

and

$$\cdots \xrightarrow{d'_{i-1}} H_i \xrightarrow{d'_i} H_{i+1} \xrightarrow{d'_{i+1}} H_{i+2} \xrightarrow{d'_{i+2}} \cdots$$

are said to be *equivalent* if there exist isomorphisms $\phi_i : G_i \to H_i$ such that

$$\begin{tikzcd}
\cdots \arrow[r, "d_{i-1}"] & G_i \arrow[r, "d_i"] \arrow[d, "\phi_i"'] & G_{i+1} \arrow[r, "d_{i+1}"] \arrow[d, "\phi_{i+1}"'] & G_{i+2} \arrow[r, "d_{i+2}"] \arrow[d, "\phi_{i+2}"'] & \cdots \\
\cdots \arrow[r, "d'_{i-1}"'] & H_i \arrow[r, "d'_i"'] & H_{i+1} \arrow[r, "d'_{i+1}"'] & H_{i+2} \arrow[r, "d'_{i+2}"'] & \cdots
\end{tikzcd}$$

commutes.

More generally, we can consider a homomorphism of exact sequences, where we don't require the ϕ_i to be isomorphisms (just homomorphisms).

Now,

$$1 \longrightarrow N \longrightarrow G$$

being exact means the map $N \to G$ is injective, and

$$G \longrightarrow H \longrightarrow 1$$

being exact means the map $G \to H$ is surjective. Thus, the short exact sequence

$$1 \longrightarrow N \longrightarrow G \longrightarrow H \longrightarrow 1$$

gives an embedding $N \hookrightarrow G$, and a surjection $G \twoheadrightarrow H$. Identifying N with its image in G, exactness at G means $H \cong G/N$. Thus, such a short exact sequence (called an *extension* of H by N) corresponds to a way of "piecing together" N and H. Additionally, any nonsimple group has a nontrivial normal subgroup, so we can write it as a nontrivial extension. Thus, understanding all groups boils down to understanding all simple groups and all extensions. Finite simple groups have been completely classified, but unfortunately, the latter is hopeless. We can, however, work with certain type of extensions.

Definition 2.4.53. A short exact sequence

$$1 \longrightarrow N \xrightarrow[\alpha]{} G \xrightarrow[\beta]{} H \longrightarrow 1$$

is said to be

- *left split* if there exists $\alpha' : G \to N$ such that $\alpha' \circ \alpha = \mathrm{id}_N$.
- *right split* if there exists $\beta' : H \to G$ such that $\beta \circ \beta' = \mathrm{id}_H$.
- *split* if it is equivalent to the short exact sequence

$$1 \longrightarrow N \longrightarrow N \times H \longrightarrow H \longrightarrow 1$$

Very often, extensions are not split, but when they are, we can determine the structure of the group G.

Proposition 2.4.54. *Let G be an extension of H by N.*

(1) If the sequence $1 \to N \to G \to H \to 1$ is split, then it is both left split and right split.

(2) If the sequence $1 \to N \to G \to H \to 1$ is left split, it is split.

(3) If the sequence $1 \to N \to G \to H \to 1$ is right split, it is equivalent to the sequence $1 \to N \to N \rtimes H \to H \to 1$ for some action of H on N.

Proof.

(1) The projection maps $\pi_N : N \times H \to N$ and $\pi_H : N \times H \to H$ show that the sequence is left and right split.

(2) Let $\alpha' : G \to N$ with $\alpha' \circ \alpha = \mathrm{id}_N$, and define

$$\phi_G : g \mapsto (\alpha'(g), \beta(g)).$$

Clearly ϕ_G is a homomorphism. Further, ϕ_G is injective: if $\phi_G(g) = (1,1)$, then $\beta(g) = 1$, so that $g \in \ker\beta = \operatorname{im}\alpha$, say $g = \alpha(n)$. Then, $n = \alpha'(\alpha(n)) = \alpha'(g) = 1$, so that $g = 1$. Further, ϕ_G is surjective: if $(n,h) \in N \times H$ with $h = \beta(g)$, then

$$\phi_G : g \cdot \alpha(\alpha'(g)^{-1}n) \mapsto \big(\alpha'(g \cdot \alpha(\alpha'(g)^{-1}n)), \beta(g \cdot \alpha(\alpha'(g)^{-1}n))\big) = (n,h).$$

It's easy to see that the relevant diagrams commute, so that ϕ_G provides an isomorphism of exact sequences from $1 \to N \to G \to H \to 1$ to $1 \to N \to N \times H \to H \to 1$.

(3) Since $\operatorname{im}\alpha = \ker\beta$, it is a normal subgroup of G. Let $\beta' : H \to G$ such that $\beta \circ \beta' = \mathrm{id}$, and let H act on N by $h \cdot n = \alpha^{-1}\big(\beta'(h)\alpha(n)\beta'(h)^{-1}\big)$ (note we may take preimages under α since $\beta'(h)\alpha(n)\beta'(h)^{-1} \in \ker\beta = \operatorname{im}\alpha$). Writing $\pi = \beta' \circ \beta$, define

$$\phi_G : g \mapsto \big(\alpha^{-1}(g\pi(g)^{-1}), \beta(g)\big),$$

where we remark that $g \cdot \pi(g)^{-1} \in \ker\beta = \operatorname{im}\alpha$. We have that

$$\begin{aligned}\phi_G(gh) &= \big(\alpha^{-1}(gh\pi(gh)^{-1}), \beta(gh)\big) = \big(\alpha^{-1}(g\pi(g)^{-1})\alpha^{-1}(\pi(g)h\pi(h)^{-1}\pi(g)^{-1}), \beta(g), \beta(h)\big)\\ &= \big(\alpha^{-1}(g\pi(g)^{-1})\big(\beta(h)\cdot\alpha^{-1}(h\pi(h)^{-1})\big), \beta(g)\beta(h)\big) = \phi_G(g)\phi_G(h)\end{aligned}$$

Similar reasoning as above shows that ϕ_G is bijective, and it is clear that the relevant diagrams commute.

■

2.5 Subnormal Series of Groups

Our primary objects of study in this section will be groups that have particularly nice subnormal series.

Definition 2.5.1. Let G be a group. A *subnormal series* is a chain of subgroups

$$G = G_0 \trianglerighteq G_1 \trianglerighteq G_2 \trianglerighteq \cdots.$$

If further we have $G_i \trianglelefteq G$ for all i, we say this is a *normal series.* We say these series have length n if $G_n = 1$.

2.5.1 Composition Series

Definition 2.5.2. Let G be a group. A *composition series* for G is a subnormal series $G = G_0 \trianglerighteq G_! \trianglerighteq \cdots \trianglerighteq G_n = 1$ such that the quotient G_i/G_{i+1} is simple for $i = 0, \ldots, n-1$.

Clearly any finite group has a composition series. However, an arbitrary group need not have a composition series. Consider for the sake of contradiction a composition series of minimal length for $\mathbb{Z}$: $\mathbb{Z} = G_0 \trianglerighteq G_1 \trianglerighteq \cdots \trianglerighteq G_n = 1$. Since each G_i/G_{i+1} is simple and abelian, we can write $G_i/G_{i+1} \cong \mathbb{Z}/p_i\mathbb{Z}$. But this then means $\infty = |\mathbb{Z}| = p_0 \cdots p_{n-1}$, which is absurd[5]. We can, however, piece together groups with composition series.

Proposition 2.5.3. *Let G be a group and $N \trianglelefteq G$. Then G has a composition series if and only if N and G/N do.*

Proof. Suppose first that G has a composition series $G = G_0 \trianglerighteq G_1 \trianglerighteq \cdots \trianglerighteq G_n = 1$. Then we have a subnormal series $N = G_0 \cap N \trianglerighteq G_1 \cap N \trianglerighteq \cdots \trianglerighteq G_n \cap N = 1$, and the map $G_i \cap N \to G_i/G_{i+1}$ has kernel $G_{i+1} \cap N$, and so $G_i \cap N/G_{i+1} \cap N$ is isomorphic to a subgroup of G_i/G_{i+1}. Further, since $G_i \cap N \trianglelefteq G_i$, its image is a normal subgroup of G_i/G_{i+1}, and thus either trivial or simple. Removing all trivial factors gives a composition series of N. Similarly, we have a subnormal series $G/N = G_0N/N \trianglerighteq G_1N/N \trianglerighteq \cdots \trianglerighteq G_nN/N = 1$ that a similar argument shows can be reduced to a composition series.

Conversely, suppose we have a composition series $N = N_0 \trianglerighteq N_1 \trianglerighteq \cdots \trianglerighteq N_n = 1$ and $G/N = G_0/N \trianglerighteq G_1/N \trianglerighteq \cdots \trianglerighteq G_m/N$. We then have a composition series

$$G = G_0 \trianglerighteq G_1 \trianglerighteq \cdots \trianglerighteq G_m = N = N_0 \trianglerighteq N_1 \trianglerighteq \cdots \trianglerighteq N_n = 1.$$

■

A natural question is how unique a composition series is, if it exists. It certainly need not be completely unique: for instance, we have

$$\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/q\mathbb{Z} \trianglerighteq \mathbb{Z}/p\mathbb{Z} \times 1 \trianglerighteq 1 \qquad \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/q\mathbb{Z} \trianglerighteq 1 \times \mathbb{Z}/q\mathbb{Z} \trianglerighteq 1.$$

However, we can hope that the simple quotients and length of the series are uniquely determined, and remarkably this turns out to be true.

Definition 2.5.4. Two subnormal series $G = G_0 \trianglerighteq G_1 \trianglerighteq \cdots \trianglerighteq G_n = 1$ and $G = H_0 \trianglerighteq H_1 \trianglerighteq \cdots \trianglerighteq H_m = 1$ are said to be *equivalent* if $n = m$ and there is a permutation π of $0, \ldots, n-1$ such that $G_i/G_{i+1} \cong H_{\pi(i)}/H_{\pi(i+1)}$.

The fact that composition series are unique up to equivalence follows immediately from the following theorem.

[5]Indeed, an abelian group has a composition series if and only if it is finite

Theorem 2.5.5 (Schreier Refinement)**.** *Let G be a group with subnormal series $(G_i)_{0\le i\le n}$ and $(H_j)_{0\le j\le m}$. Then there are equivalent subnormal series $(G'_k)_{0\le k\le N}$ and $(H'_\ell)_{0\le \ell \le < M}$ such that each G_i appears as a G'_k and each H_j appears as a H'_ℓ.*

Proof. For $0 \le i \le n-1$ and $0 \le j \le m-1$, define

$$G'_{im+j} = G_{i+1} \cdot (G_i \cap H_j).$$

We note that

$$G'_{im} = G_{i+1} \cdot (G_i \cap H_0) = G_i = G_i \cdot (G_{i-1} \cap H_m) = G'_{(i-1)m+m},$$

Similarly, define $H'_{jn+i} = (H_j \cap G_i) \cdot H_{j+1}$ for $0 \le i \le n-1$ and $0 \le j \le m-1$. Then, by the Butterfly Lemma (2.3.20),

$$\frac{G'_{im+j}}{G'_{im+j+1}} = \frac{G_{i+1} \cdot (G_i \cap H_j)}{G_{i+1} \cdot (G_i \cap H_{j+1})} \cong \frac{(H_j \cap G_i) \cdot H_{j+1}}{(H_j \cap G_{i+1}) \cdot H_{j+1}} = \frac{H'_{jn+i}}{H'_{jn+i+1}},$$

so our two subnormal series are equivalent. ■

Theorem 2.5.6 (Jordan-Hölder)**.** *Any two composition series of a group G (if any exist) are equivalent.*

Proof. Given any two composition series of G, we can construct equivalent refinements. If our composition series is $G = G_0 \trianglelefteq \cdots \trianglelefteq G_n = 1$ and we have a refinement $G_i = H_0 \trianglelefteq H_1 \trianglelefteq \cdots \trianglelefteq H_m = G_{i+1}$, then the simplicity of G_i/G_{i+1} means that exactly one quotient H_j/H_{j+1} will be isomorphic to G_i/G_{i+1}, and that all other quotients will be trivial. Thus, the quotients of the full refinement of our composition series will consist of each of the factors G_i/G_{i+1}, together with trivial groups. But this is true for both composition series, and so they must have the same factors (up to rearranging) and the same length, and are thus equivalent. ■

Definition 2.5.7. If a group G has a composition series $G = G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_n = 1$ the quotients G_i/G_{i+1} are called the *composition factors* of G.

We may hope that composition factors uniquely determine a group, but this is unfortunately not true. For instance, $S_3 \trianglelefteq \langle(123)\rangle \trianglelefteq 1$ and $\mathbb{Z}/6\mathbb{Z} \trianglelefteq \langle 2\rangle \trianglelefteq 1$ both have composition factors $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/3\mathbb{Z}$.

2.5.2 Solvability

Definition 2.5.8. A group G is said to be *solvable* if G has a subnormal series $G = G_0 \trianglerighteq G_1 \trianglerighteq \cdots \trianglerighteq G_n = 1$ such that the quotients G_i/G_{i+1} are abelian for $i = 0, \ldots, n-1$.

Example. All abelian subgroups are solvable.

Example. Recall that a finite p-group of order p^n has a chain of subgroups $P = P_n \ge P_{n-1} \ge \cdots \ge P_0 = 1$ with $|P_i| = p^i$. This means that $[P_{i+1} : P_i] = p$, so that $P_i \trianglelefteq P_{i+1}$ (by Proposition 2.4.16), and the quotient $P_{i+1}/P_i \cong \mathbb{Z}/p\mathbb{Z}$ is abelian. Thus, finite p-groups are solvable.

Recall that the commutator subgroup (also called the derived subgroup) $G' = [G, G]$ has the property that G/H is abelian if and only if $G' \le H$. We may define $G'' = [G', G']$, and inductively take $G^{(i+1)} = [G^{(i)}, G^{(i)}]$. Then the *derived series* $G = G^{(0)} \trianglerighteq G^{(1)} \trianglerighteq \cdots$ has the property that $G^{(i)}/G^{(i+1)}$ is abelian for all i. Since we know that these are "minimal" abelian quotients, one would expect that this series would terminate for a solvable group.

Proposition 2.5.9. *A group G is solvable if and only if $G^{(n)} = 1$ for some n.*

Proof. Suppose that G is solvable, and let $G = G_0 \trianglerighteq G_1 \trianglerighteq \cdots \trianglerighteq G_n = 1$ be the relevant subnormal series. We show by induction on i that $G^{(i)} \le G_i$. This is clearly true for $i = 0$, and if $G^{(i)} \le G_i$, then we have that

$$G_{i+1} \ge [G_i, G_i] \ge [G^{(i)}, G^{(i)}] = G^{(i+1)},$$

since the quotient G_i/G_{i+1} is abelian. In particular, $G^{(n)} \le G_n = 1$.

Conversely, if $G^{(n)} = 1$, then the derived series $G = G^{(0)} \trianglerighteq G^{(1)} \trianglerighteq \cdots \trianglerighteq G^{(n)} = 1$ shows that G is solvable. ■

Recall that we also have $G' \operatorname{char} G$. Iterating Proposition 2.3.30 shows that $G^{(n)} \operatorname{char} G$, that $\phi(G^{(n)}) \le H^{(n)}$ for homomorphisms $\phi : G \to H$, and that $\phi(G^{(n)}) = H^{(n)}$ when ϕ is surjective. Taking ϕ to be inclusion and quotient homomorphisms, we see that subgroups and quotient groups of solvable groups are solvable. Replicating the proof of Proposition 2.5.3 shows conversely that if N and G/N are solvable, then G is as well.

If a solvable group G has a composition series $G = G_0 \trianglerighteq G_1 \trianglerighteq \cdots \trianglerighteq G_n = 1$, then the group G_i is solvable, and thus so is the quotient G_i/G_{i+1}. But this quotient is simple, and so must be abelian, and thus cyclic of prime order. Conversely, if all composition factors of G are cyclic of prime order, then G is immediately solvable. Notice, however, that solvable groups need not have composition series (for instance, $\mathbb{Z}$). Indeed, a solvable group has a composition series if and only if it is finite (since we have verified that each composition factor must be finite). This classification of possible composition factors gives us a result quite useful for induction in finite solvable groups.

Proposition 2.5.10. *Let N be a finite minimal normal subgroup of a group G, and suppose that N is solvable. Then $N \cong (\mathbb{Z}/p\mathbb{Z})^n$ for some n (we say that N is an elementary abelian p-group).*

Proof. Recall by Proposition 2.4.50 that we can write $N \cong S^n$ for a simple group S. It is easy to see that S is the only composition factor for N, and so we must have $S_i \cong \mathbb{Z}/p\mathbb{Z}$ for some prime p, as desired. ■

We are also able to classify the so-called *chief factors* of a finite solvable group.

Definition 2.5.11. Let G be a group. A *chief series* is a normal (not subnormal!) series $G = N_0 \trianglerighteq N_1 \trianglerighteq \cdots \trianglerighteq N_n = 1$ that is fully saturated in the sense that there is no $M \trianglelefteq G$ for $N_i > M > N_{i+1}$. In other words, N_i/N_{i+1} is a minimal normal subgroup of G/N_{i+1}.

Clearly any finite group must contain a chief series. Further, if G has a chief series, then for any normal subgroup $N \trianglelefteq G$, one can find a chief series of G containing N. In the case that G is solvable, then since the chief factor N_i/N_{i+1} is a minimal normal subgroup of G/N_{i+1}, it is an elementary abelian p-group.

Proposition 2.5.12. *Let G be a finite solvable group and $M < G$ be a maximal proper subgroup. Then $[G : M]$ is a prime power.*

Proof. Let L be maximal among normal subgroups of G contained in M[6]. Since G is finite, we can find a chief series of G containing L. If K/L is a chief factor, then $K \trianglelefteq G$, and so the maximality of L means that $K \not\leq M$. Thus, $KM > M$, and so the maximality of M gives that $KM = G$. Taking orders and rearranging, we have that $[G : M] = [K : K \cap M]$. Since $L \leq K \cap M$, we see that $[G : M]$ divides $[K : L]$, which is a prime power (since K/L is a chief factor). ■

Finally, no exposition of solvable groups is complete without stating the following two theorems.

Theorem 2.5.13 (Feit-Thompson)**.** *Any group of odd order is solvable.*

Theorem 2.5.14 (Burnside)**.** *Any group of order $p^a q^b$ for primes p and q is solvable.*

We will not prove the Feit-Thompson theorem, but we will show Burnside's theorem during our study of representation theory in .

2.5.3 Hall's Theorem

The goal of this section is to prove Hall's theorem, a beautiful extension of Sylow's theorem for solvable groups. Recall that a finite p-group is one whose order is a power of p, and that a Sylow p-subgroup is a p-subgroup of maximal order. These definitions lend themselves nicely to more than one prime.

Definition 2.5.15. Let π be a set of primes. A finite group G is said to be a *π-group* if the only primes dividing $|G|$ are elements of π (that is, if the prime factorization of $|G|$ is $p_1^{e_1} \cdots p_n^{e_n}$, then $p_1, \dots, p_n \in \pi$). If G is a finite group, a π-subgroup $H \leq G$ is said to be a *Hall π-subgroup* if $[G : H]$ is not divisible by any elements of π.

Note that Hall π-subgroups $N \trianglelefteq G$ satisfy the strong condition that $\gcd(|N|, [G : N]) = 1$ (and that all nontrivial proper subgroups satisfying this are necessarily Hall π-subgroups for some π).

Definition 2.5.16. Let G be a finite subgroup and $N \trianglelefteq G$ with $\gcd(|N|, [G : N]) = 1$. We say $H \leq G$ *complements N in G* if $|H| = [G : N]$.

Note that, if H complements N in G, then $G = NH$ and $N \cap H = 1$, conversely these conditions imply that H complements N. It is the content of the following theorem that we can always find complements.

Theorem 2.5.17 (Schur-Zassenhaus)**.** *Let G be a finite solvable group and $N \trianglelefteq G$ with* $\gcd(|N|, [G : N]) = 1$*. Then there is $H \leq G$ with complementing N. Further, all complements of N are conjugate.*

[6]One can check that $L = \bigcap_{g \in G} gMg^{-1}$ works. This is sometimes called the *core* of M

Proof. We first show existence, proceeding by induction on $|G|$, with $|G| = 1$ being trivial. Let M/N be a chief factor for G, so that M/N is a p-group for some prime p by Proposition 2.5.10. Let P be a Sylow p-subgroup of M, and note $PN = M$ by order considerations.

If $X = N_G(P)$, then the Frattini argument 2.4.44 shows that $G = XM = XPN = XN$. Taking orders and rearranging, $[X : X \cap N] = [G : N]$ is coprime to $|X \cap N|$ (as the latter divides $|N|$). Note also that $N \cap X \trianglelefteq X$ and that X is solvable.

If $X < G$, then by induction, we can find $H \leq X$ with $|H| = [X : N \cap X] = [G : N]$. Since $H \leq X \leq G$, we are done, so suppose $X = G$. Then $P \trianglelefteq G$, and we may consider $NP/P \trianglelefteq G/P$ (which is solvable). We have that $|NP/P|$ divides $|N|$ and $[G/P : NP/P] = [G : NP]$ divides $[G : N]$, so by induction there is $H/P \leq G/P$ with $|H/P| = [G/P : NP/P]$. We then have that

$$|H| = |P| \cdot [G/P : NP/P] = |P| \cdot [G : NP] = |P| \cdot \frac{[G : N]}{[NP : N]}.$$

Since $\gcd(|N|, [G : N]) = 1$ and $p \mid [G : N]$, we have that $p \nmid N$. Thus, $|N \cap P| = 1$, so $|NP| = |N| \cdot |P|$, and we have $|H| = [G : N]$, as desired.

To see that all such H are conjugate, we refer to the proof given in §2.6.5. ∎

The Schur-Zassenhaus theorem in fact holds for *all* finite groups, but the proof is significantly more complicated, see §2.6.5 [7].

We're now ready to state and prove Hall's Theorem.

Theorem 2.5.18 (Hall)**.** *Let G be a finite solvable group, and let π be a set of primes.*

(1) G contains a Hall π-subgroup

(2) If K is a π-subgroup of G and K is a Hall π-subgroup of G, then K is contained in a conjugate of H.

Proof.

(1) We proceed by induction on G, with $|G| = 1$ being trivial. Let $M \trianglelefteq G$ be a minimal normal subgroup. Since G is solvable, M is a p-group for some prime p. By induction, G/M has a Hall π-subgroup H/M.

If $p \in \pi$, then H is a π-subgroup, and $[G : H] = [G/M : H/M]$ isn't divisible by any member of π, so H is a Hall π-subgroup. If $p \notin \pi$, then $\gcd(|M|, [H : M]) = 1$, so we can find a complement K of M in H. Then $|K| = |H/M|$ is a π-subgroup, and $[G : K] = [G : H] \cdot |M|$ is not divisible by any elements of π, so K is a Hall π-subgroup.

(2) We once again proceed by induction on G, with $|G| = 1$ being trivial. Let M be a minimal normal subgroup of G, say a p-group for some prime p. Note that KM/M is a π-subgroup of G/M, and HM/M is a Hall π-subgroup of G/M. By induction, they are conjugate in G/M, and so we have that $KM \leq (gHg^{-1})M$ for some $g \in G$.

If $p \in \pi$, then we must have that $(gHg^{-1})M = gHg^{-1}$, since its order would otherwise be divisible by a higher power of p than $|H|$, so that $K \leq KM \leq gHg^{-1}$. If $p \notin \pi$, then K and $gHg^{-1} \cap KM$ are complements to M in KM (the latter is true because $|gHg^{-1}|$ is relatively prime to M, and $KM \leq (gHg^{-1} \cap KM)M \leq KM$). Thus, they are conjugate in KM, so that there is some $h \in KM$ with

$$H \leq h(gHg^{-1} \cap KM)h^{-1} \subseteq (hg)H(hg)^{-1}.$$

∎

We in particular note that all Hall π-subgroups are conjugate. We also note that taking $\pi = \{p\}$ yields Sylow's theorem.

Remarkably, the converse of Hall's theorem is also true: if a (finite) group contains Hall π-subgroups for all π, then it is solvable. This proof depends on Burnside's theorem 2.5.14, but we will present all remaining parts here.

Lemma 2.5.19. *Let G be a finite group, and $H, K \leq G$ with $[G : H]$ and $[G : K]$ coprime. Then $G = HK$, $[G : H \cap K] = [G : H] \cdot [G : K]$, and $[H : H \cap K] = [G : K]$.*

[7] The reason we have chosen to delay showing that all complements are conjugate is because there is no ease to be gained by working in the solvable case, unlike in the proof of existence

Proof. Since $[G : H \cap K] = [G : H] \cdot [H : H \cap K] = [G : K][K \cdot H \cap K]$ is divisible by $[G : H]$ and $[G : K]$, it is divisible by $[G : H] \cdot [G : K]$, as they are coprime. In particular,

$$\frac{|G|^2}{|H| \cdot |K|} \leq \frac{|G|}{|H \cap K|},$$

rearranging gives

$$|G| \leq \frac{|H| \cdot |K|}{|H \cap K|} = |HK|,$$

so that $G = HK$. Since we have equality above, we must also have equality $[G : H \cap K] = [G : H] \cdot [G : K]$. Dividing by $[G : H]$ gives the last result. ■

Proposition 2.5.20. *Let G be a finite group and $H, K, L \leq G$ with $[G : H], [G : K], [G : L]$ pairwise coprime. If H, K, and L are solvable, then G is solvable.*

Proof. If $H = 1$, then $[G : H] = |G|$ is relatively prime to $[G : K]$, so that $G = K$ is solvable, so suppose $H > 1$. Let $M \trianglelefteq H$ be minimal, and let M be a p-group for some prime p. Assume without loss of generality that p doesn't divide $[G : K]$ (as it cannot divide both $[G : K]$ and $[G : L]$). Then K contains a full Sylow p-subgroup of G, and so M is contained in some conjugate of K (as it is contained in some conjugate of K's Sylow p-subgroup), say $M \leq gKg^{-1}$.

Now, $[G : gKg^{-1}]$ and $[G : H]$ are relatively prime, so $G = (gKg^{-1})H$. Now, for $x \in G$, writing $x = uv$ for $u \in gKg^{-1}$ and $v \in H$, we have that

$$xMx^{-1} = u(vMv^{-1})u^{-1} = uMu^{-1} \subseteq gKg^{-1}.$$

Thus, if $N = \langle xMx^{-1} : x \in G \rangle$ is the smallest normal subgroup of G containing M, then $N \leq gKg^{-1}$ is solvable.

Now, the quotients HN/N, KN/N, and LN/N are solvable (as the homomorphic images of solvable groups). Further, $[G/N : HN/N] = [G : HN]$ divides $[G : H]$, and so we see that the indices of HN/N, KN/N and LN/N are pairwise coprime. Thus, G/N is solvable by induction, and so G is as well. ■

We're now ready to prove the converse to Hall's theorem.

Theorem 2.5.21. *Let G be a finite group, and suppose that G has a Hall π-subgroup for all sets of primes π. Then G is solvable. Further, it suffices to show that G has a Hall π-subgroup whenever $\pi = \{p\}^c$.*

Proof. We proceed by induction on the number of primes n dividing $|G|$. p-groups are solvable, and Burnside's theorem tells us that $\{p, q\}$-groups are solvable, so we may assume $n \geq 3$. Let p, q, and r be prime factors of $|G|$, and let P, Q, and R be Hall $\{p\}^c$, $\{q\}^c$ and $\{r\}^c$-subgroups respectively. Note the indices $[G : P]$, $[G : Q]$ and $[G : R]$ are coprime (as powers of distinct primes), so if we show that P, Q and R are solvable, then G will be solvable as well.

We work (without loss of generality) with P. By induction, it suffices to show that P contains a Hall $\{s\}^c$-subgroup for all primes s. Let $S \leq G$ be a Hall $\{s\}^c$-subgroup. Then $[G : P]$ and $[G : S]$ are coprime, and so $[P : P \cap S] = [G : S]$ is the full power of s dividing $|G|$. This means that $P \cap S$ is a Hall $\{s\}^c$-subgroup of P, as desired. ■

2.5.4 Nilpotence

Definition 2.5.22. A group G is said to be *nilpotent* if there is a normal series $G = G_0 \trianglerighteq G_1 \trianglerighteq \cdots \trianglerighteq G_n = 1$ such that $G_i/G_{i+1} \leq Z(G/G_{i+1})$. Such a series is said to be a *central series*.

We immediately see that nilpotent groups are solvable. Just as with solvable groups, we can try and construct a "minimal series" that spans from 1 to G if and only if G is nilpotent. There are two possible ways to do this.

We may start using $Z_0 = 1$, and inductively defining $Z_{i+1}/Z_i = Z(G/Z_i)$. We call this the *ascending* or *upper* central series of G.

Conversely, for $G_i/G_{i+1} \subseteq Z(G/G_{i+1})$, elements of G_i must commute with elements of G modulo G_{i+1}, which means that $[G_i, G] \subseteq G_{i+1}$ (and note conversely that if $[G_i, G] \subseteq G_{i+1}$, then $G_i/G_{i+1} \leq Z(G/G_{i+1})$). Thus, we may define $G^0 = G$ and inductively define $G^{i+1} = [G^i, G]$. This is the *descending* or *lower* central series of G.

Proposition 2.5.23. *If G is a group, then $Z_n = G$ if and only if $G^n = 1$, and G is nilpotent if and only if $Z_n = G$ and $G^n = 1$ for some n.*

Proof. If $Z_n = G$ or $G^n = 1$, then we clearly have that G is nilpotent. Suppose conversely that G is nilpotent, and let $G = N_0 \trianglerighteq N_1 \trianglerighteq \cdots \trianglerighteq N_k$ be a central series.

We show by induction that $G^i \leq N_i$, with $i = 0$ being clear. Since $N_i/N_{i+1} \leq Z(G/N_{i+1})$, elements of N_i commute with elements of G up to an element of N_{i+1}, which is to say that

$$N_{i+1} \geq [N_i, G] \geq [G^i, G] = G^{i+1}.$$

In particular, we have that $G^k = 1$.

We also show by induction that $N_{k-i} \leq Z_i$, with $i = 0$ being clear. Now, if $Z_i \geq N_{k-i} \geq [N_{k-i-1}, G]$, then $N_{k-i-1}Z_i/Z_i \leq Z(G/Z_i) = Z_{i+1}/Z_i$, so that $N_{k-i-1} \leq Z_{i+1}$. In particular, $Z_k = G$.

Once one has established that $Z_n = G$ or $G^n = 1$, one can then use the corresponding central series to show that the other is true as well. ■

We can see that, if $Z(G/M) > 1$ for all $M \lhd G$, then the upper central series is strictly increasing. So if G is finite, then this forces G to be nilpotent. In particular, finite p-groups are nilpotent.

Proposition 2.5.24. *Subgroups and quotient groups of nilpotent groups are nilpotent.*

Proof. If $H \leq G$, then $H^n \leq G^n$. If $\phi : G \to G/N$ is a surjective homomorphism, then $\phi(G^n) = (G/N)^n$. ■

Proposition 2.5.25. *If $G_1, \ldots, G_n$ are nilpotent, then $\prod_{i=1}^n G_i$ is nilpotent.*

Proof. Since elements of G_i and G_j commute with each other for $i \neq j$, we have that $(\prod_{i=1}^n G_i)^m = \prod_{i=1}^n G_i^m$. ■

The nilpotence condition is quite strict, and so we are able to determine the structure of a nilpotent group quite well.

Theorem 2.5.26. *Let G be a finite group. The following are equivalent.*

(1) G is nilpotent

(2) $N_G(H) > H$ for all $H < G$

(3) For all $H \leq G$, there exists a subnormal series $G = H_0 \trianglerighteq H_1 \trianglerighteq \cdots \trianglerighteq H_k = H$

(4) All maximal subgroups of G are normal

(5) Every Sylow subgroup of G is normal

(6) G is isomorphic to the direct product of its Sylow subgroups

Proof.

(1)⇒(2) Let $G = N_0 \trianglerighteq N_1 \trianglerighteq \cdots \trianglerighteq N_n = 1$ be a central series and $H < G$. Take k to be minimal such that $N_k \leq H$ (in particular, $k > 0$). Then $[H, N_{k-1}] \leq [G, N_{k-1}] \leq N_k \leq H$, which shows that $H < N_{k-1} \leq N_G(H)$.

(2)⇒(3) Let $H \leq G$. We induct on $[G : H]$, with the case of $[G : H] = 1$ being clear. For any H, we have $H < N_G(H)$, and induction applied to $N_G(H)$ gives a series $G = H_0 \trianglerighteq H_1 \trianglerighteq \cdots \trianglerighteq H_k = N_G(H) \trianglerighteq H$.

(3)⇒(4) If $M < G$ is not normal, then a subnormal series of subgroups $M \trianglelefteq M_1 \trianglelefteq \cdots \trianglelefteq G$ with at least 1 intermediate term exists. Thus, M is not maximal.

(4)⇒(5) Let P be a Sylow p-subgroup of G, and suppose that $N_G(P) < G$. Let $M \geq N_G(P)$ be a maximal proper subgroup of G. Then $M \trianglelefteq G$, and so the Frattini argument gives $G = N_G(P)M \leq M < G$, a contradiction. Thus, $N_G(P) = G$, and so $P \trianglelefteq G$.

(5)⇒(6) Let $P_1, \ldots, P_n$ be the distinct Sylow subgroups of G, and note that they each correspond to distinct primes. Then order considerations show $G = P_1 \cdots P_n$ and $P_i \cap \prod_{i \neq j} P_j$ for $1 \leq i \leq n$. Thus, $G \cong P_1 \times \cdots \times P_n$ by Proposition 2.4.49.

(6)⇒(1) G is isomorphic to the direct product of p-groups, each of which is nilpotent. Thus, G is nilpotent as well.

■

Another quite important result is the following.

Proposition 2.5.27. *Let $M, N \trianglelefteq G$ be finite and nilpotent. Then MN is nilpotent.*

Proof. Fix a prime p, and let P be a Sylow p-subgroup of M and Q a Sylow p-subgroup of N. Then P and Q are characteristic in M and N respectively, and thus normal in G, so that PQ is a normal p-subgroup of MN. Using a subscript to denote the p-part, we have that

$$|MN|_p = \frac{|M|_p \cdot |N|_p}{|M \cap N|_p} = \frac{|P| \cdot |Q|}{|M \cap N|_p}.$$

Now, $P \leq PQ \cap M$ and $Q \leq PQ \cap M$, and $|M \cap N|_p \geq |PQ \cap M \cap N|$, since the latter is a p-group. Thus,

$$|MN|_p \leq \frac{|PQ \cap M| \cdot |PQ \cap N|}{|PQ \cap M \cap N|} = |(PQ \cap M)(PQ \cap N)| \leq |PQ|,$$

so PQ is a (normal) Sylow p-subgroup of MN. Since p was arbitrary, this means that MN is nilpotent. ∎

In particular, G must contain a unique largest normal nilpotent subgroup.

Definition 2.5.28. The *Fitting subgroup* $F(G)$ is largest normal nilpotent subgroup of G.

Another quite important subgroup is the Frattini subgroup.

Definition 2.5.29. The *Frattini subgroup* $\Phi(G)$ is the intersection of all maximal subgroups of G.

It is not hard to see that the conjugate of a maximal subgroup is again a maximal subgroup, so that $\Phi(G) \trianglelefteq G$. In fact, for the same reason, $\Phi(G)$ is characteristic in G.

Proposition 2.5.30. *Let G be a finite group. The following are equivalent.*

(1) G is nilpotent

(2) $G/\Phi(G)$ is abelian

(3) $G/\Phi(G)$ is nilpotent

Proof.

(1)⇒(2) We want to show $G' \leq \Phi(G)$, so it suffices to show $G' \leq M$ for every maximal $M < G$. Since G is nilpotent, $M \trianglelefteq G$, and G/M is simple. But since G is nilpotent (hence solvable), this means that G/M has prime order, and so is abelian, so that $G' \leq M$, as desired.

(2)⇒(3) Abelian groups are nilpotent.

(3)⇒(1) If $M < G$ is maximal, then $\Phi(G) \leq M < G$. Thus, $M/\Phi(G) \leq G/\Phi(G)$ is maximal, hence normal, so that $M \trianglelefteq G$. Thus, G is nilpotent.

∎

We can in fact generalize the last implication.

Proposition 2.5.31. *Let G be finite and $\Phi(G) \leq N \trianglelefteq G$. Then N is nilpotent if and only if $N/\Phi(G)$ is nilpotent.*

Proof. If N is nilpotent, then so is $N/\Phi(G)$. Conversely, suppose that $N/\Phi(G)$ is nilpotent, and let $P \in \mathrm{Syl}_p(N)$. Then $P\Phi(G)/\Phi(G)$ is a Sylow p-subgroup of $N/\Phi(G)$, and hence unique, so characteristic. Thus, $P\Phi(G)/\Phi(G) \trianglelefteq G/\Phi(G)$, so that $P\Phi(G) \trianglelefteq G$.

Applying the Frattini argument 2.4.44 to $P \leq P\Phi(G)$, we have that $G = N_G(P)P\Phi(G) = N_G(P)\Phi(G)$. This means that $N_G(P) = G$ (if $N_G(P) \leq M < G$ for M maximal, then $G = N_G(P)\Phi(G) \leq M < G$), so that P is normal in G, thus in N, as desired. ∎

Corollary 2.5.32. *If G is finite, $\Phi(G)$ is nilpotent, and in particular $\Phi(G) \leq F(G)$.*

Corollary 2.5.33. *If G is finite, $F(G/\Phi(G)) = F(G)/\Phi(G)$.*

2.6 Miscellaneous Topics in Group Theory

2.6.1 Presentations

Suppose we have a set S of symbols and a set R of relations between those symbols (e.g. $ab = ba$ or $c^{-1} = c$). The goal of this section is to construct the group $\langle S|R\rangle$ which is the "smallest and most general" group containing S subject to R. To make sure that we get the "smallest" possible group, we it to be generated by S. That is, every element should be a word in S. To make this the "most general" possible group, we want to only impose the constraints given by R and their consequences, nothing else.

Given a set X, a *word* in X is a string $x_1 \cdots x_n$ for $x_i \in X$ (where the empty word ϵ is a valid word). Typically, we will condense words into the form $x_1^{e_1} \cdots x_m^{e_m}$ where $x_i \neq x_{i+1}$ and $e_i \geq 1$, and say that the word has length m. Our strategy will to be to consider equivalence classes of words in $X = S \sqcup S^{-1}$, where $S^{-1} = \{s^{-1} : s \in S\}$ is a set of formally inverted elements of S (which will end up serving as actual inverses once we form our group).

Writing $s^{-k} = (s^{-1})^k$ for $k \geq 1$, every word in X can then be written as $s_1^{e_1} \cdots s_n^{e_n}$ for $s_i \in S$ and $e_i \in \mathbb{Z}$ nonzero (though we cannot guarantee that $s_i \neq s_{i+1}$ since the word ss^{-1} is distinct from the empty word: we're just using this as shorthand). Given words $w_1 = s_1^{e_1} \cdots s_n^{e_n}$ and $w_2 = t_1^{f_1} \cdots t_m^{f_m}$, we define the inverse of w_1 to be

$$w_1^{-1} = s_n^{-e_n} \cdots s_1^{-e_1}$$

and the concatenation of w_1 and w_2 to be

$$w_1 w_2 = s_1^{e_1} \cdots s_n^{e_n} t_1^{f_1} \cdots t_m^{f_m}.$$

Given a relation of the form $w_1 = w_2$, we may equivalently impose the constraint that $w_1 w_2^{-1}$ is trivial, so we'll assume that our set R consists of words in X that we want to declare to be trivial, and we'll call the elements of R *relators*. An *elementary transformation* of words is an insertion or deletion of some element of

$$R' = R \cup \{ss^{-1} : s \in S\} \cup \{s^{-1}s : s \in S\},$$

and we say two words are equivalent if one can be obtained from the other by a sequence of elementary transformations. Since undoing an elementary transformation is itself an elementary transformation, this is an equivalence relation, and we'll denote the equivalence class of w by $[w]$. Words in the same equivalence class are then precisely the words that we want to be equal in our group.

Proposition 2.6.1. *The equivalence classes of words in X form a group under $[w_1][w_2] = [w_1 w_2]$, denoted $\langle S|R\rangle$.*

Proof. To see that this operation is well-defined, suppose $w_1 \sim w_1'$ and $w_2 \sim w_2'$. Then, by applying the elementary transformations to get from w_1 to w_1' followed by the elementary transformations to get from w_2 to w_2', we have that $w_1 w_2 \sim w_1' w_2 \sim w_1' w_2'$. The identity is given by $[\epsilon]$, since $[w][\epsilon] = [w\epsilon] = [w]$, and the inverse of $[w]$ is given by $[w]^{-1} = [w^{-1}]$, since repeatedly deleting occurrences of the form ss^{-1} or $s^{-1}s$ shows that ww^{-1} is equivalent to the empty word. Associativity follows from the associativity of concatenation. ■

Note that we have a map $S \to \langle S|R\rangle$ given by $s \mapsto [s]$. That $\langle S|R\rangle$ is the most general group subject to R is then captured by the following universal property.

Proposition 2.6.2 (Universal Property of Presentations)**.** *Let S be a set, R a set of words in $S \sqcup S^{-1}$, G a group, and $f : S \to G$ a set map. Suppose that $f(s_1)^{e_1} \cdots f(s_n)^{e_n} = 1$ for each word $s_1^{e_1} \cdots s_n^{e_n} \in R$. Then there is a unique group homomorphism $\phi : \langle S|R\rangle \to G$ such that $[s] \mapsto f(s)$.*

Proof. Let $F : s_1^{e_1} \cdots s_n^{e_n} \mapsto f(s_1)^{e_1} \cdots f(s_n)^{e_n}$ from the set of words on $S \sqcup S^{-1}$ to G. Note that such a ϕ must be unique, since we must have

$$\phi([s_1^{e_1} \cdots s_n^{e_n}]) = \phi([s_1]^{e_1} \cdots [s_n]^{e_n}) = \phi([s_1])^{e_1} \cdots \phi([s_n])^{e_n} = f(s_1)^{e_1} \cdots f(s_n)^{e_n} = F(s_1^{e_1} \cdots s_n^{e_n}).$$

This map ϕ will clearly be a group homomorphism once we show that it is well-defined. But, by assumption, F takes on the same value on words that differ by an elementary transformation, and so takes on the same value on equivalent words. ■

As an immediate consequence of our work, we can construct the coproduct in Grp.

Definition 2.6.3. Let G and H be groups. Their *free product* (or *free amalgamation*) is

$$\left\langle G \sqcup H \middle| \{g_1 g_2 (g_1 \cdot g_2)^{-1} : g_1, g_2 \in G\} \cup \{h_1 h_2 (h_1 \cdot h_2)^{-1} : h_1, h_2 \in H\} \right\rangle.$$

Proposition 2.6.4. *The free product forms the coproduct in* Grp.

Proof. Given group homomorphisms $G, H \to N$, these give a set map $G \sqcup H \to N$. Since we started off with group homomorphisms, each of our relators must map to the identity, and so we get a unique group homomorphism $G * H \to N$ by the universal property of presentations. ■

Note that, by abstract nonsense, the group $\langle S|R\rangle$ is the unique group satisfying its universal property up to isomorphism[8]. Thus, if we can find a subset $S \subseteq G$ and a set of relations R such that G satisfies the universal property of presentations with respect to S and R, then $G \cong \langle S|R\rangle$.

Definition 2.6.5. A *presentation* for a group G is an isomorphism $G \cong \langle S|R\rangle$. We say a G is *finitely generated* if there exists such a presentation with S finite, *finitely related* if there exists such a presentation with R finite, and *finitely presented* if there exists such a presentation with both S and R finite.

Example. For any group G, we have a presentation

$$\langle G|\{gh(g \cdot h)^{-1} : g, h \in G\}\rangle.$$

Indeed, the universal property follows immediately, since any set map $f : G \to H$ such that $f(g)f(h)f(g \cdot h)^{-1} = 1$ is automatically a group homomorphism. This shows that finite groups are finitely presented.

Example. If $G = \langle S|R\rangle$ and $N \trianglelefteq G$, then we get a presentation

$$G/N \cong \langle S|R \cup N\rangle,$$

where we add in one representative of each element of N. Indeed, if we have a set map $S \to H$ such that each element of R and N maps to the identity, then we get a group homomorphism $G = \langle S|R\rangle \to H$ such that each element of N maps to the identity, which then gives a group homomorphism $G/N \to H$ by the universal property of quotients.

Example. For any group G and subset S such that $G = \langle S\rangle$, we have a surjective homomorphism $\langle S|\varnothing\rangle \to G$. If K is the kernel of this homomorphism, the previous example shows $G = \langle S|K\rangle$.

Example. $\mathbb{Z} \cong \langle a|\varnothing\rangle$ and $\mathbb{Z}/n\mathbb{Z} \cong \langle a|a^n\rangle$. In particular, finitely presented groups need not be finite.

Just because a group is finitely generated (resp. finitely related), every presentation need not have a finite generating set (resp. finite set of relators). For instance, the presentation of $\mathbb{Z}$ from Example 2.6.1 has an infinite set of generators and relators, but we know that $\mathbb{Z}$ is finitely presented. Of course, this presentation has a lot of dead weight.

Definition 2.6.6. In the group $\langle S|R\rangle$, we say that $s \in S$ is *redundant* if $[s] = [w]$ for some word w in $S \setminus \{s\}$. We say that $w \in R$ is *redundant* if $[w] = [\epsilon]$ in $\langle S|R \setminus \{w\}\rangle$.

Redundant elements in a presentation are essentially those that can be removed without changing the group. A redundant relator can simply be deleted from the presentation, and a redundant symbol can be removed after replacing all appearances of that symbol in relators with the corresponding word. While we can't ensure that arbitrary presentations of finitely generated (resp. finitely related) groups have a finite number of generating symbols (resp. a finite number of relators), we can get something almost as good.

Proposition 2.6.7. *Let G be a group and fix a presentation $G \cong \langle S|R\rangle$.*

(1) If G is finitely generated, then all but finitely many elements of S are redundant.

(2) If G is finitely related, then all but finitely many elements of R are redundant.

Proof.

(1) Suppose that $G = \langle x_1, \ldots, x_n|R'\rangle$. Then each x_i can be expressed as a word in S. If S_i is the set of elements of S that are used to express x_i, then $S_1 \cup \cdots \cup S_n$ is a finite generating set of G. Thus, every element of $S \setminus S_1 \cup \cdots \cup S_n$ is redundant.

(2) Suppose that $G = \langle S'|w_1, \ldots, w_n\rangle$. By writing the elements of S' in terms of elements of S and substituting these into the w_i, we get a presentation $\langle S|w_1', \ldots, w_n'\rangle$. Now, each w_i' can be written as a word in R, say using the finitely many elements in R_i. Similarly, each element of R can be written as a word in the w_i'. Thus, each element of $R \setminus R_1 \cup \cdots \cup R_n$ is redundant.

[8] For instance, it is in the initial object in the category whose objects are groups G together with maps $S \to G$ such that the elements of R become trivial

■

It is easy to see that a finitely related group need not be finitely presented (i.e. finitely generated). For instance, take $G = \langle X|\varnothing\rangle$ for any infinite set X. For any finite collection of words $w_1, \ldots, w_n \in G$, each word involves finitely many elements of X, and so each element of the subgroup $\langle w_1, \ldots, w_n\rangle$ can only contain words on some finite subset of X: in particular $G \neq \langle w_1, \ldots, w_n\rangle$, so G is not finitely generated. We now turn to the slightly harder task of producing a group that is finitely generated yet not finitely related.

Lemma 2.6.8. *Let $j \geq 7$ and*

$$\alpha = (1 \cdots j) \qquad \beta = (12)$$

be elements of S_j. Then $\alpha^k\beta\alpha^{-k}$ commutes with β for $2 \leq k \leq j-2$, but not for $k = j-1$.

Proof. For $2 \leq k \leq j-2$,

$$\alpha^k\beta\alpha^{-k} = (\alpha^k(1), \alpha^k(2)) = (k+1, k+2),$$

which is disjoint from $\beta = (12)$, and so commutes with it. For $k = j-1$, $\alpha^k\beta\alpha^{-k} = (1, j)$, and we can compute

$$(12)(1j) = (1j2) \qquad (1j)(12) = (12j).$$

■

Proposition 2.6.9. *The group*

$$G = \big\langle a, b \big| \big\{[a^kba^{-k}, b] : k \geq 2\big\}\big\rangle$$

is not finitely related.

Proof. Write $w_k = [a^kba^{-k}, b]$. If G were finitely related, all but finitely many of the w_k would be redundant, and so in particular we could find K such that each w_k for $k \geq K$ is expressible as a word in $w_2, \ldots, w_{K-1}$, and we may take $K \geq 3$. Now, taking $j = K+1$, the map $a \mapsto \alpha, b \mapsto \beta$ (where $\alpha, \beta \in S_j$ are as in the previous lemma) sends each of $w_2, \ldots, w_{K-1}$ to the trivial permutation. Thus, we get an extension $G \to S_j$, and in particular each w_k for $k \geq K$ must map to the trivial permutation (as it can be expressed as a word in $w_2, \ldots, w_{K-1}$). However, w_K does not map to the trivial permutation, as $\alpha^K\beta\alpha^{-K}$ does not commute with β. This is a contradiction, and so G is not finitely related. ■

2.6.2 Free Groups

A very important class of groups are the free groups.

Definition 2.6.10. Let S be a set. The *free group on S* is

$$F_S = \langle S|\varnothing\rangle.$$

Applying the universal property of presentations in the case $R = \varnothing$ gives the following.

Proposition 2.6.11 (Universal Property of Free Groups)**.** *Let G be a group and $f : S \to G$ be a map of sets. Then there exists a unique homomorphism $\phi : F_S \to G$ such that $\phi|_S = f$.*

The lack of nontrivial relations means that we can obtain a canonical representative for each equivalence class in F_S.

Definition 2.6.12. A word in $S \sqcup S^{-1}$ is said to be *reduced* if it does not contain ss^{-1} or $s^{-1}s$ as a subword for any $s \in S$.

Proposition 2.6.13. *Every word in $S \sqcup S^{-1}$ is equivalent to a unique reduced word.*

Proof. If $S = \varnothing$ then this is trivially true (the only word is the empty word ϵ), so suppose S is nonempty. To see that each word w is equivalent to a reduced word, we induct on the length of w. If w has length 0, then $w = \epsilon$ is reduced. Now, if w does not contain any subwords of the form ss^{-1} or $s^{-1}s$, then it is reduced and we are done. Otherwise, we may delete such a subword to get a word of strictly smaller length, which is equivalent to a reduced word by induction.

Suppose now that w and w' are equivalent reduced words, and take a sequence $w \to w_1 \to \cdots \to w_n \to w'$ where each step is an elementary transformation and n is minimal. We'll show that $n = 0$, which forces $w = w'$. Suppose otherwise that $n \geq 1$. Since w is a reduced word, the step $w \to w_1$ must be an insertion. Since w' is a reduced word, the step $w_n \to w'$ must be a deletion. Thus, we may find the first deletion $w_i \to w_{i+1}$. Suppose we delete an occurrence of ss^{-1} in this step (the case of $s^{-1}s$ is similar). Since w is reduced and this is the first deletion, we must insert at least one of s or s^{-1} at some step $w_j \to w_{j+1}$. In the event that both of these are inserted, take $w_j \to w_{j+1}$ to be the later insertion.

Suppose first that we insert s by inserting ss^{-1} in $w_j \to w_{j+1}$. Then we delete this same pair in $w_i \to w_{i+1}$ (since by assumption $w_j \to w_{j+1}$ is the latest relevant insertion, and so we couldn't have inserted terms between the s and s^{-1}). By deleting this pair from each of $w_{j+1}, \dots, w_i$ to get $w'_{j+1}, \dots, w'_i$, we get a strictly shorter sequence

$$w \to w_1 \to \cdots \to w_j = w'_{j+1} \to w'_{j+2} \to \cdots \to w'_{i-1} \to w'_i = w_{i+1} \to \cdots \to w_n \to w',$$

contradicting the minimality of n.

Now suppose that we insert s by inserting $s^{-1}s$ in $w_j \to w_{j+1}$. Since this is the last relevant insertion, $s^{-1}s$ must be inserted directly to the left of an existing s^{-1}. Just as before, by deleting ss^{-1} from each of $w_{j+1}, \dots, w_i$, we obtain a strictly shorter sequence.

Finally, if we insert s^{-1} by inserting $s^{-1}s$ in $w_j \to w_{j+1}$, it must be directly to the right of an existing s, and we may delete ss^{-1} from each of $w_{j+1}, \dots, w_i$ to obtain a strictly shorter sequence.

Since we can always contradict the minimality of n, we must have $n = 0$ and $w = w'$, as desired. ■

Remark 2.6.14. If we have a set function $A \to B$, then composing with the inclusion $B \to F_B$ gives a set map $A \to F_B$. The universal property of free groups then gives a homomorphism $F_A \to F_B$. Thus, taking the free group is a functor $\mathsf{Set} \to \mathsf{Grp}$. Now, we have the "forgetful functor" $U : \mathsf{Grp} \to \mathsf{Set}$ that sends a group to its underlying set. The universal property of free groups says that there is a bijection between set maps $A \to U(G)$ and homomorphisms $F_A \to G$. Thus, we see that the free functor is the left adjoint of the forgetful functor.

Since functors preserve isomorphisms, and sets of the same cardinality are isomorphic in Set, free groups on sets of order n are isomorphic. We'll write F_n for this group. It is not hard to see that $F_1 \cong \mathbb{Z}$, but for $n \geq 2$, these free groups have quite complicated structure.

Proposition 2.6.15. *Every group is a quotient of a free group.*

Proof. The identity map $G \to G$ lifts to a surjective group homomorphism $F_G \to G$. ■

Proposition 2.6.16. *If $F_n \cong F_m$, then $n = m$.*

Proof. Group homomorphisms $F_n \to \mathbb{Z}/2\mathbb{Z}$ are in bijection with set maps $[n] \to \mathbb{Z}/2\mathbb{Z}$, of which there are 2^n. Thus, $2^n = 2^m$, and so $n = m$. ■

2.6.3 The Krull-Schmidt Theorem

The goal of this section is to study how we can decompose groups as direct products. Recall that a group has *finite length* if it has a composition series. We'll write $\ell(G)$ for the minimal length of a composition series of G, and note that $\ell(G) = \ell(N) + \ell(G/N)$ for $N \trianglelefteq G$.

Definition 2.6.17. We say a group is *indecomposable* if it is not the (internal) direct product of two nontrivial subgroups. Otherwise, we say the group is *decomposable.*

Proposition 2.6.18. *Let G be a group of finite length. Then G is isomorphic to the direct product of some finite collection of indecomposable subgroups.*

Proof. If G is indecomposable, we are done. Otherwise, let $G \cong H \times K$ with H and K nontrivial. Then they are both proper subgroups of G, so we have that $\ell(H), \ell(K) < \ell(G)$. Thus, induction yields the result. ■

We may of course ask how unique these decompositions are. If we have a decomposition $G \cong H_1 \times \cdots \times H_n$, then we get projection maps $\pi_i : G \to H_i$. There is a lot we can say about the structure of the π_i: $\pi_i \circ \pi_j = \delta_{i,j}\,\mathrm{id}$, $\sum_{i=1}^n \pi_i = \mathrm{id}$, and π_i sends normal subgroups to normal subgroups. Taking advantage of these properties will allow us to prove uniqueness.

Recall that an endomorphism $\phi : G \to G$ is nilpotent if $\phi^n = 1$ for some n, and normal if ϕ sends normal subgroups to normal subgroups.

Theorem 2.6.19 (Fitting)**.** *Let G be a group of finite length, and let ϕ be a normal endomorphism of G. Then we can write G as the internal direct product of H and K, where H and K are ϕ-invariant, $\phi|_H$ is an automorphism, and $\phi|_K$ is nilpotent. In particular, if G is indecomposable, all normal endomorphisms are automorphisms or nilpotent.*

Proof. We have a normal series
$$G \trianglerighteq \phi(G) \trianglerighteq \phi^2(G) \trianglerighteq \cdots.$$
In particular, we have $\ell(G) \geq \ell(\phi(G)) \geq \ell(\phi^2(G)) \geq \cdots$, and so there is some n with $\ell(\phi^n(G)) = \ell(\phi^{n+1}(G))$. If $H = \phi^n(G)$, we see that $H = \phi(H)$. Also, since $\phi(H) \cong H/H \cap \ker\phi$, we have that $H \cap \ker\phi = 1$ (by length considerations), so that $\phi|_H$ is an automorphism.

Now let $K = \ker(\phi^n)$. Then $\phi^n(\phi(K)) = \phi(\phi^n(K)) = 1$, so $\phi(K) \leq K$. Since $\phi^n(K) = 1$, the restriction $\phi|_K$ is nilpotent.

Since $K \trianglelefteq G$, it remains to show $G = HK$ and $H \cap K = 1$. Since $\phi^n|_H$ is an automorphism, we have that $1 = H \cap \ker\phi^n = H \cap K$. Additionally, since $\phi^n(G) = H = \phi^n(H) = \phi^n(HK)$ and both sides contain $\ker\phi^n$, so that $G = HK$. ■

We now turn to taking advantage of the fact that the π_i commute with each other.

Definition 2.6.20. If α and β are endomorphisms of g, we define $(\alpha + \beta)(g) = \alpha(g)\beta(g)$.

This isn't generally an endomorphism of G, and we generally don't have $\alpha + \beta = \beta + \alpha$, but if the images $\alpha(G)$ and $\beta(G)$ commute with each other (i.e. centralize one another), then these are both true.

Lemma 2.6.21. *Let α, β be endomorphisms of G whose images commute with each other, and suppose that $\alpha + \beta$ is surjective. Then α and β are normal.*

Proof. Let $N \trianglelefteq G$, so that $\alpha(N) \trianglelefteq \alpha(G)$. Then $\beta(G)$ commutes with $\alpha(N)$, and so $\alpha(G), \beta(G) \subseteq N_G(\alpha(N))$. Thus, $G = \alpha(G)\beta(G) \subseteq N_G(\alpha(N))$, so α is normal. Similarly, β is normal. ■

Lemma 2.6.22. *Let G be an indecomposable group of finite length, and let $\alpha_1, \dots, \alpha_n$ be endomorphisms of G whose images pairwise commute. If $\alpha_1 + \cdots + \alpha_n$ is an automorphism, then at least one α_i is an automorphism.*

Proof. We induct on n, with $n = 1$ being clear. We consider the case $n = 2$. If $\alpha_1 + \alpha_2 = \sigma$, write $\beta_i = \alpha_i\sigma^{-1}$. Then the β_i are automorphisms, and $\beta_1 + \beta_2 = \mathrm{id}$. Further, the $\beta_i(G)$ centralize each other (as $\beta_i(G) = \alpha_i(\sigma^{-1}(G)) = \alpha_i(G)$), so that the β_i are normal. Since G is indecomposable, the β_i are either nilpotent or automorphisms. If either β_i is an automorphism, we are done, so suppose they are both nilpotent.

Let $1 \neq g \in \ker\beta_1$. Then $g = (\beta_1 + \beta_2)(g) = \beta_2(g)$, and so $\beta_2^m(g) = g$ for all m. But β_2 is nilpotent, so this is impossible, and so we've dealt with the $n = 2$ case.

Now, if $n \geq 3$, write $\alpha = \alpha_1 + \alpha_2$. Then $\alpha(G) \subseteq \alpha_1(G)\alpha_2(G)$ centralizes the $\alpha_i(G)$ for $i \geq 3$, so the $n - 1$ case shows either α is an automorphism or α_i for some $i \geq 3$. If α is the automorphism, then the $n = 2$ case shows that α_1 or α_2 is an automorphism. ■

We're now ready to apply this theory to projection maps.

Theorem 2.6.23 (Factor Replacement)**.** *Let G be a group of finite length with normal subgroups H, N, K_i such that $G \cong H \times N \cong K_1 \times \cdots \times K_n$ realized by internal direct products, where H and the K_i are indecomposable. Then there is some i such that $H \cong K_i$ and G is the internal direct product of K_i and N.*

Proof. Let σ be the projection map onto H, and π_i be the projection map onto K_i, and view each of these as endomorphisms of G. Then $\sum_{i=1}^n \pi_i = \mathrm{id}$, and so $\sum_{i=1}^n \sigma\pi_i = \sigma$. Let $\alpha_i = (\sigma\pi_i)|_H$, so that $\sum_{i=1}^n \alpha_i = \sigma|_H = \mathrm{id}_H$. Since the $\pi_i(G)$ centralize each other, the $\sigma\pi_i(G)$ centralize each other, and so the $\alpha_i(G)$ centralize each other. Thus, some α_i, say α_1, is an automorphism of H.

Let $U = \pi_1(H) \trianglelefteq \pi_1(G) = K_1$. Since the K_i centralize K_1, we have that $U \trianglelefteq G$. We claim that G is the internal direct product of U and N. If $u \in U \cap N$, then we can write $u = \pi_1(h)$ for $h \in H$. Since $u \in N$, $1 = \sigma(u) = \alpha_1(h)$, so $h = 1$, since α_1 is an automorphism. Now, $\sigma(U) = \alpha_1(H) = H$, so that $\sigma(UN) = H = \sigma(G)$ and $UN = G$, since both sides contain $N = \ker\sigma$.

Since $G \cong U \times N \cong H \times N$, we have that $U \cong G/N \cong H$, so it remains to show that $U = K_1$. Since $UN = G$ and $U \subseteq K_1$, we have that $K_1 = U(N \cap K_1)$. Since $U \cap N = 1$, we have that $U \cap (N \cap K_1) = 1$, and so this is an internal direct product. But K_1 is indecomposable, and $U \cong H$ is nontrivial. Thus, $N \cap K_1 = 1$ and $U = K_1$. ■

Theorem 2.6.24 (Krull-Schmidt)**.** *Let G be a group of finite length, and suppose $G = H_1 \cdots H_n = K_1 \cdots K_m$, where these are internal direct products. Then $n = m$ and $H_i \cong K_i$ after rearrangement.*

Proof. We apply factor replacement with $H = H_1$ and $N = H_2 \cdots H_n$ to get (after rearranging) $H_1 \cong K_1$ and $G = K_1H_2 \cdots H_n$. Using $H = H_2$ and $N = H_1H_3 \cdots H_n$, we get (up to rearrangement) $H_2 \cong K_2$ and $G = K_1K_2H_3 \cdots H_n$ (note we won't have $H_2 \cong K_1$, as if $H_2 \cong K_j$, then $K_j \cap N = 1$, so $j \neq 1$). Continuing in this manner gives the theorem. ■

We end with some applications of the Krull-Schmidt Theorem.

Proposition 2.6.25. *Let G, H, and K be groups of finite length.*

(1) If $G \times H \cong G \times K$, then $H \cong K$.

(2) If $G^n \cong H^n$, then $G \cong H$.

Proof. Write each group as the direct product of indecomposable groups and apply Krull-Schmidt. ∎

2.6.4 Transfer Theory

Let $H \leq G$, and let T be a left transversal of H (that is, let T consist of one representative of each left coset of H). Then we can act G on T by letting $g \cdot t$ be the element of H corresponding to the coset gtH. We also note that $(g \cdot t)^{-1}gt \in H$ for all $g \in G$ and $t \in T$. Thus, in the case that T is finite (i.e. $[G : H] < \infty$), we can define $\pi : G \to H$ by

$$\pi(g) = \prod_{t \in T} (g \cdot t)^{-1} gt.$$

This map of course need not be well-defined or a homomorphism in the case that H is not abelian. However, if $M \trianglelefteq H$ with H/M abelian, then composing π with the projection $H \to H/M$ gives a well-defined map $v : G \to H/M$. We call this the *transfer* from G to H/M, and denote it by $v(g)$.

We first show that the the transfer map is indeed a homomorphism. Using the fact that $g \cdot t$ ranges over T as t does, we have that (working modulo M)

$$\pi(g) \equiv \prod_{t \in T} (g \cdot t)^{-1} gt \equiv \prod_{t \in T} (g \cdot (h \cdot t))^{-1} g(h \cdot t) \equiv \prod_{t \in T} (gh \cdot t)^{-1} g(h \cdot t),$$

so that

$$\pi(g)\pi(h) \equiv \prod_{t \in T} (gh \cdot t)^{-1} g(h \cdot t) \prod_{t \in T} (h \cdot t)^{-1} ht \equiv \prod_{t \in T} (gh \cdot t)^{-1} ght \equiv \pi(gh),$$

where we've rearranged elements of H (which is allowable since H/M is abelian).

Proposition 2.6.26. *Let $H \leq G$ have finite index, and $M \trianglelefteq H$ with H/M abelian. The transfer map $G \to H/M$ is independent of the choice of transversal.*

Proof. Fix two transversals S and T. Let $h_t \in H$ so that $th_t \in S$ for $t \in T$. We then have that $(g \cdot t)h_{g \cdot t} = g \cdot (th_t)$. Thus, working modulo M,

$$\begin{aligned} \prod_{s \in S} (g \cdot s)^{-1} gs &\equiv \prod_{t \in T} ((g \cdot t)h_{g \cdot t})^{-1} gth_t \equiv \prod_{t \in T} h_{g \cdot t}^{-1} (g \cdot t)^{-1} gth_t \\ &\equiv \prod_{t \in T} (g \cdot t)^{-1} gt \prod_{t \in T} h_{g \cdot t}^{-1} \prod_{t \in T} h_t \equiv \prod_{t \in T} (g \cdot t)^{-1} gt. \end{aligned}$$

∎

The main result we leverage is the following transver evaluation lemma.

Lemma 2.6.27 (Transfer Evaluation)**.** *Let $M \trianglelefteq H \leq G$ with $[G : H] < \infty$ and H/M abelian, and let T be a transversal for H. Then for each $g \in G$, there is $T_0 \subseteq T$ and positive integers n_t for $t \in T_0$ such that*

(1) $\sum n_t = [G : H]$

(2) $t^{-1} g^{n_t} t \in H$ for $t \in T_0$

(3) $\pi(g) \equiv \prod_{t \in T_0} t^{-1} g^{n_t} t$ modulo M

(4) If g has finite order, then n_t divides $|g|$

Proof. Let $\langle g\rangle$ act on T (via our action of G on T), and decompose T into orbits. Let T_0 consist of a representative of each orbit, and let n_t be the size of the orbit containing t (so that (1) and (4) are immediate). For $t \in T_0$, g acts as a n_t-cycle on the orbit of t, so that $g^{n_t} \cdot t = t$. Thus, $t^{-1}g^{n_t}t \in H$.

Now, the elements of T in the same orbit as t contribute

$$\prod_{i=0}^{n_t-1} (g^{i+1} \cdot t)^{-1} g (g^i \cdot t) = t^{-1} g^{n_t} t$$

to the product $\pi(g)$. ■

For example, the transfer evaluation lemma can cleanly be applied when dealing with $Z(G)$.

Proposition 2.6.28. *If $[G : Z(G)] = m < \infty$, then $g \mapsto g^m$ is a homomorphism $G \to Z(G)$.*

Proof. We show that this is equal to the transfer map. For $g \in G$, we have that $t^{-1}g^{n_t}t \in Z(G)$, so that $t^{-1}g^{n_t}t = g^{n_t}$ (since conjugating an element of $Z(G)$ doesn't change that element). Thus,

$$v(g) = \prod_{t\in T_0} t^{-1}g^{n_t}t = \prod_{t\in T_0} g^{n_t} = g^{\sum n_t} = g^m.$$

■

We're often concerned with the kernel of the transfer map. It often turns out to be connected to the focal subgroup

$$\mathrm{Foc}_G(H) = \{x^{-1}y : x, y \text{ are } G\text{-conjugate}\}.$$

Since elements that are H-conjugate are in particular G-conjugate, we have that $x^{-1}(hxh^{-1}) \in \mathrm{Foc}_G(H)$, so that $H' \subseteq \mathrm{Foc}_G(H)$. Note that this inclusion may be strict, since G-conjugate elements need not be H-conjugate.

Proposition 2.6.29. *Let G be finite and $H \leq G$ a Hall subgroup. If $v : G \to H/H'$ is the transfer map, then*

$$\mathrm{Foc}_G(H) = H \cap G' = H \cap \ker v.$$

Proof. Note that $x^{-1}(gxg^{-1}) \in G'$, so that $\mathrm{Foc}_G(H) \subseteq H \cap G'$. Since v maps into an abelian group, $G' \subseteq \ker v$, so $H \cap G' \subseteq H \cap \ker v$. Finally, let $g \in H \cap \ker v$. Write $m = [G : H]$. By the transfer evaluation lemma,

$$\pi(g) \equiv \prod_{t\in T_0} tg^{n_t}t^{-1} \equiv g^m \prod_{t\in T_0} g^{-n_t}tg^{n_t}t^{-1},$$

where we work modulo H' and use the fact that $g \in H$. Since each factor of the product lies in $\mathrm{Foc}_G(H)$ (since $tg^{n_t}t^{-1} \in H$) and $\pi(g) \in H' \subseteq \mathrm{Foc}_G(H)$, we have that $g^m \in \mathrm{Foc}_G(H)$. Since m is relatively prime to $|H|$, it is relatively prime to $|g|$, and so $g \in \langle g^m\rangle \leq \mathrm{Foc}_G(H)$. ■

We end with some final applications of transfer theory.

Lemma 2.6.30. *Let $P \in \mathrm{Syl}_p(G)$, and let $x, y \in Z_G(P)$ be conjugate in G. Then x and y are conjugate in $N_G(P)$.*

Proof. Let $y = gxg^{-1}$. We have that $P \subseteq Z_G(x), Z_G(y)$, and

$$gPg^{-1} \subseteq gZ_G(x)g^{-1} = Z_G(gxg^{-1}) = Z_G(y),$$

so that P and gPg^{-1} are Sylow p-subgroups of $Z_G(y)$. Thus, there is $c \in Z_G(y)$ with $P = (cg)P(cg)^{-1}$. Thus, $cg \in N_G(P)$ and $(cg)x(cg)^{-1} = cyc^{-1} = y$, as desired. ■

We note the similarity between the proof of this lemma and the Frattini Argument 2.4.44.

Theorem 2.6.31 (Burnside)**.** *Let $P \in \mathrm{Syl}_p(G)$, and suppose $P \subseteq Z(N_G(P))$. Then G has a Hall $\{p\}^c$ subgroup.*

Proof. Let $v : G \to P = P/P'$ be the transfer map, so that $P \cap \ker v = \mathrm{Foc}_G(P)$. Let $x, y \in P$ be G-conjugate. Since P is abelian, $x, y \in Z_G(P)$, and so x and y are $N_G(P)$-conjugate by the above lemma, say $y = nxn^{-1}$. But since $P \subseteq Z(N_G(P))$, we have that $x = y$, so that $x^{-1}y = 1$. Thus, $\mathrm{Foc}_G(P) = 1$, so $\ker(v) \cap P = 1$. Thus, $p \nmid |\ker v|$ and $[G : \ker v] = |\operatorname{im} v|$ is a power of p, so that $\ker v$ is a Hall $\{p\}^c$-subgroup. ■

Theorem 2.6.32. *If all Sylow subgroups of G are cyclic, then G is solvable.*

Proof. Let p be the smallest prime divisor of $|G|$, and let $P \in \mathrm{Syl}_p(G)$. We have that $[N_G(P) : Z_G(P)]$ divides $|\operatorname{Aut}(P)|$ (as $N_G(P)$ acts on P by conjugation with kernel $Z_G(P)$). Now, if $|P| = p^a$, then $|\operatorname{Aut}(P)| = \varphi(|P|) = (p-1)p^{a-1}$, so that $[N_G(P) : Z_G(P)]$ is not divisible by any prime larger than P. Since $P \subseteq Z_G(P)$, it is not divisible by p, and since p is minimal, it is not divisible by any prime smaller than p. Thus, $[N_G(P) : Z_G(P)] = 1$, so that $P \subseteq Z(Z_G(P))$. Thus, we have a Hall $\{p\}^c$-subgroup $N < G$. By induction, N is solvable. Further, since G/N is a p-group, and thus solvable, G is solvable as well. ■

2.6.5 The Schur-Zassenhaus Theorem

In this section, we will provide a full proof of the Schur-Zassenhaus theorem. We first deal with the case when the subgroup $N \trianglelefteq G$ is abelian.

Typically, our most powerful tool for finding subgroups is taking the kernel of a homomorphism. However, kernels are normal subgroups, and it is very possible that a complement is not normal. As such, we need to look at a more broad class of maps.

Definition 2.6.33. Let G and N be groups, and let G act on N (via automorphisms). A map $\phi : G \to N$ is said to be a *crossed homomorphism* if

$$\phi(gh) = \phi(g)(g \cdot \phi(h)).$$

If G acts on N trivially, then ϕ is a regular homomorphism, but groups may have more interesting crossed homomorphisms. For instance, if G acts on $N \trianglelefteq G$ by conjugation, then $g \mapsto [n, g]$ (where $n \in N$ is fixed) is a crossed homomorphism. Just as with regular homomorphisms, we define $\ker \phi = \{g \in G : \phi(g) = 1\}$.

Proposition 2.6.34. *Let G and N be groups, and let G act on N via automorphisms. Let $\phi : G \to N$ be a crossed homomorphism.*

(1) $\phi(1) = 1$

(2) $\ker \phi \leq G$

(3) $\phi(g) = \phi(h)$ *if and only if* $h^{-1}g \in \ker \phi$

(4) $[G : \ker \phi] = |\operatorname{im} \phi|$

Proof.

(1) We have that

$$\phi(1) = \phi(1 \cdot 1) = \phi(1)(1 \cdot \phi(1)) = \phi(1)^2,$$

so that $\phi(1) = 1$.

(2) Let $g, h \in \ker \phi$. Then

$$1 = \phi(g-1g) = \phi(g^{-1})(g^{-1} \cdot \phi(g)) = \phi(g^{-1})(g^{-1} \cdot 1) = \phi(g^{-1})$$

and

$$\phi(gh) = \phi(g)(g \cdot \phi(h)) = g \cdot 1 = 1.$$

(3) Suppose that $g = hk$ for $k \in \ker \phi$. Then

$$\phi(g) = \phi(hk) = \phi(h)(h \cdot \phi(k)) = \phi(h)(h \cdot 1) = \phi(h).$$

Conversely, if $\phi(g) = \phi(h)$, then

$$\phi(h^{-1}g) = \phi(h^{-1})(h^{-1} \cdot \phi(g)) = \phi(h^{-1})(h^{-1} \cdot \phi(h)) = \phi(h^{-1}h) = 1.$$

(4) This follows from part (3).

■

Recall that a transversal T of N consists of a representative of each coset of N (since N is normal, left and right cosets coincide). Write $x \equiv y$ when $xN = yN$. If S and T are transversals of N, then there is exactly one $t \in T$ with $s \equiv t$ for each $s \in S$. We can then define

$$d(S,T) = \prod_{\substack{s \equiv t \\ s \in S, t \in T}} st^{-1},$$

and note that $d(S,T) \in N$. Since we are working in the case when N is abelian, this is a well-defined value. We immediately see that $d(S,T)d(T,U) = d(S,U)$, $d(gS,gT) = gd(S,T)g^{-1}$, and $d(nS,S) = n^{[G:N]}$ for all $n \in N$ (where we use the fact that N is normal to see that $nsN = sN$). This function will give rise to our necessary crossed homomorphism.

Lemma 2.6.35. *Let $N \trianglelefteq G$ be abelian with* $\gcd(|N|, [G:N]) = 1$.

(1) N is complemented in G

(2) All complements of N are conjugate

Proof.

(1) Fix a transversal T of N, and let $\theta : G \to N$ be given by $\theta(g) = d(gT,T)$. We first show that θ is indeed a crossed homomorphism with respect to the action of G on N via conjugation. We have that

$$\theta(gh) = d(ghT,T) = d(ghT,gT)d(gT,T) = gd(hT,T)g^{-1}d(gT,T) = (g \cdot \theta(h))\theta(g) = \theta(g)(g \cdot \theta(h)),$$

where we use the fact that N is abelian. Now, for $n \in N$, $\theta(n) = d(nT,T) = n^{[G:N]}$. Since $[G:N]$ and $|N|$ are relatively prime, the map $n \mapsto n^{[G:N]}$ is a bijection from N to itself, so that θ is surjective. If $H = \ker\theta$, then $|N| = |\operatorname{im}\theta| = [G:H]$, so that $|H| = [G:N]$, as desired.

(2) Let K complement N. Since $NK = G$ and $N \cap K = 1$, we see that K is a transversal of N. Let $m = d(T,K)$. Since $\theta : N \to N$ is surjective, we can find $\theta(n) = m$ for $n \in N$.

We show that $nKn^{-1} \le H$. Note that, if $k \in K$, then

$$kmk^{-1} = kd(T,K)k^{-1} = d(kT,kK) = d(kT,K) = d(kT,T)d(T,K) = \theta(k)m.$$

Thus,

$$\theta(nkn^{-1}) = \theta(n)(n \cdot \theta(kn^{-1})) = \theta(n)\theta(kn^{-1}) = m\theta(k)(k \cdot \theta(n^{-1})) = (kmk^{-1})(km^{-1}k^{-1}) = 1,$$

where we use the fact that N is abelian to ignore conjugation by n and to commute m and $\theta(k)$.

∎

Theorem 2.6.36 (Schur-Zassenhaus)**.** *Let G be a finite group and $N \trianglelefteq G$ such that* $\gcd(|N|, [G:N]) = 1$.

(1) N is complemented in G

(2) If N is solvable or G/N is solvable, then all complements of N are conjugate

Proof.

(1) We proceed by induction on $|G|$, with $|G| = 1$ being trivial. If there is $K < G$ with $NK = G$, then $[K : K \cap N] = [G:N]$ is relatively prime to $|N|$, and so to $|K \cap N|$. By induction, $K \cap N \trianglelefteq K$ is complemented in K, say by $H \le K$. Then $|H| = [K : K \cap N] = [G:N]$, so N is complemented in G.

Thus, we may assume that $NK < G$ for all $K < G$. In particular, $NM = M$ for all maximal $M < G$, so that $N \le \Phi(G)$ the Frattini subgroup. Since $\Phi(G)$ is nilpotent, N is as well. Assume $N > 1$ (since 1 is complemented by G). Then $1 < Z(N) \trianglelefteq G$. Now, $[G/Z(N) : N/Z(N)]$ is coprime to $|N|$, and so to $|N/Z(N)|$, so induction gives $K/Z(N)$ with $G/Z(N) = (N/Z(N))(K/Z(N)) = NK/Z(N)$. Thus, $G = NK$, so that $K = G$. Since $K/Z(N)$ is a complement for $N/Z(N)$, this means that $K/Z(N) \cap N/Z(N) = 1$, so that $K \cap N = Z(N)$. But $K = G$, so $N = Z(N)$ is abelian, and we are done.

(2) We proceed by induction on $|G|$. Let $H, K \leq G$ with $|H| = |K| = [G : N]$. Suppose first that we have $H, K \leq U < G$ for some U. If N is solvable, then $U \cap N$ is solvable, and if G/N is solvable, then $U/U \cap N \cong UN/N \leq G/N$ is solvable, so we may apply induction to U. We claim that H complements $U \cap N \trianglelefteq U$ in U. Indeed, $|H|$ divides $|U|$ and not $|U \cap N|$, and therefore divides $[U : U \cap N]$. But $[U : U \cap N] = [UN : N]$ divides $[G : N] = |H|$, so $|H| = [U : U \cap N]$. Symmetrically, $|K| = [U : U \cap N]$. By the inductive hypothesis on U, H and K are conjugate in U, and thus in G.

Now, let $1 < L \trianglelefteq G$. If N is solvable, then so is $NL/L \cong N/N \cap L$, and if G/N is solvable, then so is $(G/L)/(NL/L) \cong G/NL \cong (G/N)/(NL/N)$, so we may apply induction to G/N. We have that $HL/L \cap NL/L = 1$ and $(HL/L)(NL/L) = (HN)/L = G/L$, so HL/L complements NL/L in G/L, and similarly KL/L complements NL/L in G/L. By induction, HL/L and KL/L are conjugate in G/L, and so HL And KL are conjugate in G, say $HL = gKLg^{-1}$. Thus, HL contains H and gKg^{-1}. If $HL < G$, we are done taking $U = HL$. Thus, we may assume $HL = G$ for all $1 < L \trianglelefteq G$.

Suppose first that G/N is solvable. If $N = G$, then $H = K = 1$, so suppose that $N < G$. Let $M/N \trianglelefteq G/N$ be minimal, so that M/N is a p-group for some prime p. In particular, p divides $[G : N]$, and so doesn't divide $|N|$. Since $G = NH$ and $N \subseteq M$, we see that $M = N(M \cap H)$, so that $M \cap H$ complements N in M (as $M \cap H \cap N = 1$). Thus, $|M \cap H| = [M : N]$ and p doesn't divide $|N| = [M : M \cap H]$, so that $M \cap H$ is a Sylow p-subgroup of M. Similarly, $M \cap K$ is a Sylow p-subgroup of M, and so $M \cap H = m(M \cap K)m^{-1} = M \cap mKm^{-1}$ for some $m \in M$.

If $L = M \cap H = M \cap mKm^{-1}$, then $L \trianglelefteq H, mKm^{-1}$, so that H and mKm^{-1} are complements for N contained in $N_G(L)$. If $N_G(L) < G$, we are done. Otherwise, if $N_G(L) = G$, then $L \trianglelefteq G$, and so $G = HL = H$, and so $G = K$ as well.

Suppose now that N is solvable, and take $N > 1$ (as otherwise $H = K = G$). Let $L \leq N$ be a minimal normal subgroup of G. Then L is solvable and isomorphic to a direct product of simple groups, and thus L is abelian. Further, $HL = G$. Since $L \leq N$ and $N \cap H = 1$, we must have that $L = N$, so that N is abelian. Thus, in this case, all complements of N are conjugate.

■

Note that, since N and G/N must have relatively prime orders, at least one of them must be odd. Thus, by the Feit-Thompson theorem, at least one of them is solvable, and so all complements of N are always conjugate.

2.6.6 Wreath Products

Before defining the wreath product, we will provide a motivating example. Let S be a finite nonabelian simple group, and consider $\mathrm{Aut}(S^n)$. If $\sigma \in \mathrm{Aut}(G)$, we have n distinct automorphisms given by acting σ on each coordinate. For each permutation in S_n, we also have a distinct automorphism by permuting the coordinates. We claim that these are essentially the only things we can do (that is, that every automorphism of G^n comes from applying automorphisms to each coordinate, and then permuting the coordinates)[9].

Suppose that we have $\sigma \in \mathrm{Aut}(G)$. Write G_i for the ith copy of G, and let $\pi_i : G^n \to G_i$ be the projection onto the ith coordinate. Suppose that $\sigma(G_1) \neq G_i$ for all i (so that σ does not perform automorphisms on each coordinate and then permute the coordinates). Then $\pi_1 \circ \sigma^{-1}$ is a surjection $G^n \to G$ that is not of the form $\rho \circ \pi_i$ for some i and some $\rho \in \mathrm{Aut}(G)$, so it suffices to show that this cannot happen.

Let $\pi : G^n \to G$ be any surjective homomorphism. Then we have restrictions $\pi|_{G_i} : G \to G$, and at least one of these must be nontrivial (as otherwise π would be trivial), say $\pi|_{G_1}$. Since $\pi|_{G_1} : G \to G$ is a nontrivial map from a simple group to itself, it must be an automorphism (in particular, surjective). Now, for $g \in G_i$ with $i \neq 1$, g commutes with all elements of G_1, and so $\pi(g)$ commutes with all elements of $\pi(G_1) = G$. Since G is nonabelian and simple, $Z(G) = 1$, and so $\pi(g) = 1$. Thus, $\pi = \pi|_{G_1} \circ \pi_1$ is a projection of the first coordinate followed by an automorphism of G, as desired.

Thus, we have seen that $\mathrm{Aut}(G^n)$ contains a copy of $\mathrm{Aut}(G)^n$ and a copy of S_n that acts on $\mathrm{Aut}(G)^n$ by permutation, and that $\mathrm{Aut}(G^n)$ is generated by these subgroups. This idea of having one group act on another by permutation is the key notion of a *wreath product*.

Definition 2.6.37. Let G and H be groups, and let G act on a set X. The *wreath product of G and H* is

$$H \wr_X G = H^X \rtimes G,$$

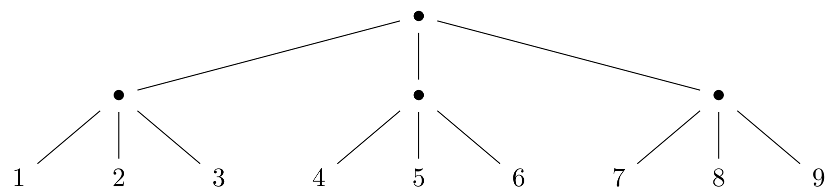
where G acts on H^X by permuting the components according to the action of G on X.

[9]Note this is not true when G is abelian. For instance, $\mathrm{Aut}(\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}) \cong S_3$ is not of this form

Example. We have just seen that
$$\mathrm{Aut}(G^n) = \mathrm{Aut}(G) \wr S_n$$
whenever G is a nonabelian simple group.

Commonly, $H \wr G$ means the wreath product $H \wr_G G$, where G is acting on itself by left multiplication. The exception to this is when $G = S_n$ or $G = A_n$, in which case G will be acting on the set $\{1, \dots, n\}$.

We will provide another example of wreath products by constructing the Sylow p-subgroup of S_{p^n}. We note that there are $1 + p + \cdots + p^{n-1}$ factors of p dividing $|S_{p^n}| = p^n!$. We group $\{1, \dots, p^n\}$ into p^{n-1} groups of p, then collect these groups of p into p^{n-2} collections each containing p groups, then collect these into p^{n-3} collections each containing p of the previous collections, and so on, as below (for $p = 3$ and $n = 2$).



Now, we may cyclically permute (i.e. act $\mathbb{Z}/p\mathbb{Z}$ on) each of groups of size p, and we may cyclically permute up one more layer on each of the collections of groups of size p, and then cyclically permute again on each of the collections of collections of groups of size p, and so on[10]. This shows us that we have a copy of

$$((\mathbb{Z}/p\mathbb{Z} \wr \mathbb{Z}/p\mathbb{Z}) \wr \mathbb{Z}/p\mathbb{Z}) \wr \cdots \wr \mathbb{Z}/p\mathbb{Z},$$

where we take $n-1$ wreath products (we'll denote this now by $\mathbb{Z}/p\mathbb{Z} \wr^{n-1} \mathbb{Z}/p\mathbb{Z}$). Since $|H \wr_X G| = |H|^{|X|} \cdot |G|$, we can see that $|\mathbb{Z}/p\mathbb{Z} \wr^{n-1} \mathbb{Z}/p\mathbb{Z}| = p^{p^{n-1}+\cdots+p+1}$, so this is indeed a Sylow p-subgroup.

More generally, if the base p expansion of m is $a_n p^n + \cdots + a_1 p + a_0$, we can construct the Sylow p-subgroup of S_m by partitioning $\{1, \dots, m\}$ to have a_i sets of size p^i, and then constructing a wreath product on each of these sets of size p^i, so that a Sylow p-subgroup of S_m is isomorphic to

$$(\mathbb{Z}/p\mathbb{Z} \wr^{n-1} \mathbb{Z}/p\mathbb{Z})^{a_n} \times (\mathbb{Z}/p\mathbb{Z} \wr^{n-2} \mathbb{Z}/p\mathbb{Z})^{a_{n-1}} \times \cdots \times (\mathbb{Z}/p\mathbb{Z} \wr \mathbb{Z}/p\mathbb{Z})^{a_2} \times (\mathbb{Z}/p\mathbb{Z})^{a_1}.$$

Further, since all Sylow p-subgroups are conjugate, they all arise in this way for an appropriate grouping of $\{1, \dots, m\}$.

We end with a result that becomes quite useful in finite group theory.

Theorem 2.6.38 (Mattarei's Lemma)**.** *Let G be a finite group containing no nontrivial solvable normal subgroups. Then there are nonabelian simple groups $T_1, \dots, T_r$ and $n_1, \dots, n_r \geq 1$ such that G contains an isomorphic copy of $T_1^{n_1} \times \cdots \times T_r^{n_r}$ and is isomorphic to a subgroup of*

$$(\mathrm{Aut}(T_1) \wr S_{n_1}) \times \cdots \times (\mathrm{Aut}(T_r) \wr S_{n_r}) \cong \mathrm{Aut}(T_1^{n_1}) \times \cdots \times \mathrm{Aut}(T_r^{n_r}) \leq \mathrm{Aut}(T_1^{n_1} \times \cdots \times T_r^{n_r}).$$

Proof. Let $N_1, \dots, N_r$ be the minimal normal subgroups of G, and let $N = N_1 \cdots N_r$. Recall by Proposition 2.4.50 that we may write $N_i \cong T_i^{n_i}$ for some simple group T_i, and note that T_i must be nonabelian since N_i is not solvable. In particular, $Z(N_i) = 1$. Now, for $i \neq j$, $N_i \cap N_j = 1$, and so elements of N_i and N_j commute, meaning that elements of N_i and $\prod_{j \neq i} N_j$ commute. Thus, $N_i \cap \prod_{j \neq i} N_j \leq Z(N_i) = 1$, so that N is the internal direct product of the N_i: $N \cong N_1 \times \cdots \times N_r$.

Now, $N \trianglelefteq G$, and so we have a homomorphism $G \to \mathrm{Aut}(N)$ with kernel $Z_G(N)$, where g maps to $\varphi_g : n \mapsto gng^{-1}$. Since elements of $Z_G(N)$ commute with elements of N,

$$Z_G(N) \cap N = Z(N) \cong Z(N_1) \times \cdots \times Z(N_r) = 1.$$

Note that $Z_G(N)$ is normal in G, as it is a kernel of a homomorphism. Thus, if $Z_G(N) \neq 1$, then it would contain a minimal normal subgroup, and so we would have $Z_G(N) \cap N \neq 1$, an impossibility. This means, $Z_G(N) = 1$, and so we can embed G into $\mathrm{Aut}(N)$.

Finally, since each N_i is normal in G, it is fixed by the action of G, so that the image of G lies in

$$\mathrm{Aut}(N_1) \times \cdots \times \mathrm{Aut}(N_r) \cong (\mathrm{Aut}(T_1) \wr S_{n_1}) \times \cdots \times (\mathrm{Aut}(T_r) \wr S_{n_r}),$$

as desired. ∎

[10]So that we have a copy of $\mathbb{Z}/p\mathbb{Z}$ at each node permuting the branches

Note that the T_i need not be pairwise nonisomorphic in the above result, and so we in fact embed G into a proper subgroup of $\mathrm{Aut}(N)$. In the case that the T_i are pairwise nonisomorphic, we have

$$\mathrm{Aut}(T_1^{n_1} \times \cdots \times T_r^{n_r}) \cong \mathrm{Aut}(T_1^{n_1}) \times \cdots \times \mathrm{Aut}(T_r^{n_r}).$$

Indeed, composing any automorphism of $T_1^{n_1} \times \cdots \times T_r^{n_r}$ with inclusion and projection maps gives a homomorphism $T_i \to T_j$ whose image is a normal subgroup of T_j, which is necessarily the trivial homomorphism (forcing the automorphism to restrict to automorphisms of $T_i^{n_i}$ for all i).

2.6.7 The 5/8ths Theorem

We first state a quite useful result.

Proposition 2.6.39. *If $G/Z(G)$ is cyclic then G is abelian.*

Proof. Suppose that the coset $xZ(G)$ generates $G/Z(G)$. Then, for any $g \in G$, we can write $g = x^i z$ for some $z \in Z(G)$. But elements of this form commute. ■

Example. This result shows that all groups of order p^2 are abelian. If $|Z(G)| = p^2$, we are done. We cannot have $Z(G) = 1$, and if $|Z(G)| = p$, then $|G/Z(G)| = p$, so $G/Z(G)$ is cyclic, meaning that G is abelian.

The key point in the proof of the 5/8ths theorem is that, if G is nonabelian, then $[G : Z(G)] \geq 4$ (as otherwise $G/Z(G)$ would be cyclic). It is also a fact of interest of itself that it is impossible for $[G : Z(G)] = 2, 3$.

Theorem 2.6.40 (5/8ths Theorem)**.** *Let G be a finite group. If the probability of any two elements of G commuting is $> 5/8$, then G is abelian.*

Proof. Let G be a nonabelian finite group, and let $g, h \in G$. If $g \in Z(G)$, then g and h automatically commute. If $g \notin Z(G)$, then g and h commute if and only if $h \in Z_G(g)$, which happens with probability $1/[G : Z_G(g)] \leq 1/2$ (since $Z_G(g)$ is a proper subgroup of G). Since the probability that $g \in Z(G)$ is $1/[G : Z(G)]$, the probability of g and h commuting is at most

$$\frac{1}{[G : Z(G)]} + \left(1 - \frac{1}{[G : Z(G)]}\right) \cdot \frac{1}{2} = \frac{1}{2} + \frac{1}{2[G : Z(G)]} \leq \frac{1}{2} + \frac{1}{2 \cdot 4} = \frac{5}{8}.$$

■

The quaternion group Q_8 shows that this bound is sharp: $[Q_8 : Z(Q_8)] = 4$, and each noncentral element has centralizer of size 4.